

LECTURE 2 ANATOMY

-Skin-fascia-serous Sacs-muscles-joints

-Digestive System

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Faculty of medicine.

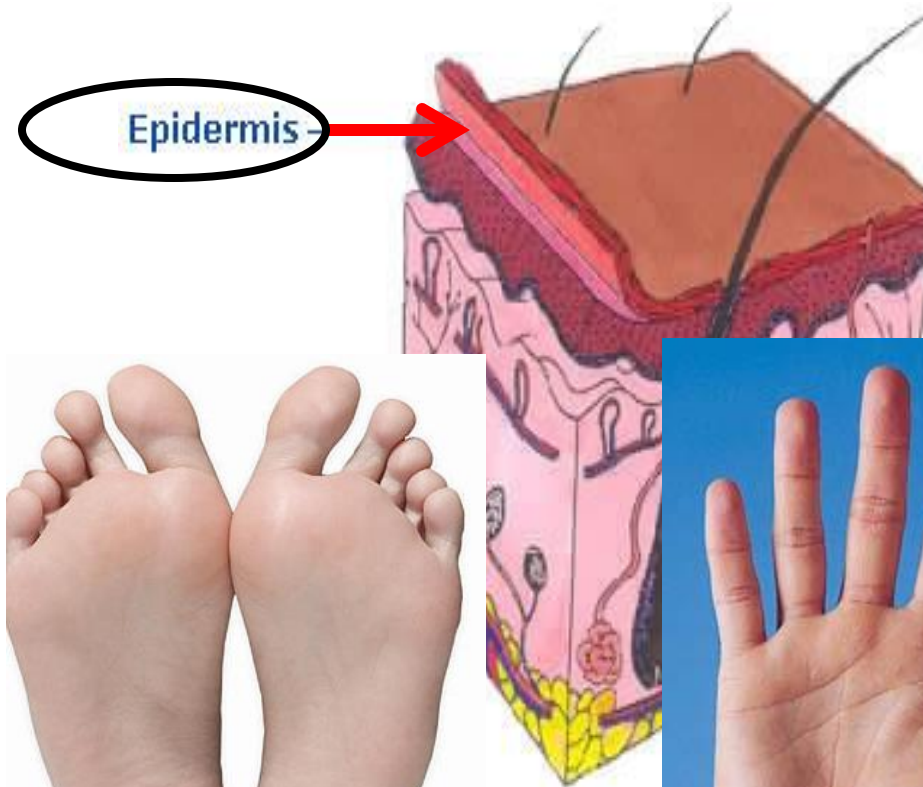
Ain Shams University

The Skin, Fascia & Serous Membranes

THE SKIN

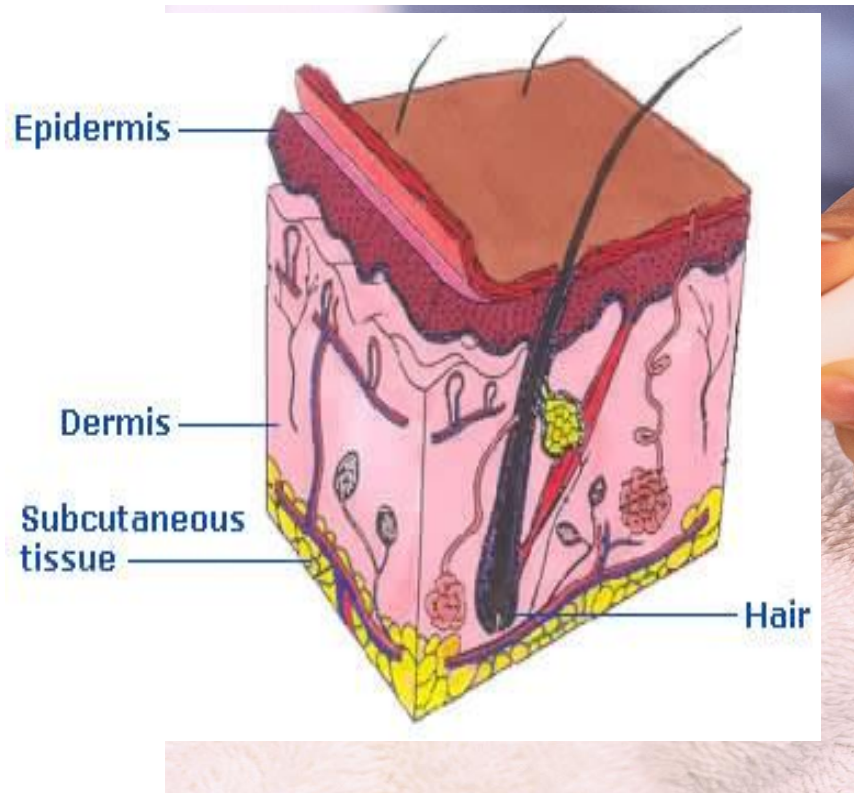
To withstand wear & tear that occur in these regions

- **Epidermis** is the superficial layer & is extremely thick on palms of hands & soles of feet . **WHY??**



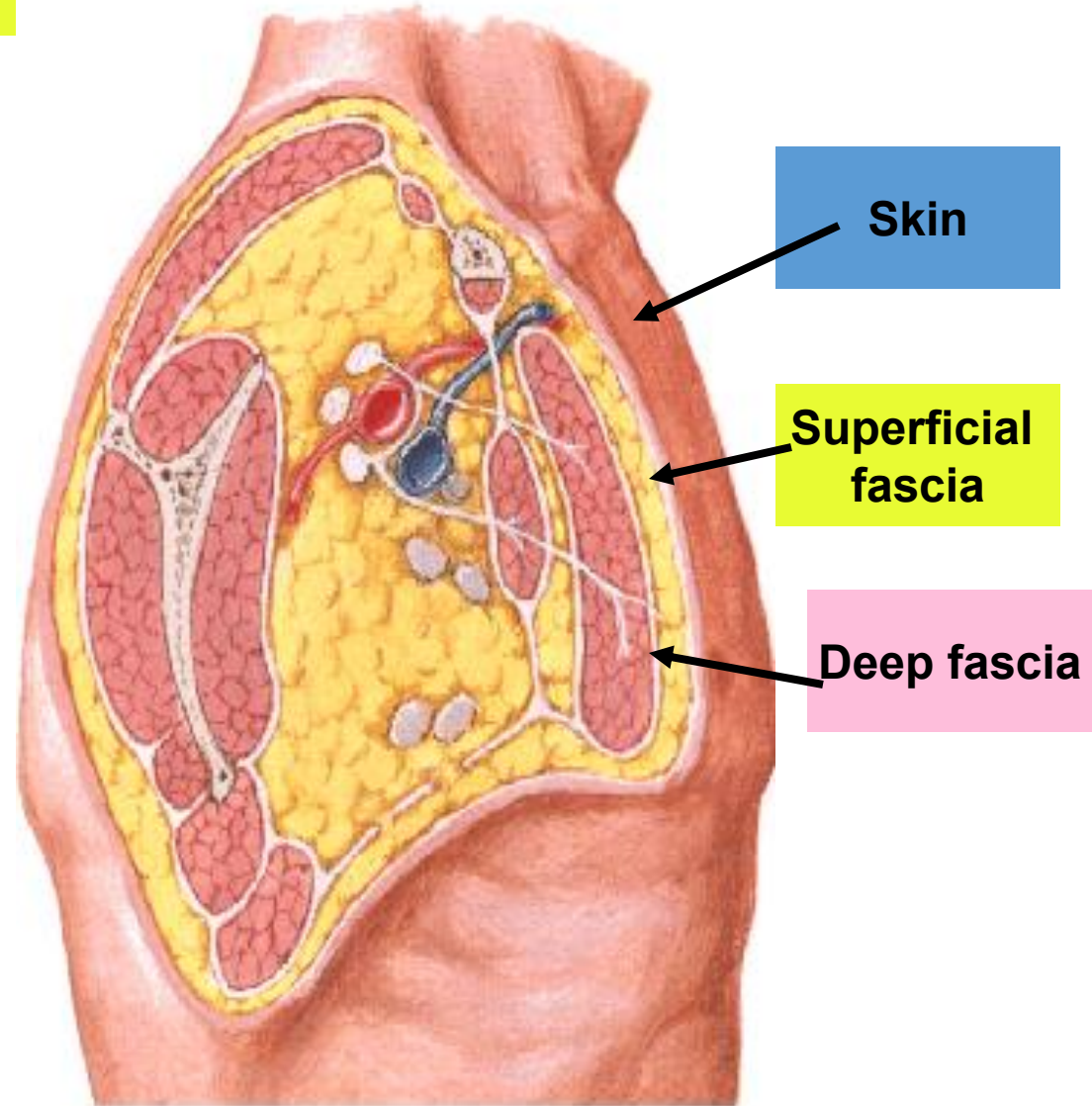
SKIN APPENDAGES

- These include:
nails, hair
follicles,
sebaceous glands
& sweat glands.



The FASCIA

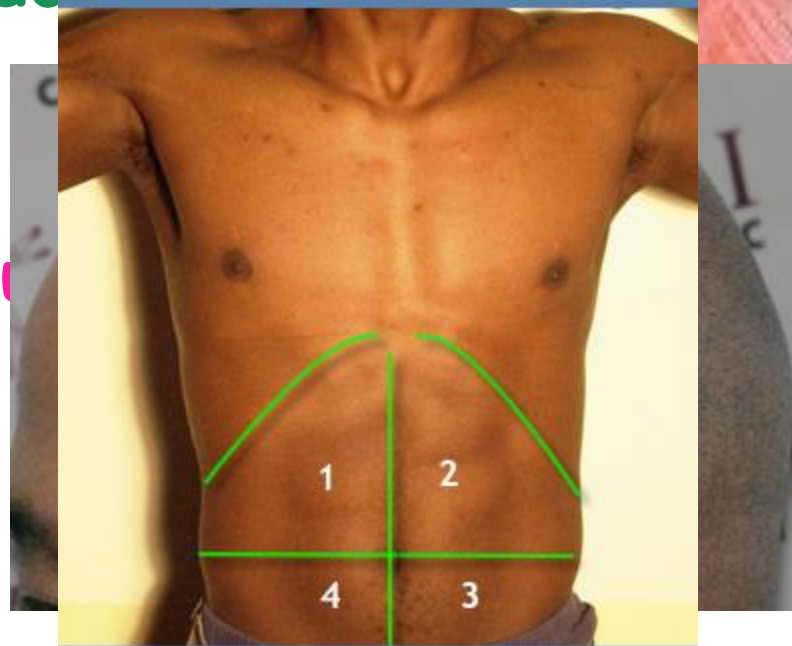
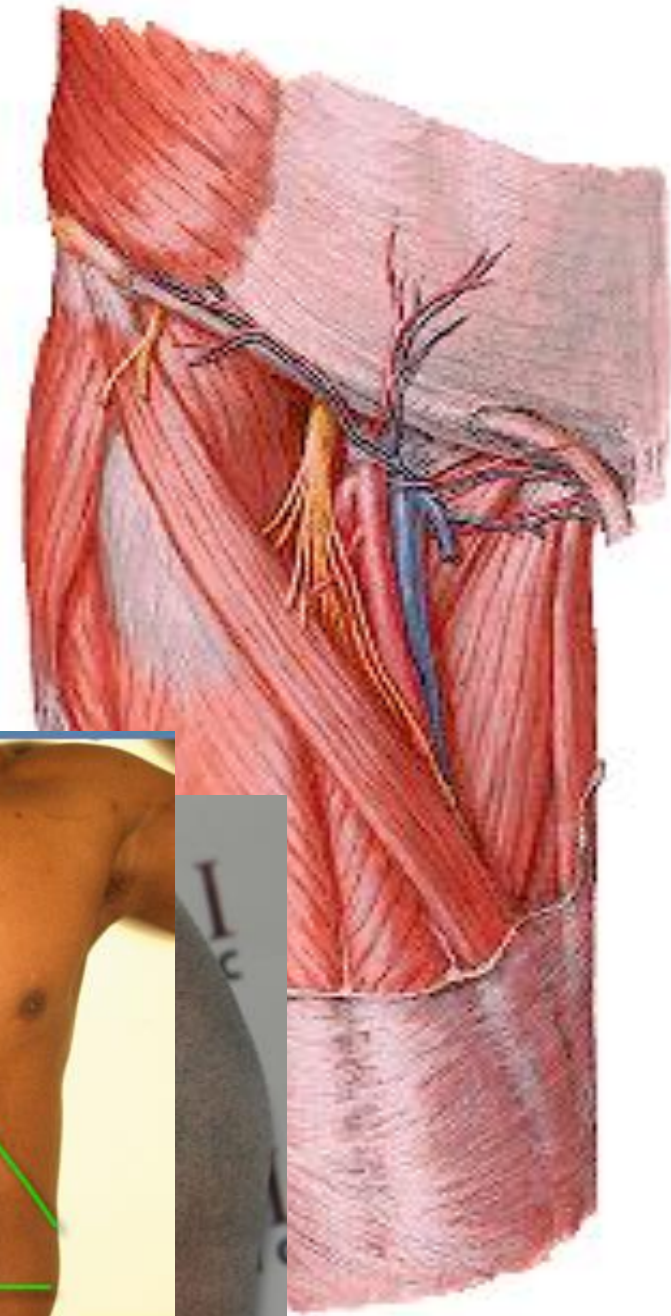
- It is the tissue which lies immediately under the skin.
- It is divided into:
 - (a) Superficial fascia
 - (b) Deep fascia



DEEP FASCIA

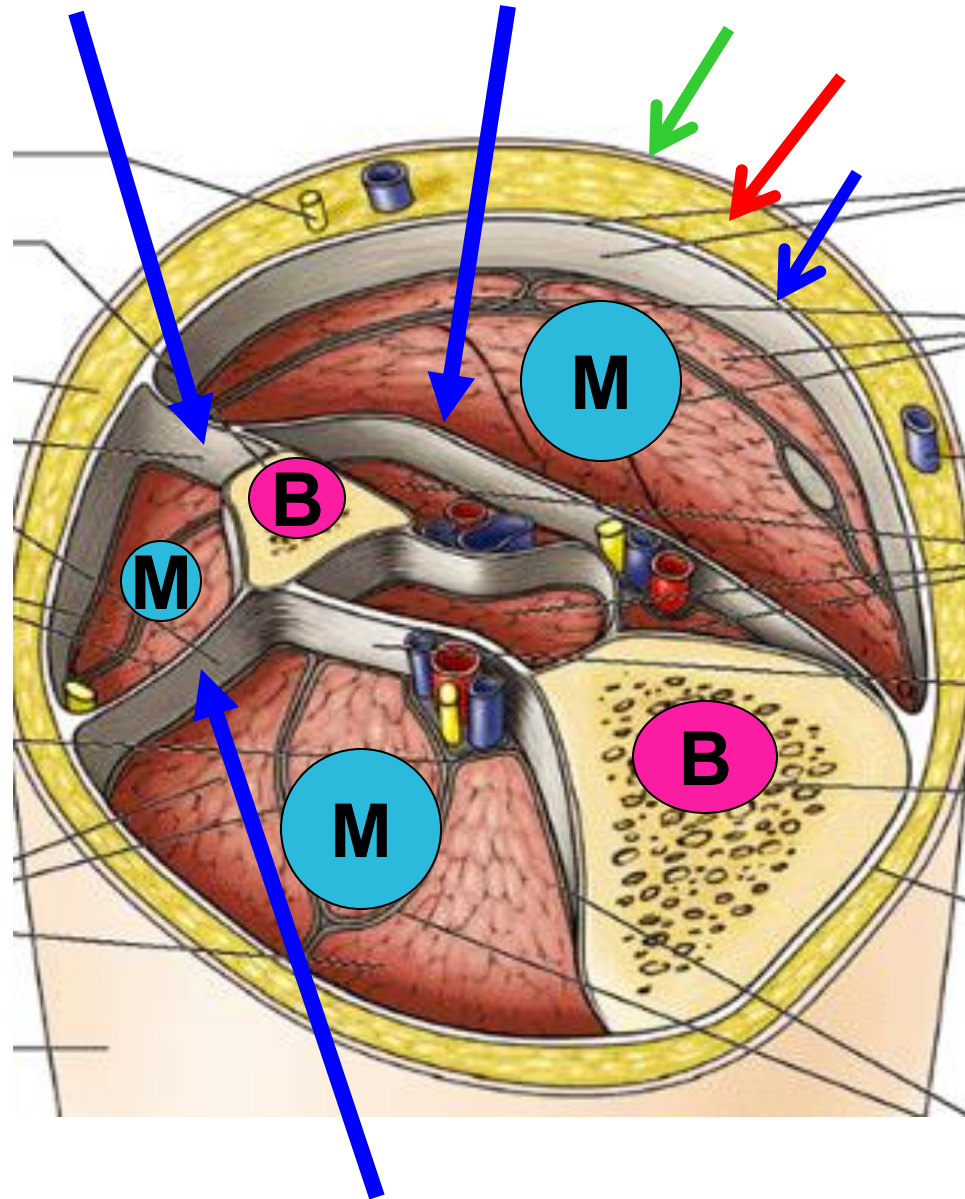
- Dense fibrous membrane that covers most of the body, **except: face, scalp and anterior abdominal wall**

- It covers the m



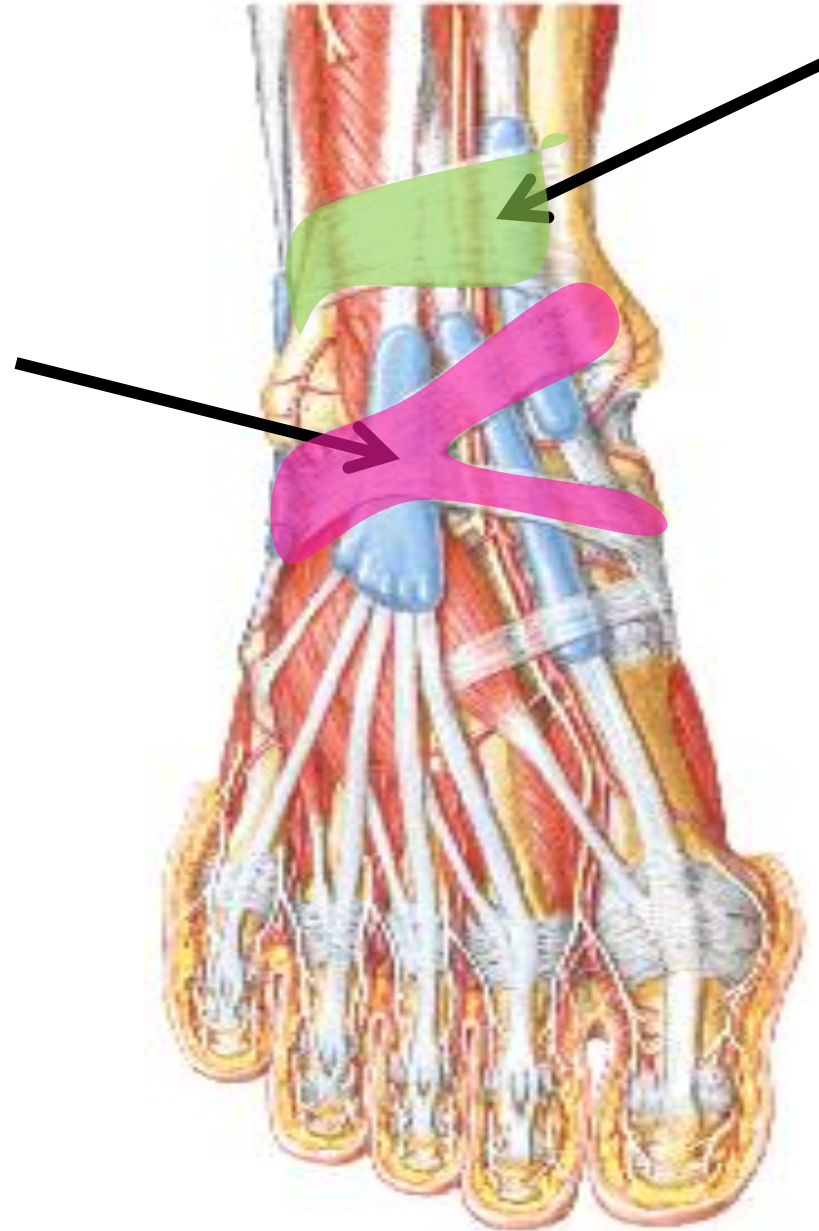
DEEP FASCIA

- In limbs deep fascia sends **intermuscular septa** which attach to the bones & so divide the muscles into different functional & regional groups



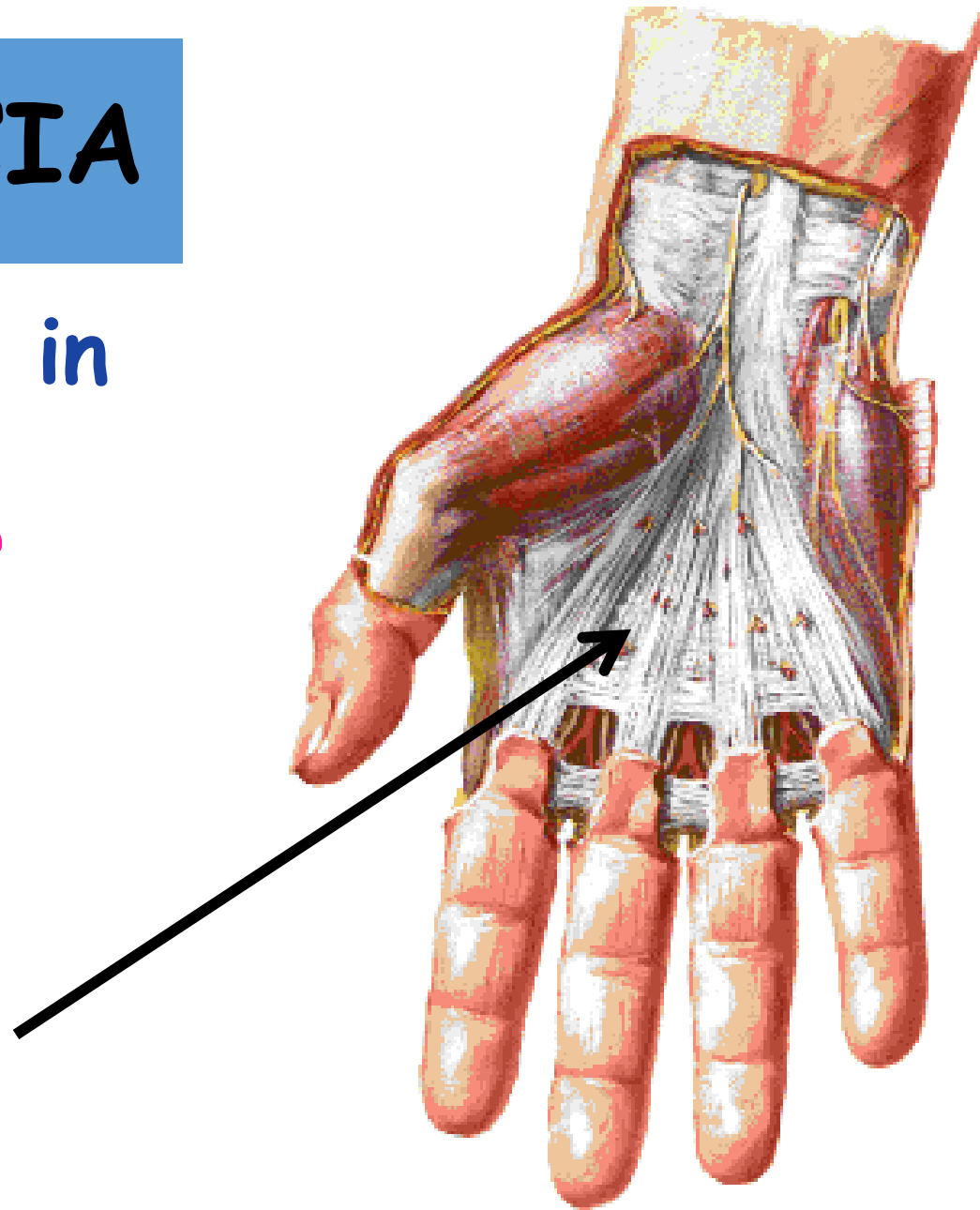
DEEP FASCIA

- It is thickened around distal joints to form **retinacula**.
- Retinacula help to fix muscle tendons in position



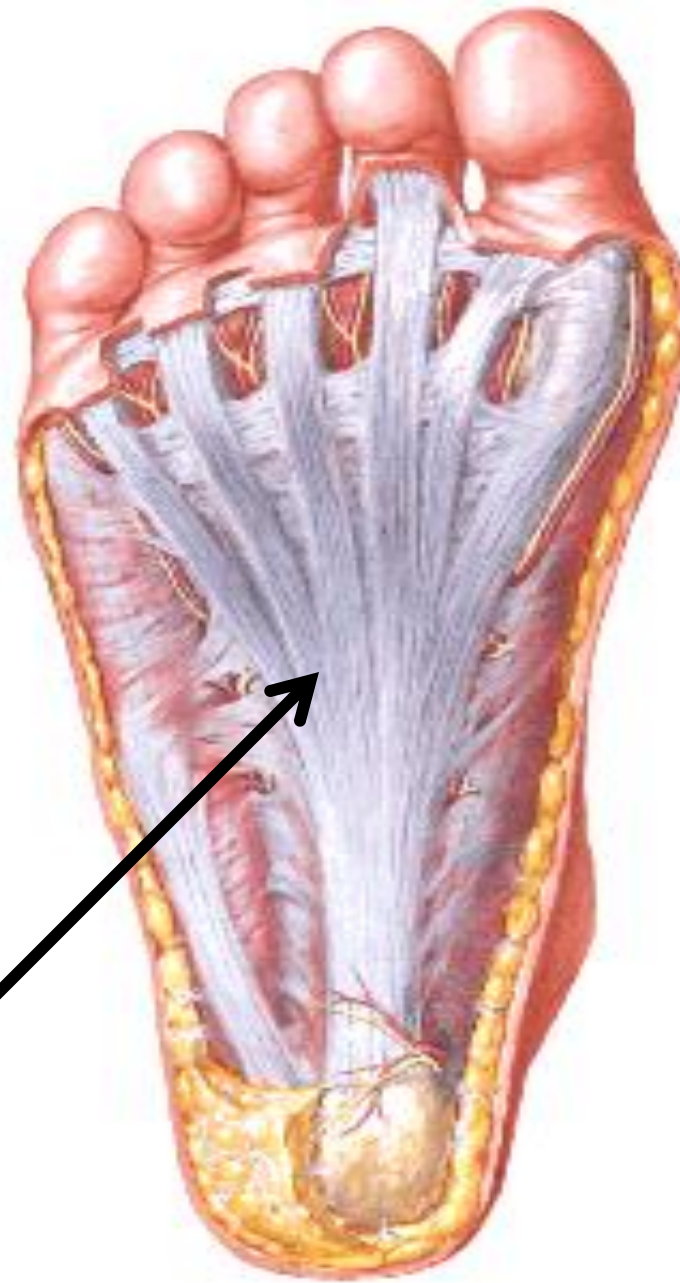
DEEP FASCIA

- It is thickened in palms of hands to form **palmar aponeurosis**.



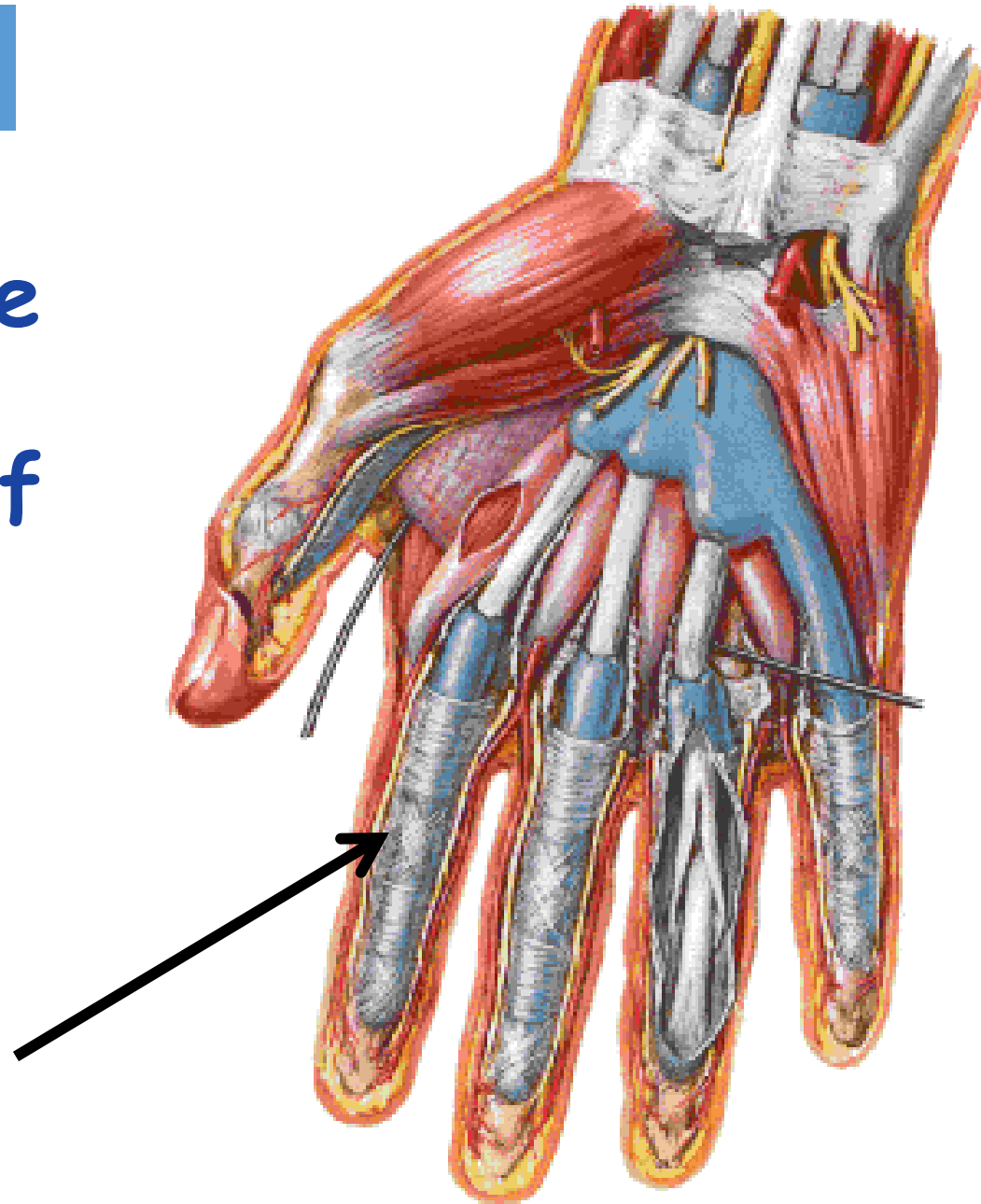
DEEP FASCIA

- It is thickened in soles of feet to form **plantar aponeurosis**.



DEEP FASCIA

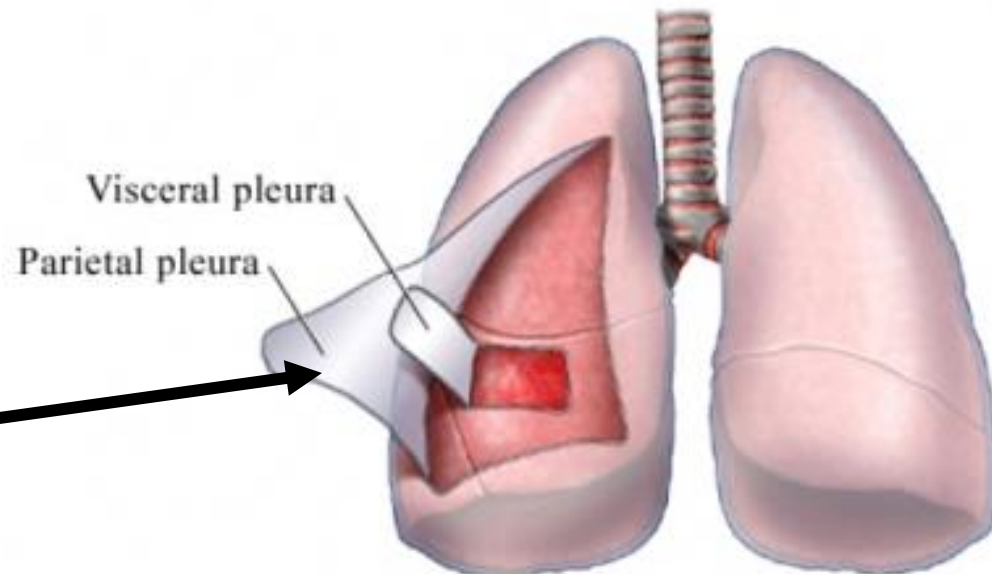
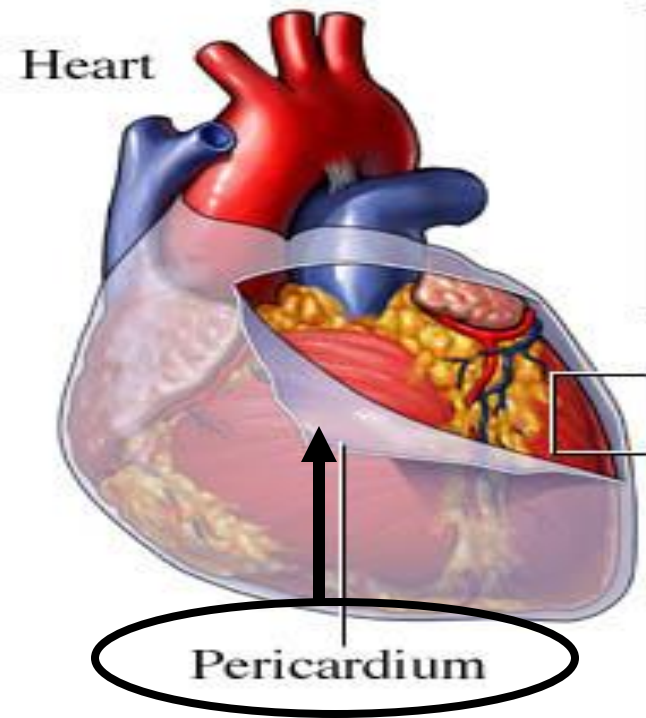
- Flexor tendons are fixed in the fingers & toes by thickening of deep fascia called **fibrous flexor sheaths**.

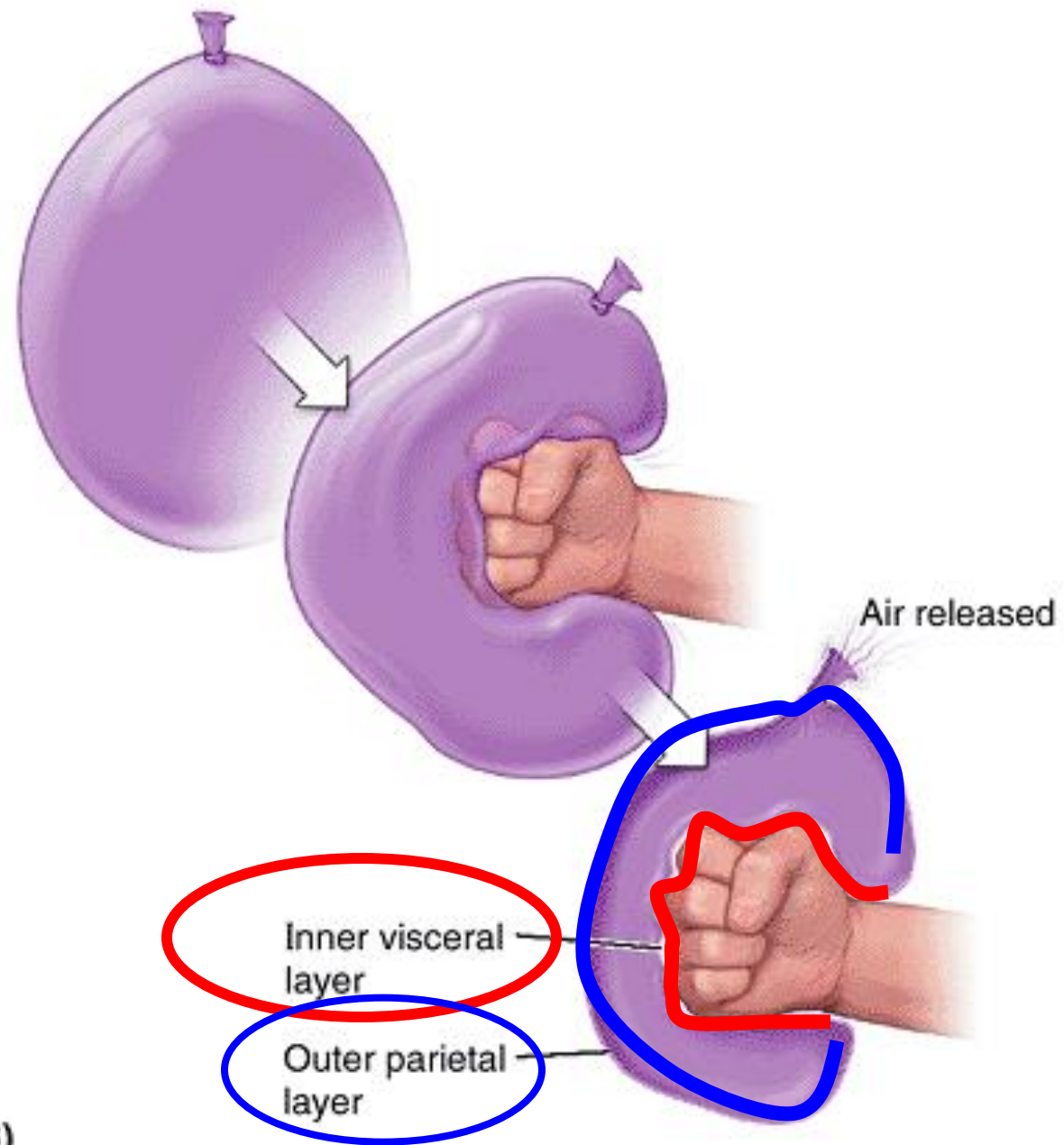


SEROUS MEMBRANES

- Closed spaces lined by a thin membrane & contain a thin film of serous fluid. E.g.
 1. **Pericardium** → surrounds heart
 2. **Pleura** → surrounds lungs

Pleura



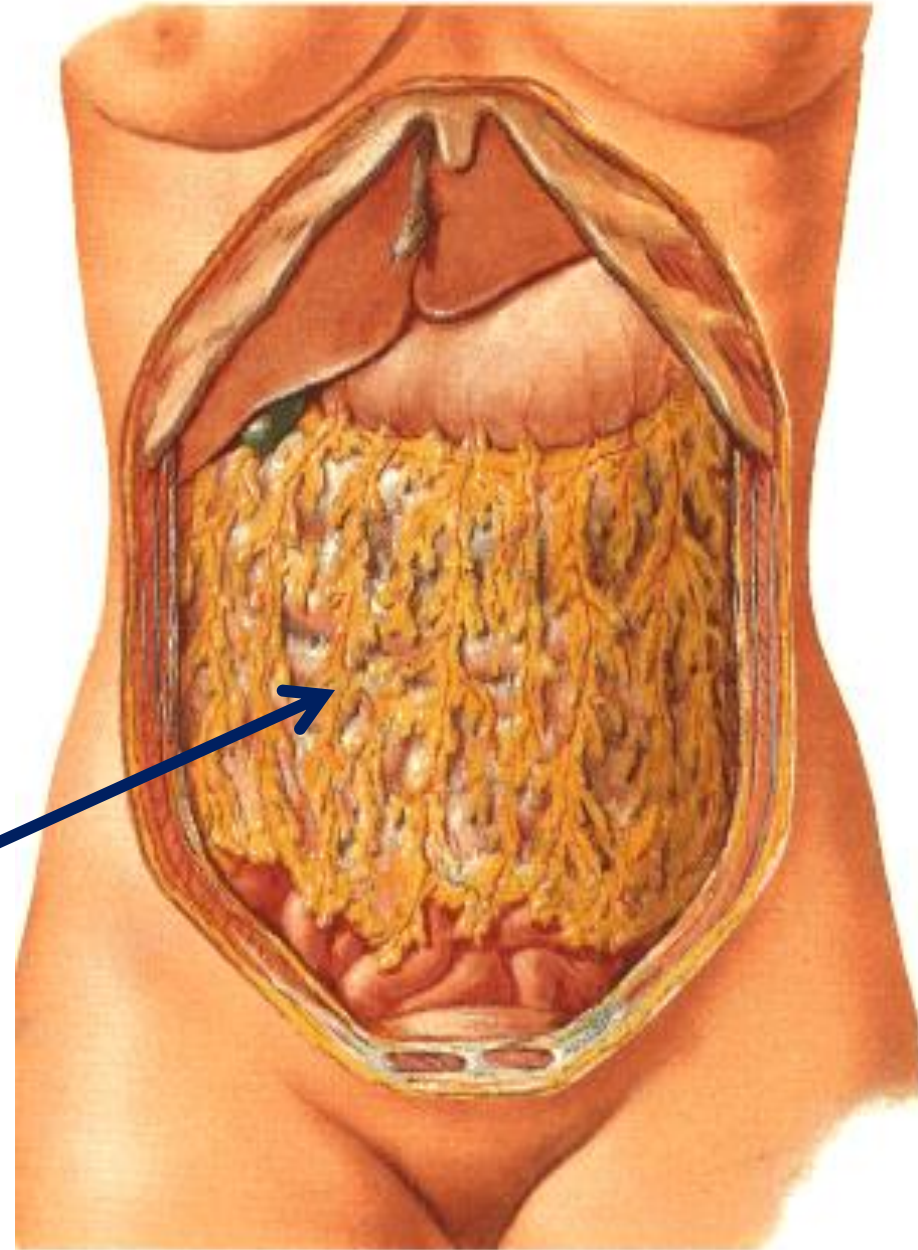


(B)

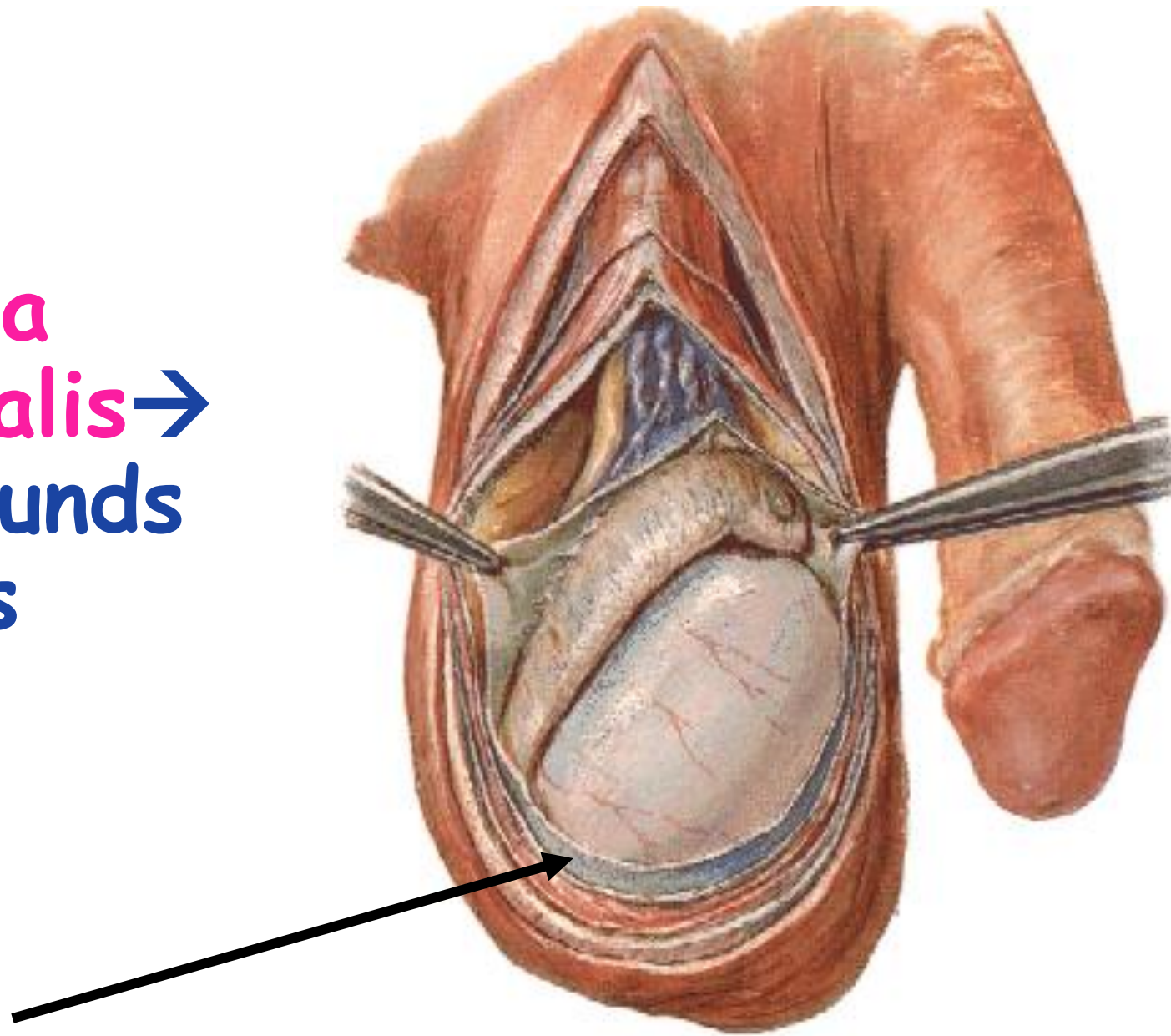
3. Peritoneum



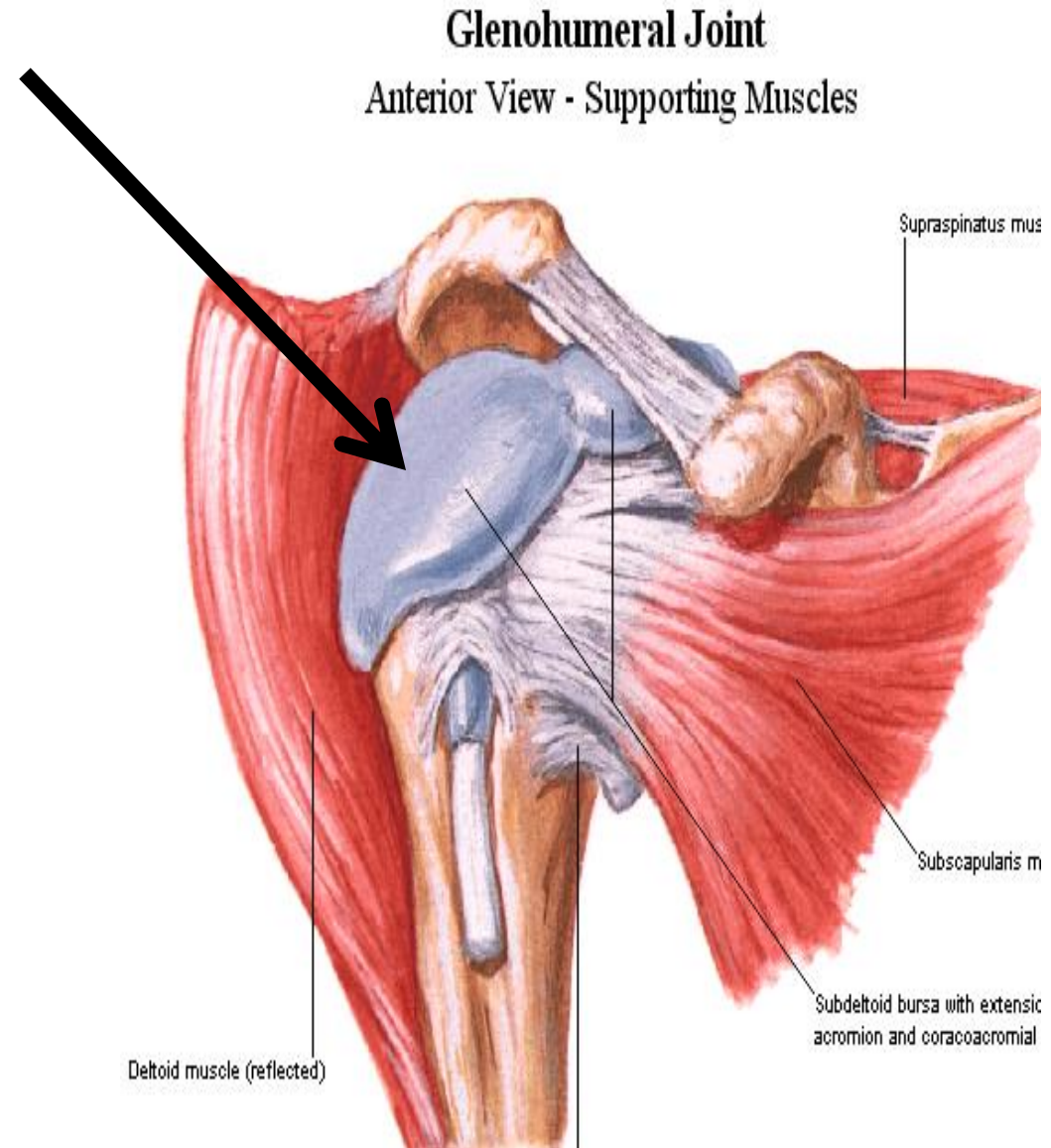
surrounds
abdominal
contents



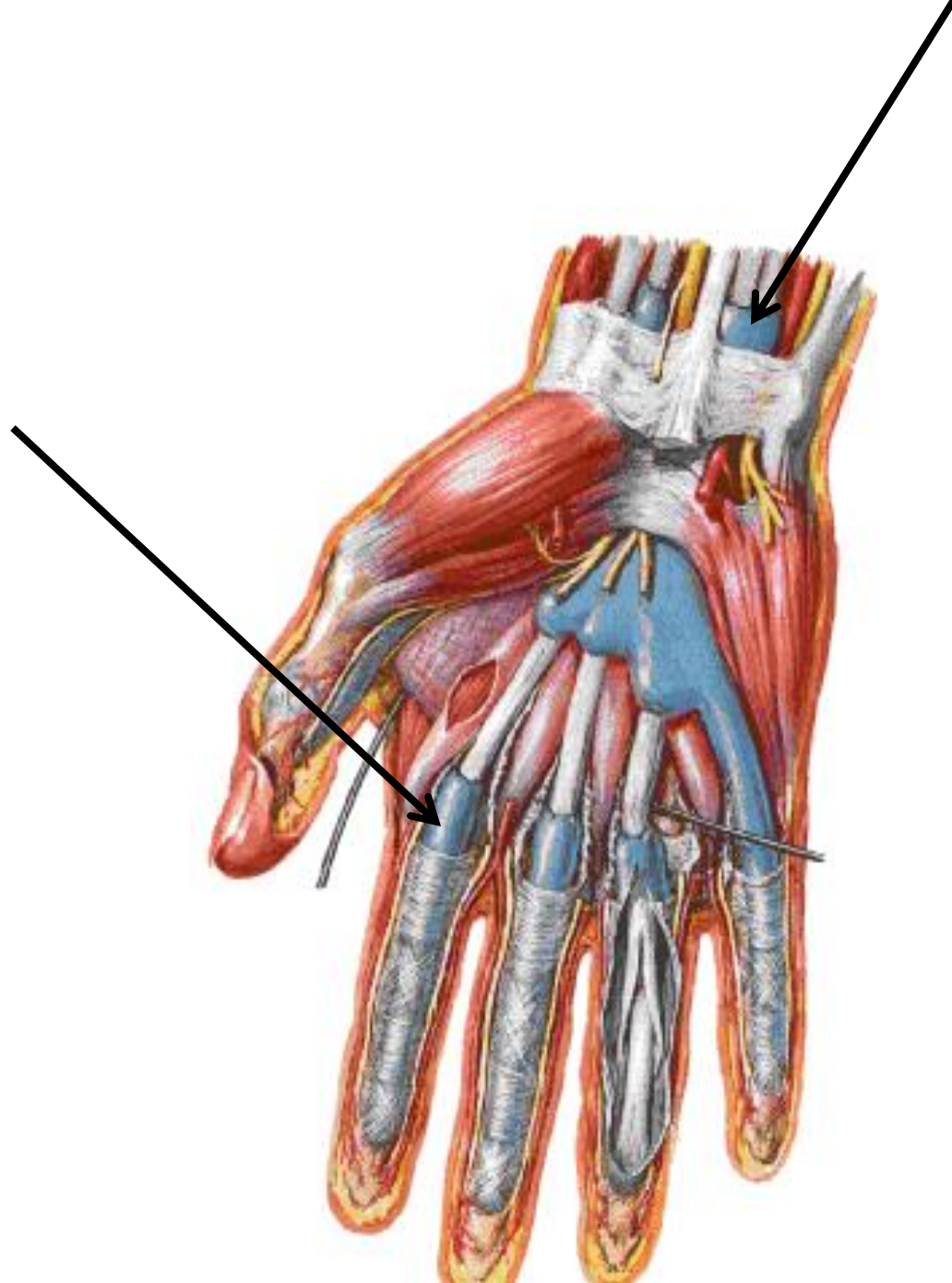
4. Tunica
vaginalis →
surrounds
testis



5. Bursae →
prevents
friction
between
muscles &
bones



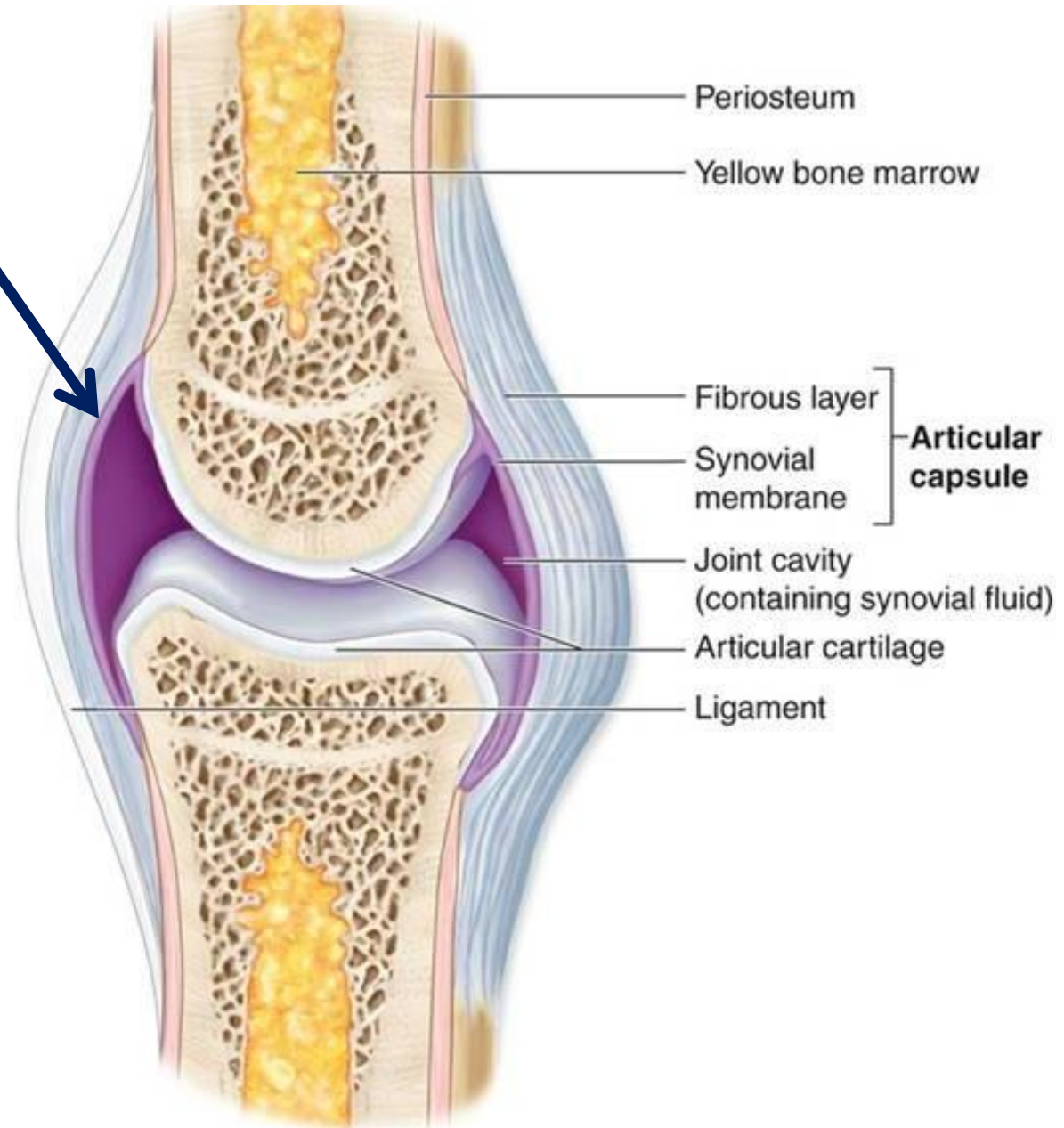
6. Synovial
sheaths→
covers
muscle
tendons as
they pass
under their
retinaculum



- 7. Synovial membrane



- Serous sacs that line the interior of joints



Typical synovial joint

Anatomy of joints and muscles

- A 23 years old football player had **severe trauma** to his **shoulder**. After examination, the doctor told him that he had a **shoulder joint dislocation**.



Joints

Definition:

- the site of meeting between two or more bones

Classification

- Joints are classified according to their structure
- (**the tissues that lie between the bones**) into:
 1. fibrous joints
 2. cartilaginous joints
 3. synovial joints

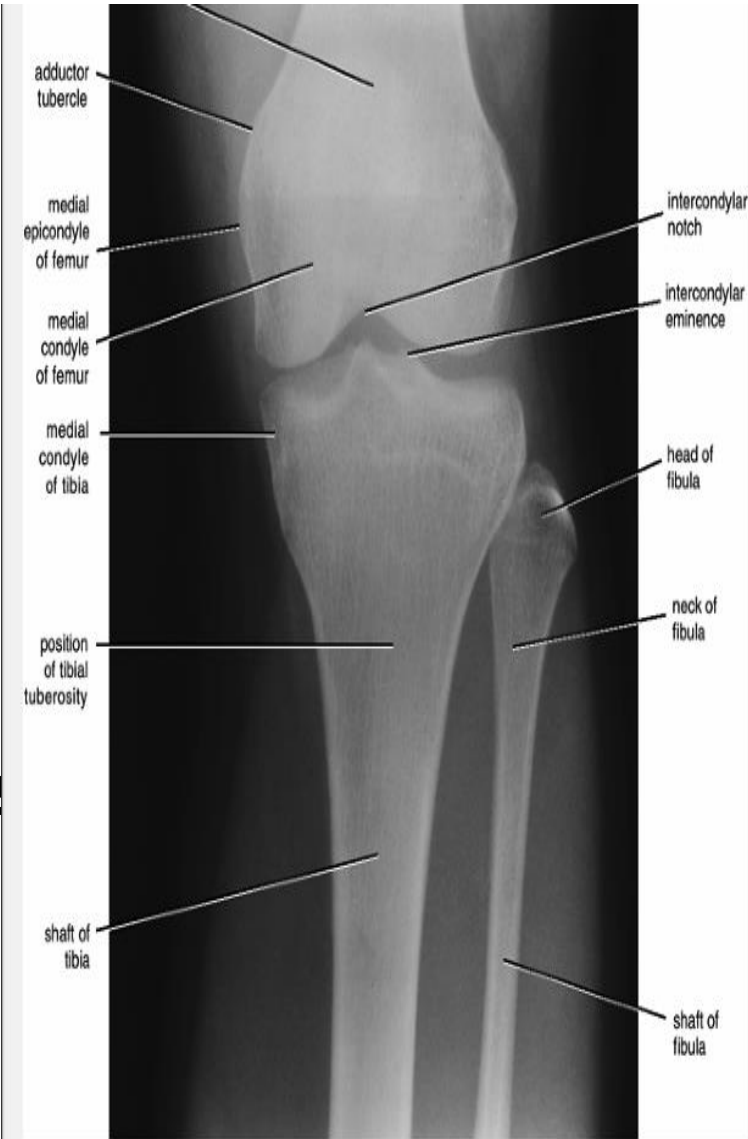
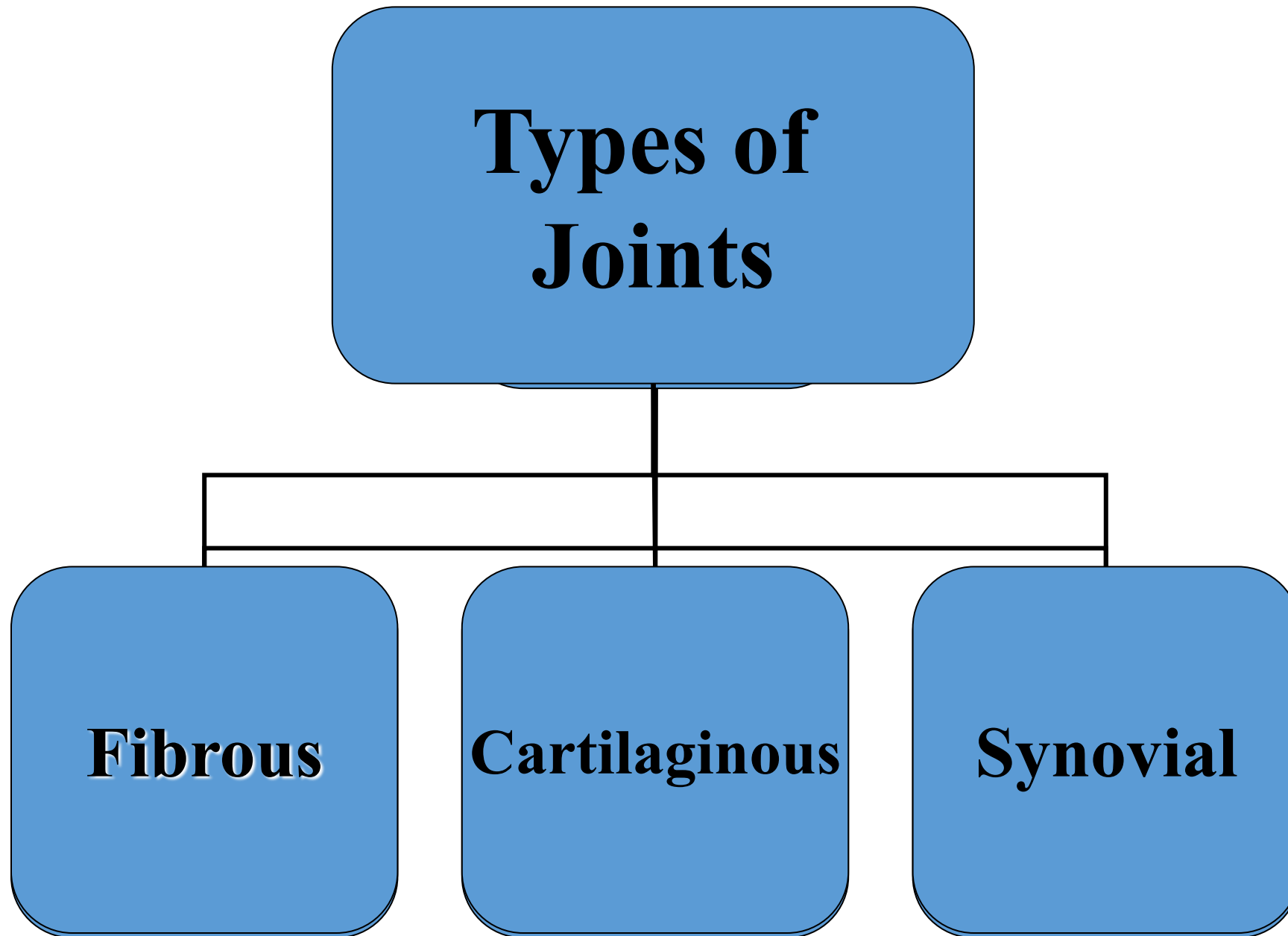


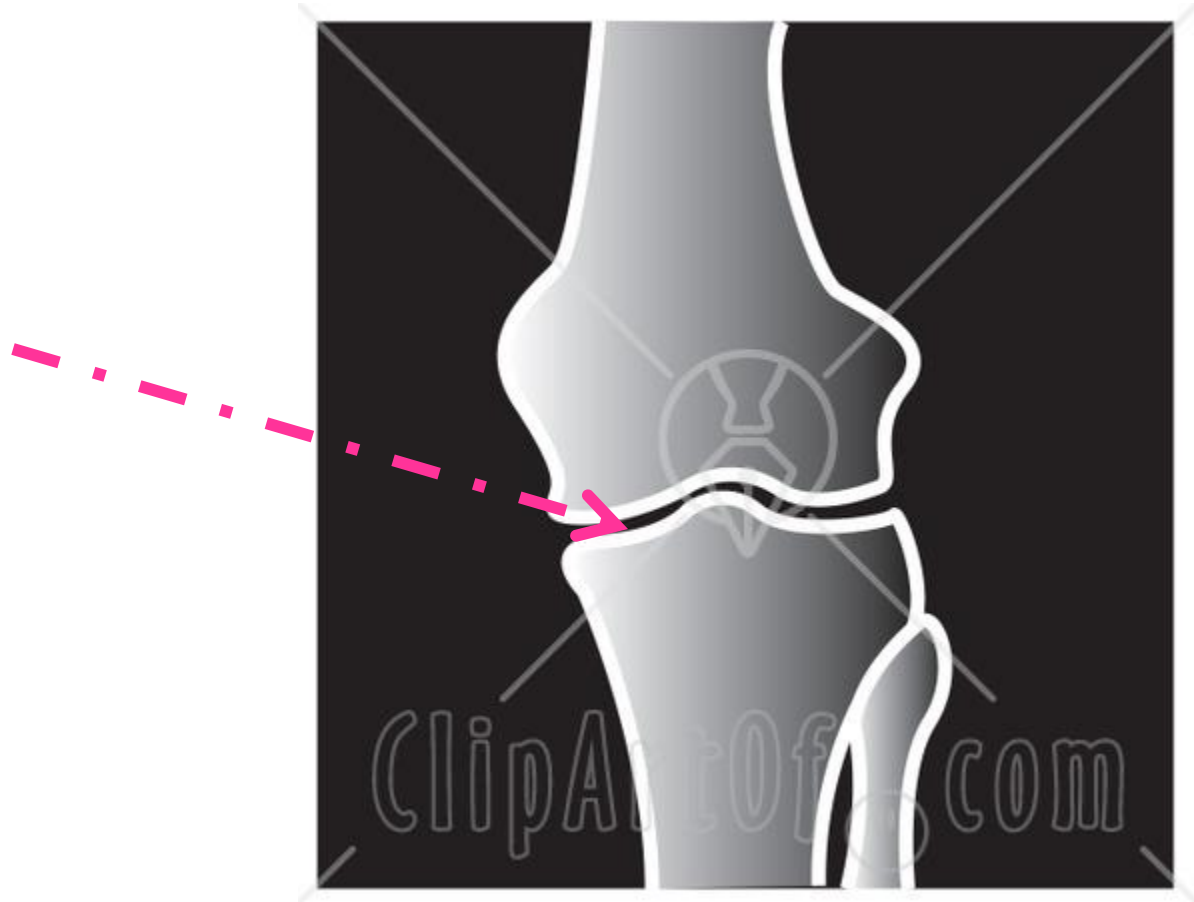
Figure 10-73 Anteroposterior radiograph of the adult knee.



Joints

The site where 2 or more bones meet.

Joints are classified according to their structure and mobility.



The articular surfaces are connected together **by strong fibrous tissue & No movements are allowed.**

1-Fibrous joints

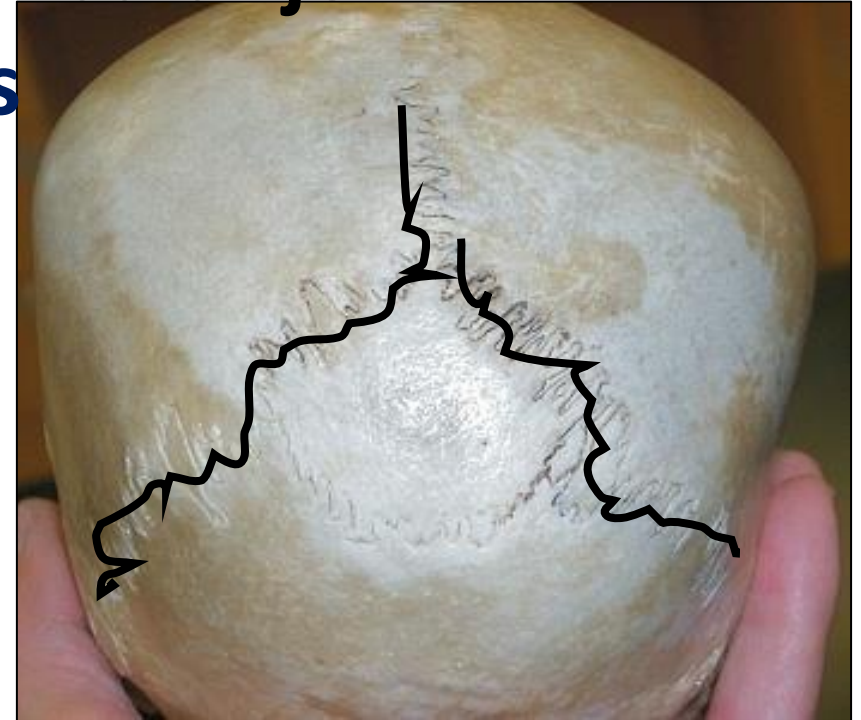
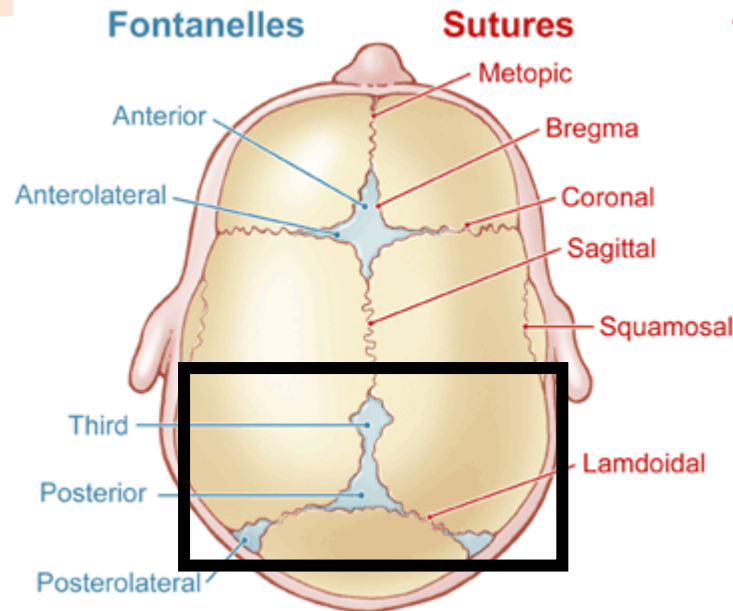
Eg :

1-suture

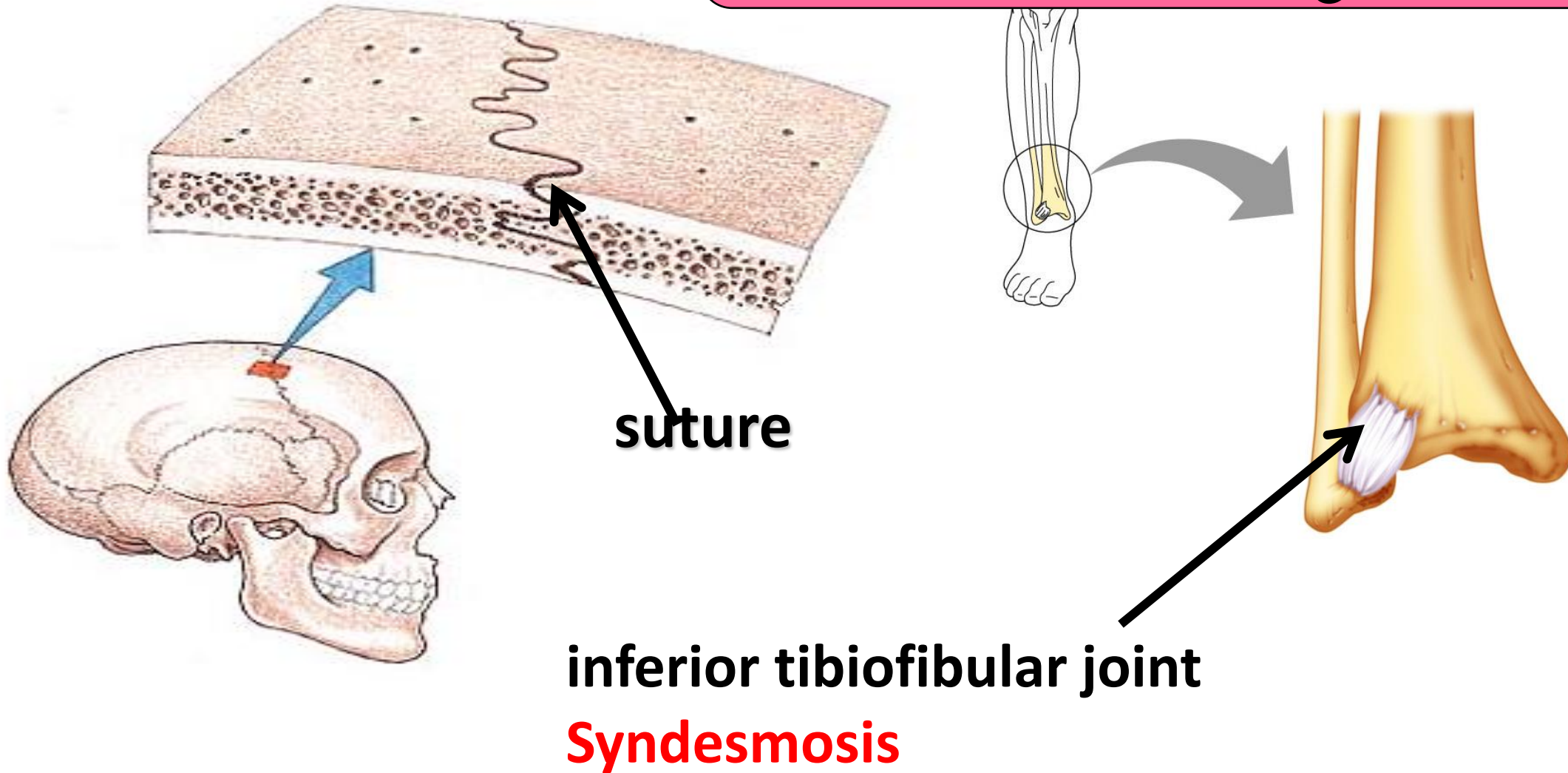
2-Syndesmosis

inferior tibiofibular joint

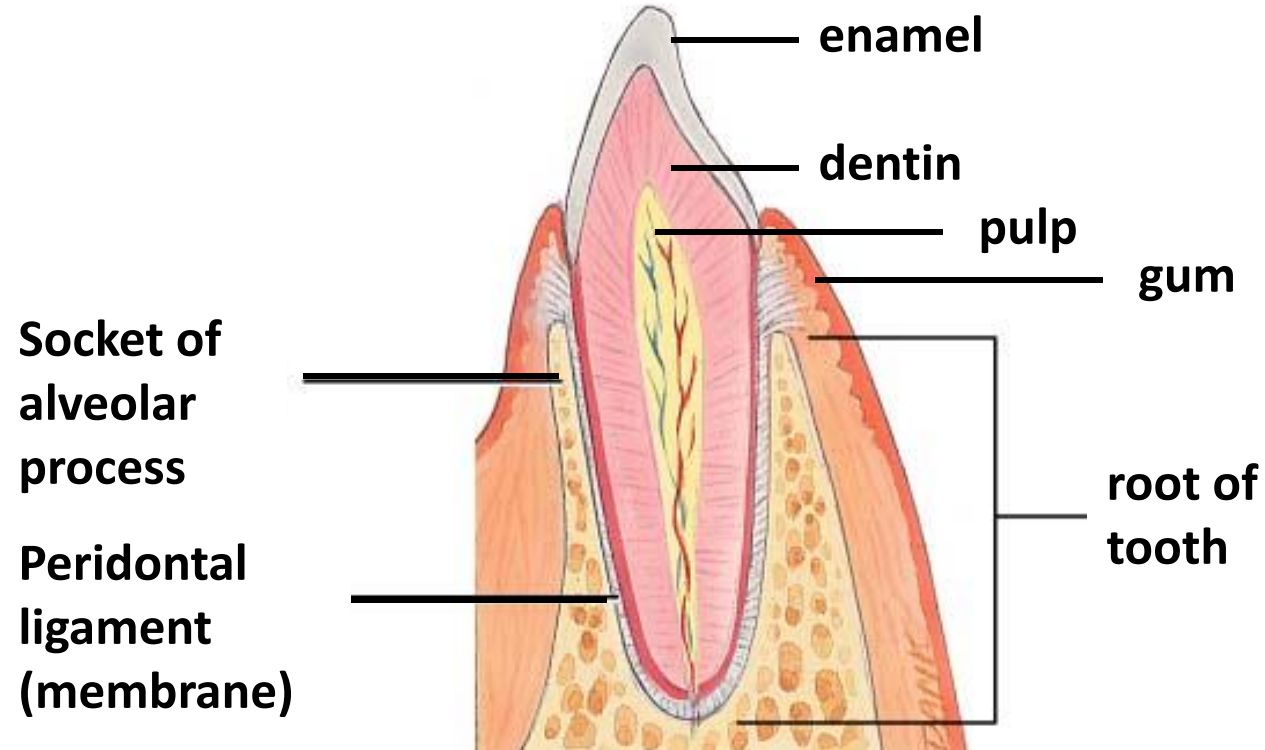
3-Gomphosis



1-Fibrous joints



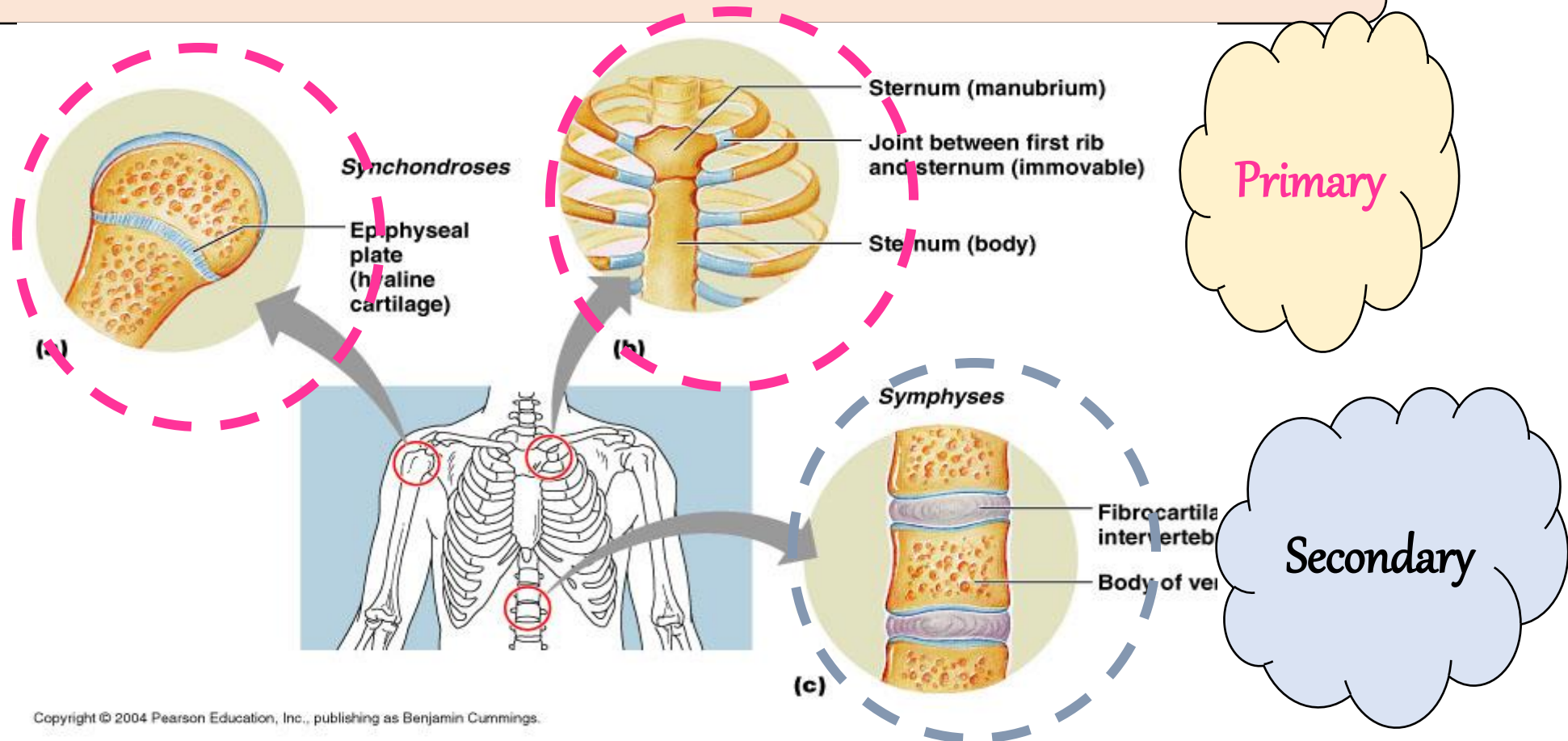
Gomphosis



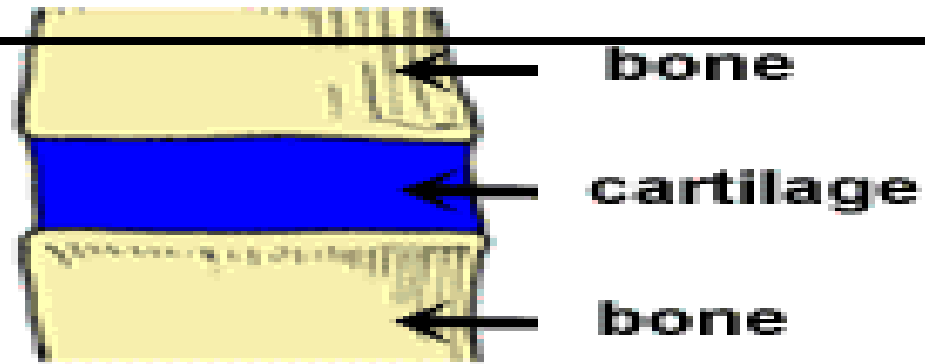
- Ligaments hold tooth in bony socket
- Immovable joint

2-Cartilaginous joints

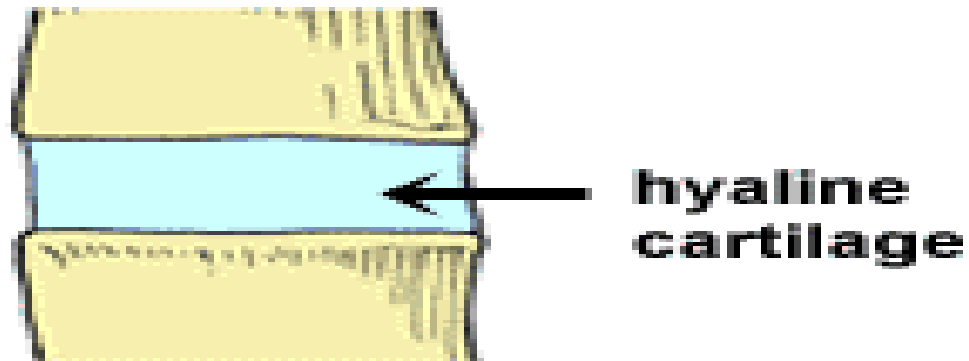
The bones are connected by cartilage



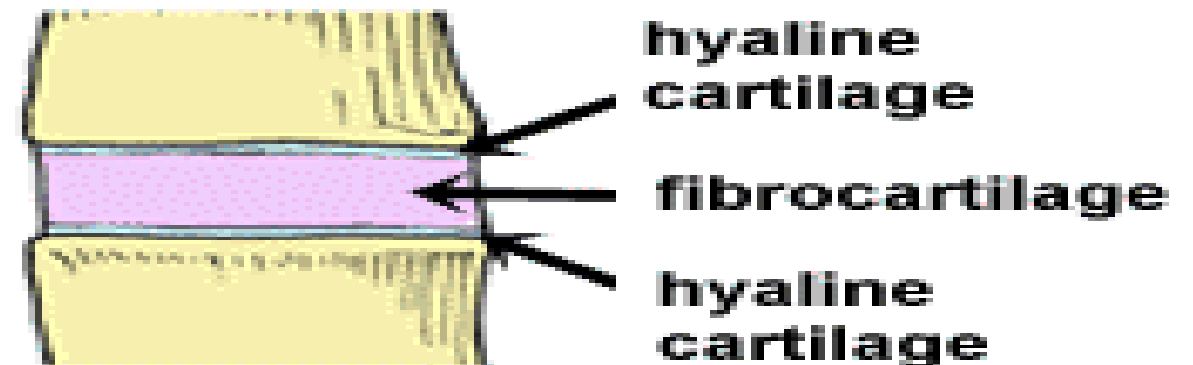
CARTILAGINOUS JOINTS: cartilage - no joint cavity



**PRIMARY
(= SYNCHONDROSIS)
hyaline cartilage only**



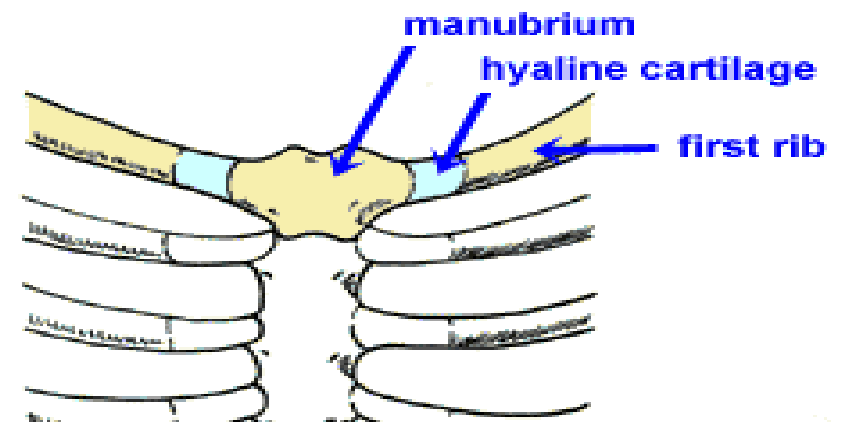
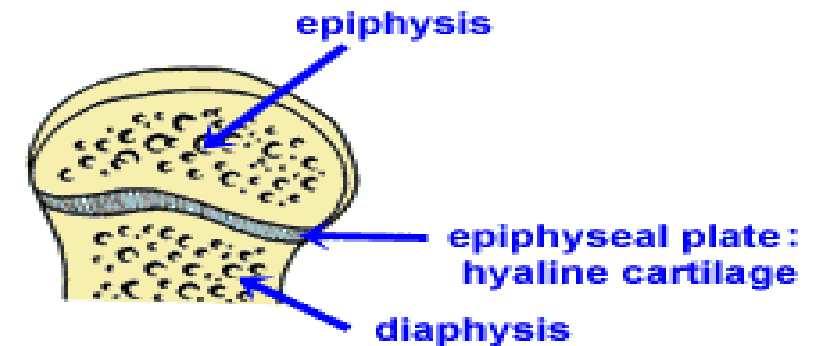
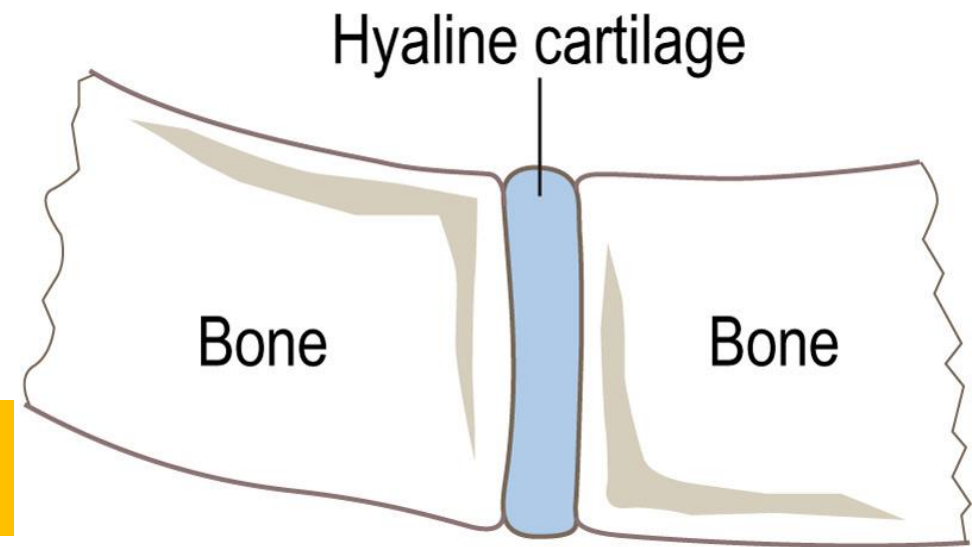
**SECONDARY
(= SYMPHYSIS)
fibrocartilage sandwiched
between hyaline cartilage**



2-Cartilaginous joints

2-Primary Cartilaginous joints

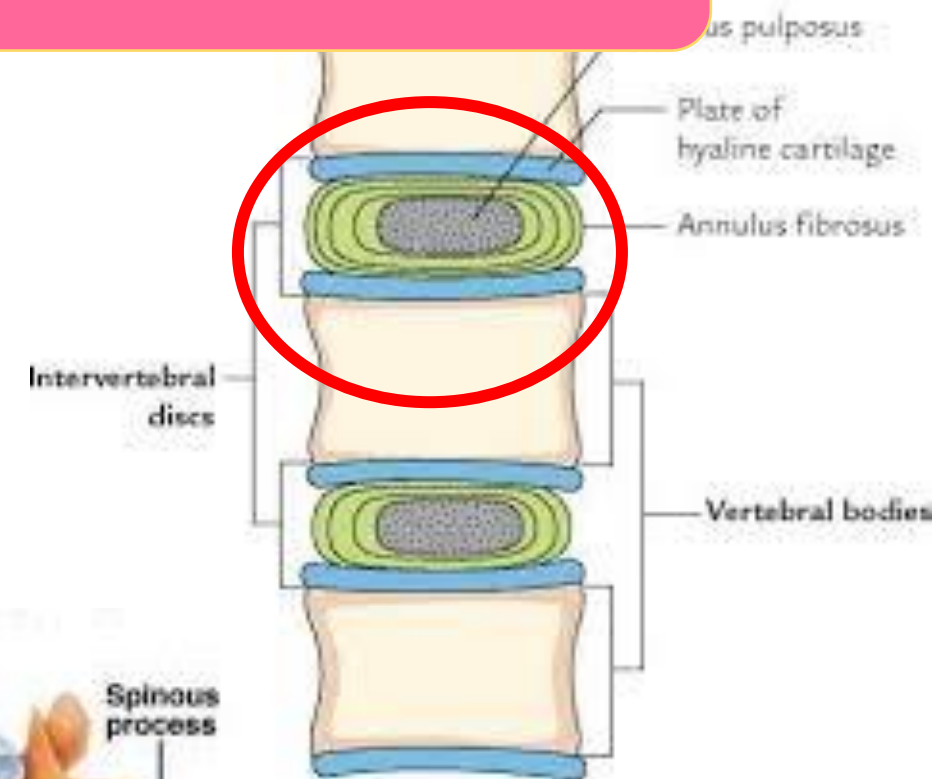
The articular surface of bones are connected by **hyaline cartilage** which does **not** allow movements and **ossifies** with age



2-Cartilaginous joints

2-Secondary Cartilaginous joints = Mid line joints

The articular surface of bones are connected by **fibrocartilage** which **allow limited movements**.
It ossifies rarely.



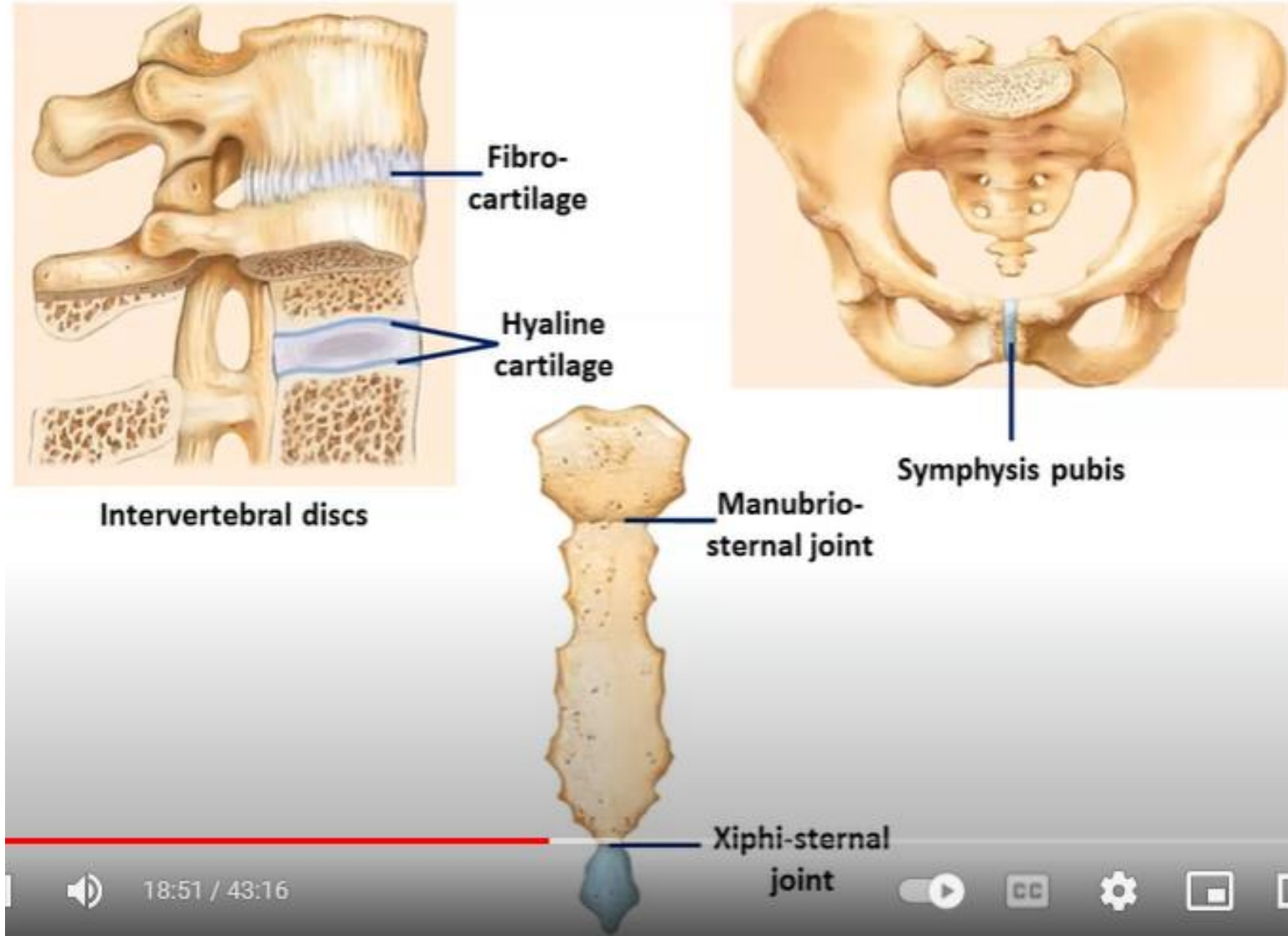
2-Secondary Cartilaginous joints

= Mid line joints

Mid line joints:

1-Intervertebral joints

2-Symphysis pubis



Test you self

➤ *The following applies to features of fibrous joints:*

- a. They are freely movable.
- b. Bones are connected by cartilage.
- c. They can be seen in the skull.
- d. They contain joint cavity.

➤ *The following is a primary cartilaginous joint:*

- a. Sutures of the skull.
- b. Epiphyseal cartilage.
- c. Intervertebral disc.
- d. Symphysis pubis.

3-Synovial joints

They are freely movable joints .

5 Structure of synovial joint:

1-Articular cartilage

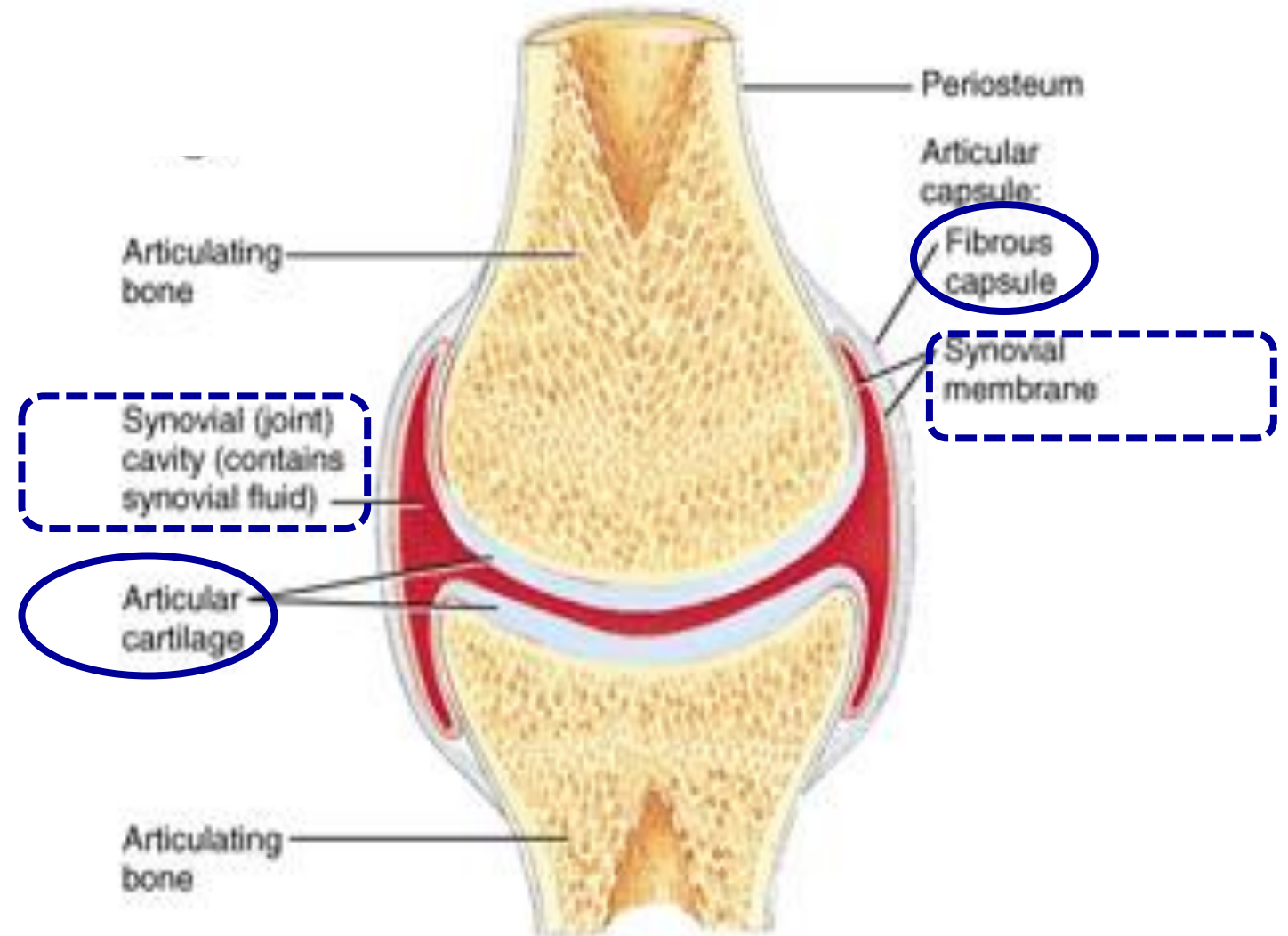
2-Fibrous capsule

3-Synovial membrane

4-Synovial fluid

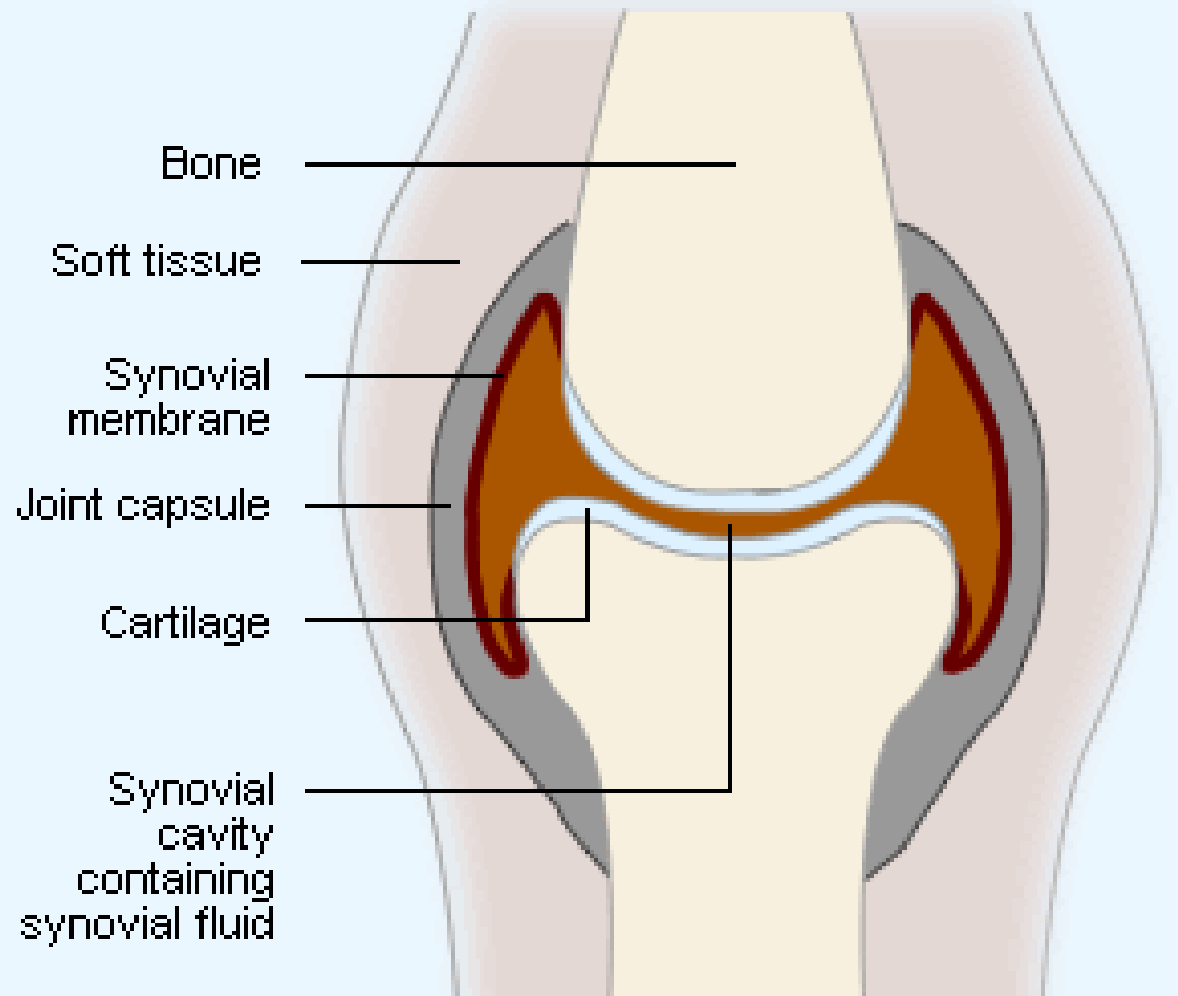
5-joint cavity

Some synovial joints may contain additional intra capsular structures (disc ,tendon of a muscle)

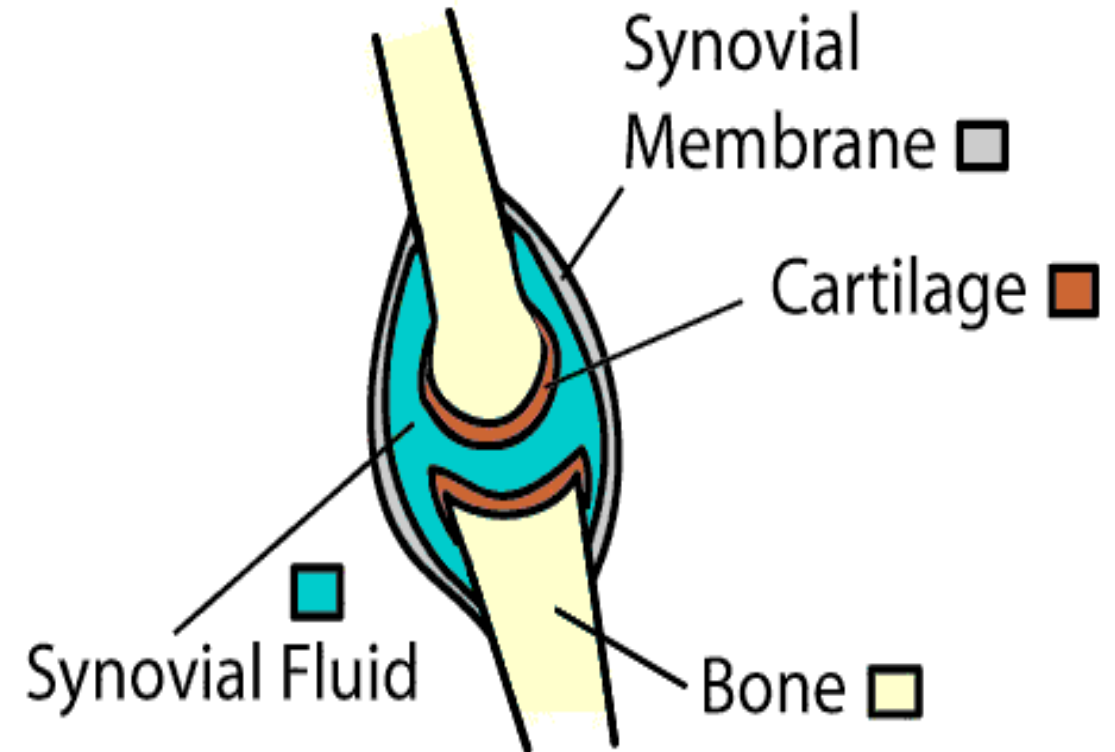


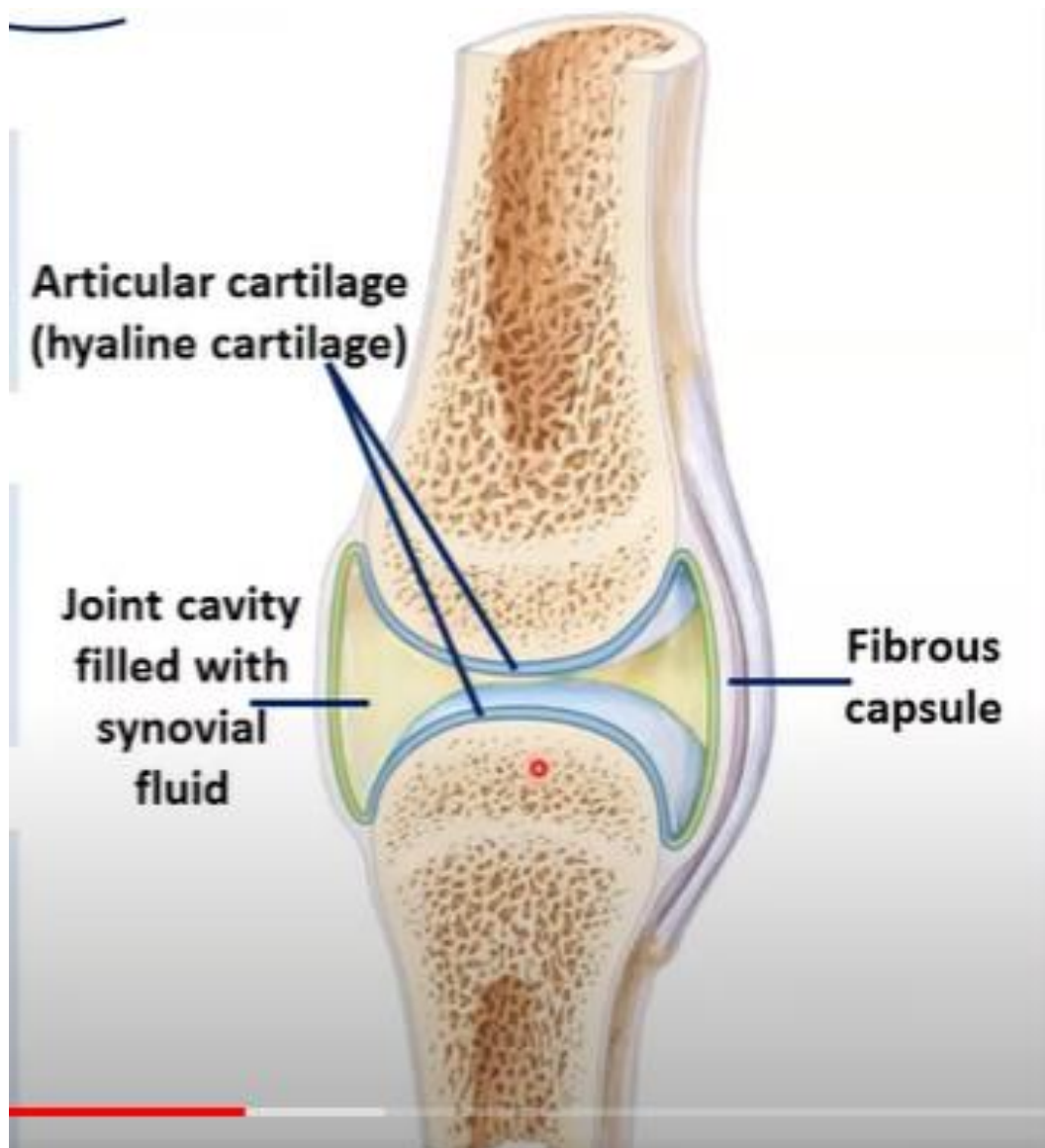
Structure of synovial joint

SYNOVIAL JOINT



A Healthy Joint





Types of synovial joints

They are classified according to the **axes** around which the movements occur & **shape of articular** surface into:

- 1- Uniaxial(hinge-pivot)
- 2-Biaxial (ellipsoid,saddle,condylar)
- 3-Multiaxial
- 4-Plane

يتم تقسيمها علي حسب:

1. عدد المحاور التي يشغل عليها المفصل
2. الشكل الهندسي لنهاية العظم
3. الحركة التي يسمح بها
4. المثال

Multiaxial Joints

All types of movements are allowed.

الكرة و الحق

e.g

(1)Upper limbs :shoulder .

(2)Lower limbs: hip joint .

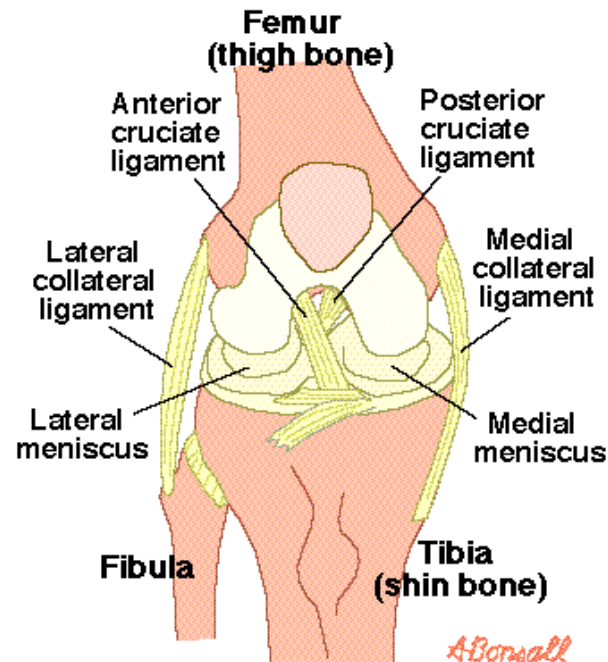


Factors affecting stability of joints

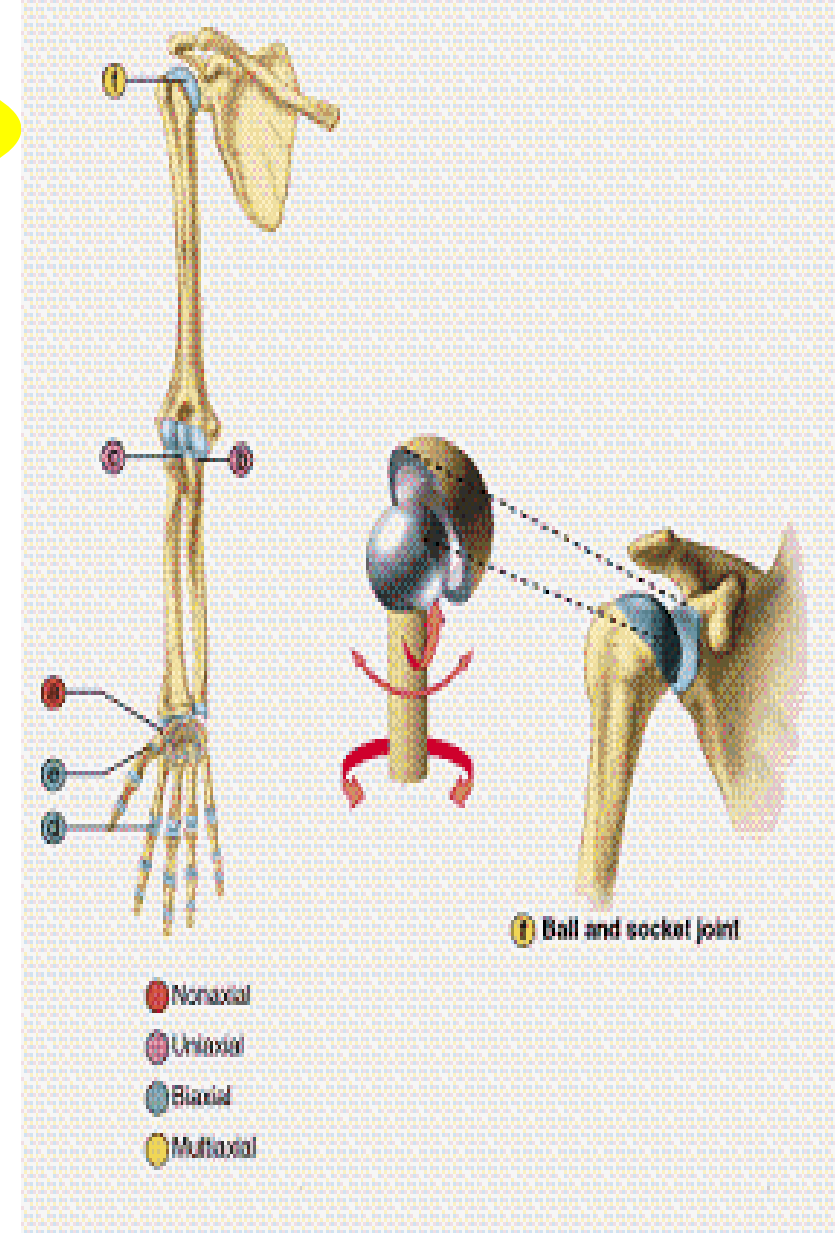
1-Shape of the **articular surface**

2-**Strength of the capsule and ligaments**

3-**The tone and action of the surrounding muscles**



KNEE JOINT front view



POP Quiz

- 1-Mention the types of synovial joints?
- 2-List factors affecting joint stability?

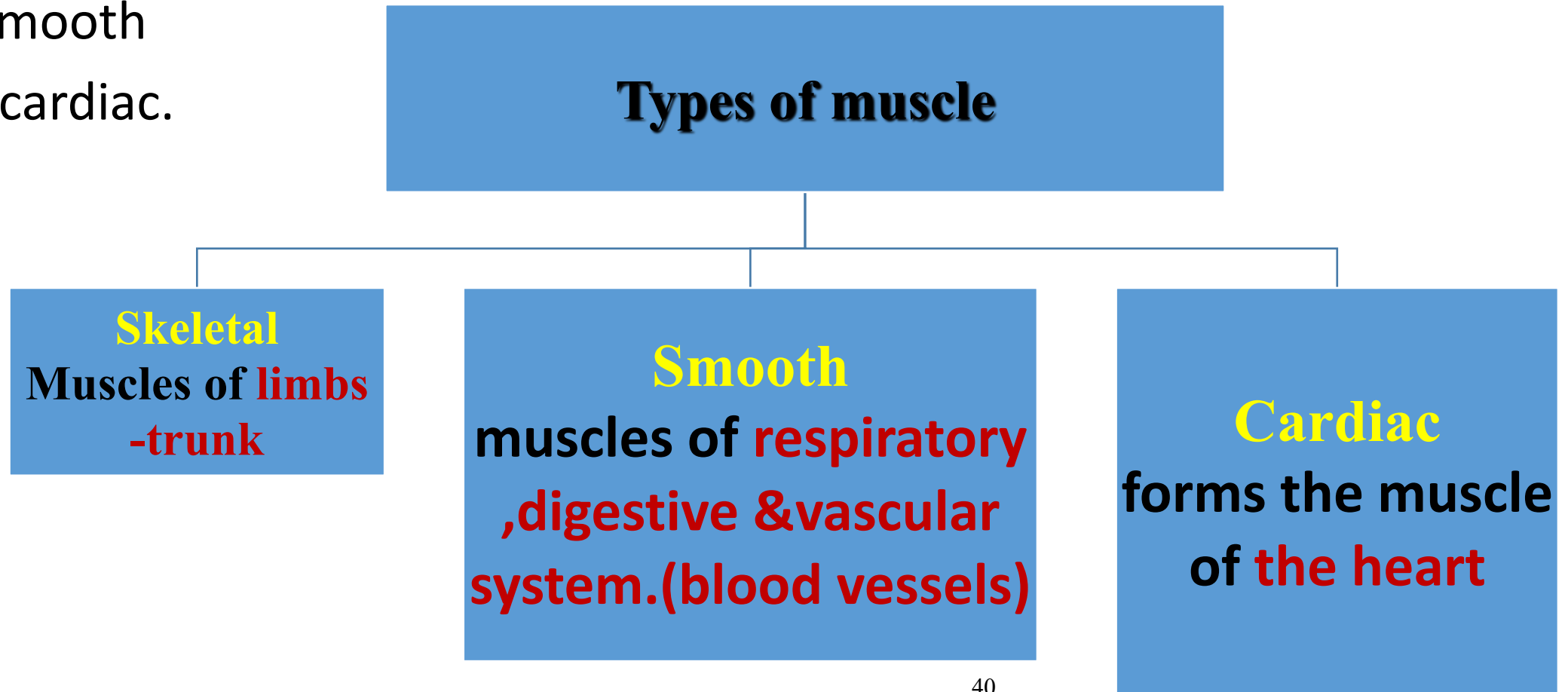
➤ *Movement around one axis only occurs atjoint.*

- a. Wrist
- b. Elbow
- c. Shoulder
- d. Inferior tibiofibular

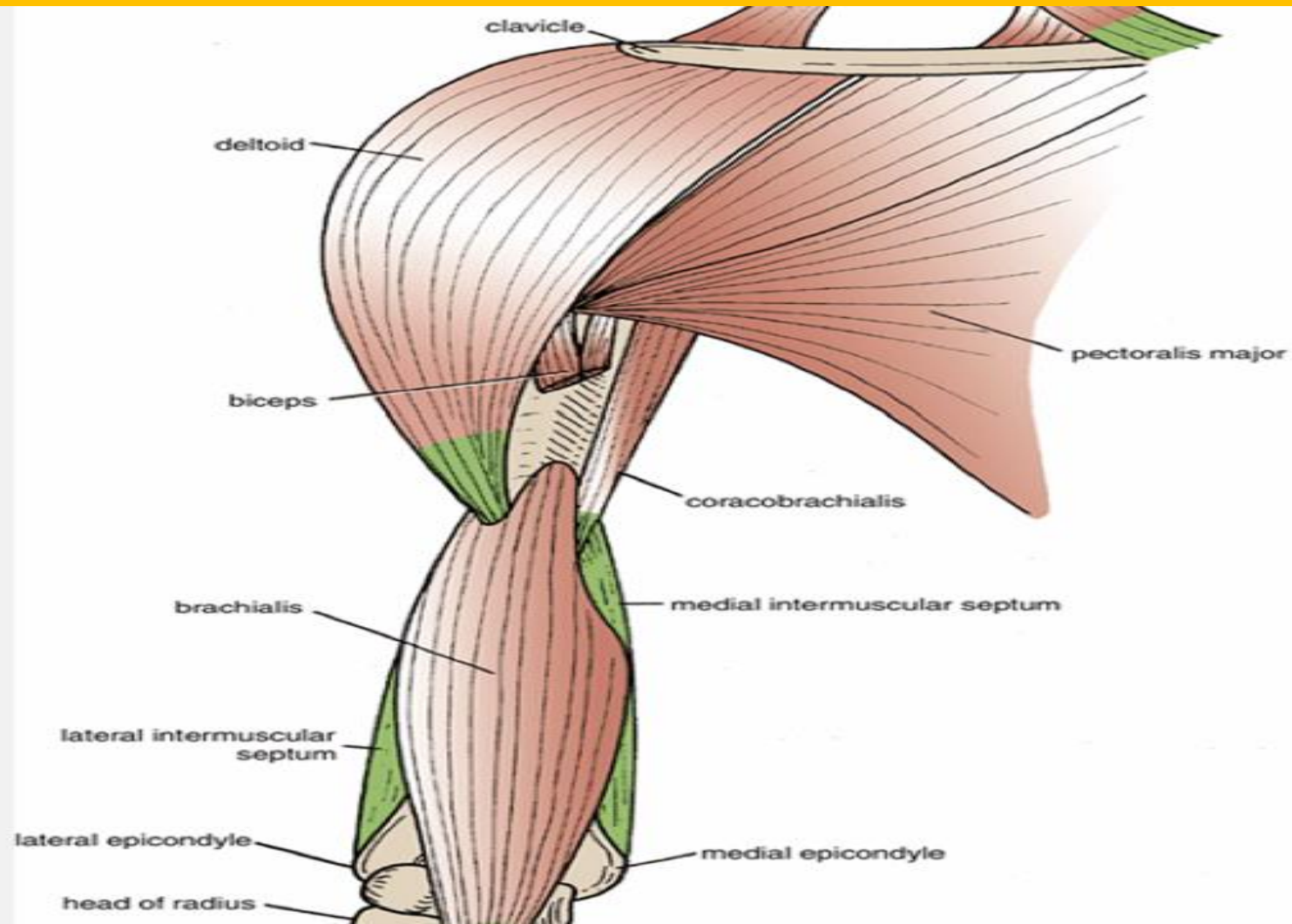
MUSCLES

- Types of muscles There are **three types** of muscles

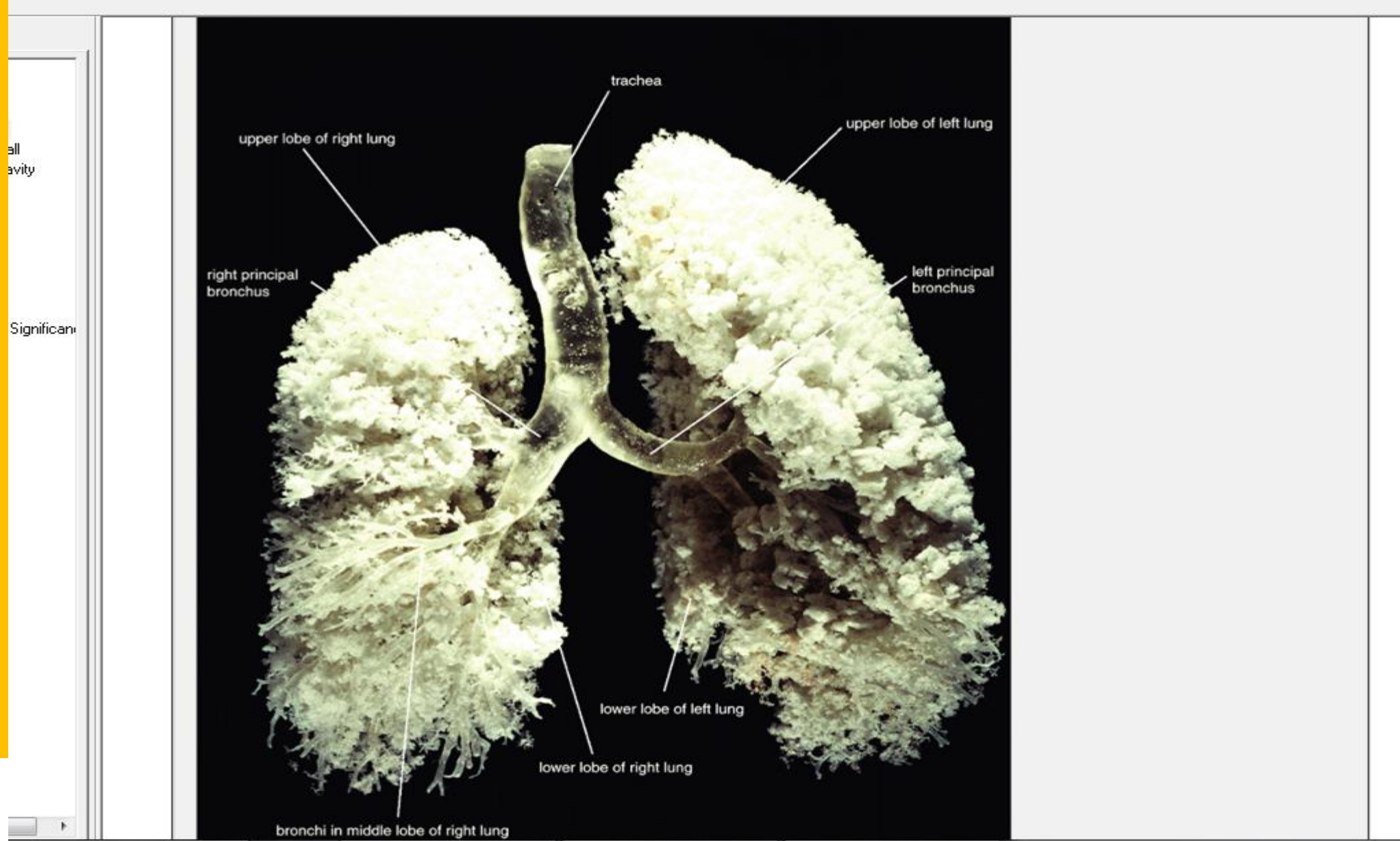
1. skeletal
2. smooth
3. cardiac.



Skeletal muscles



- Smooth muscles in the respiratory system



Smooth muscles of the digestive system

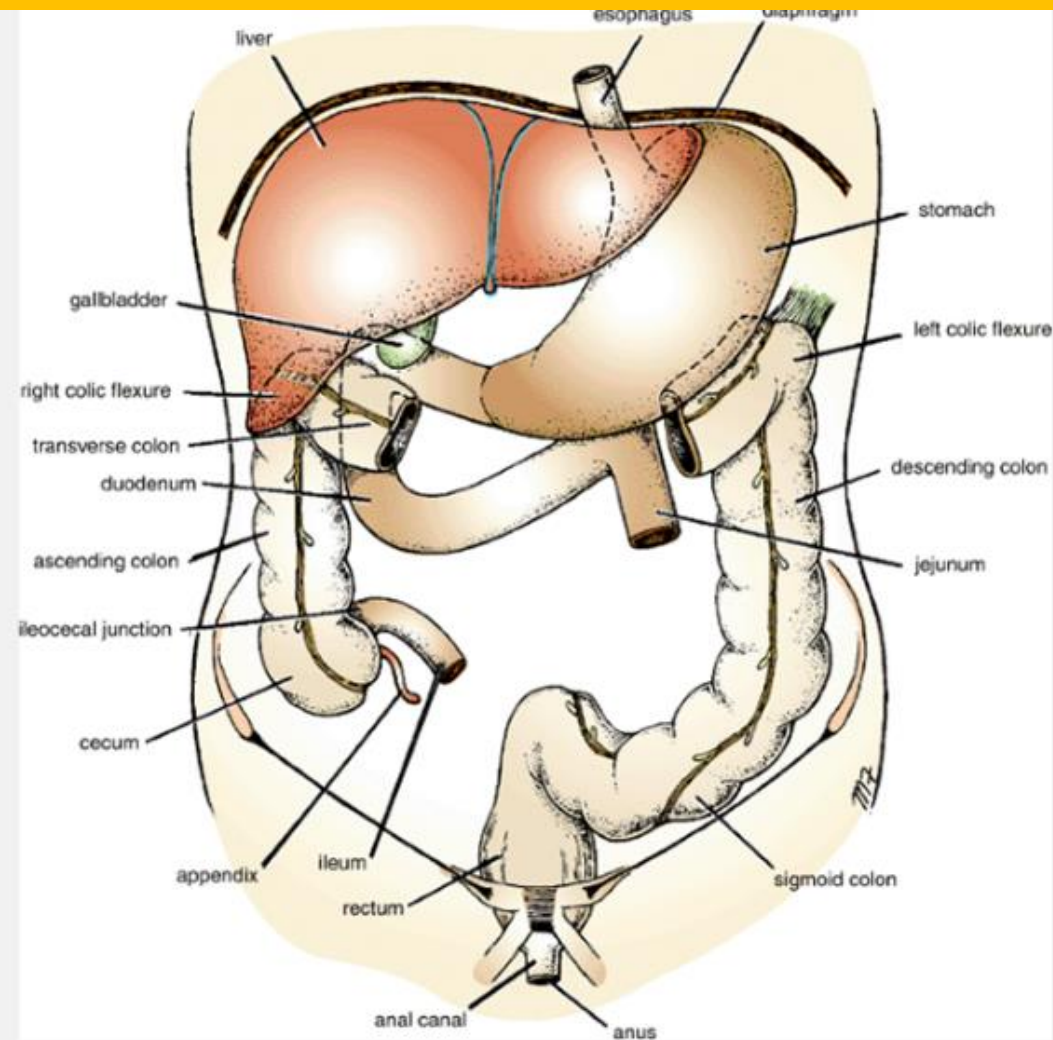
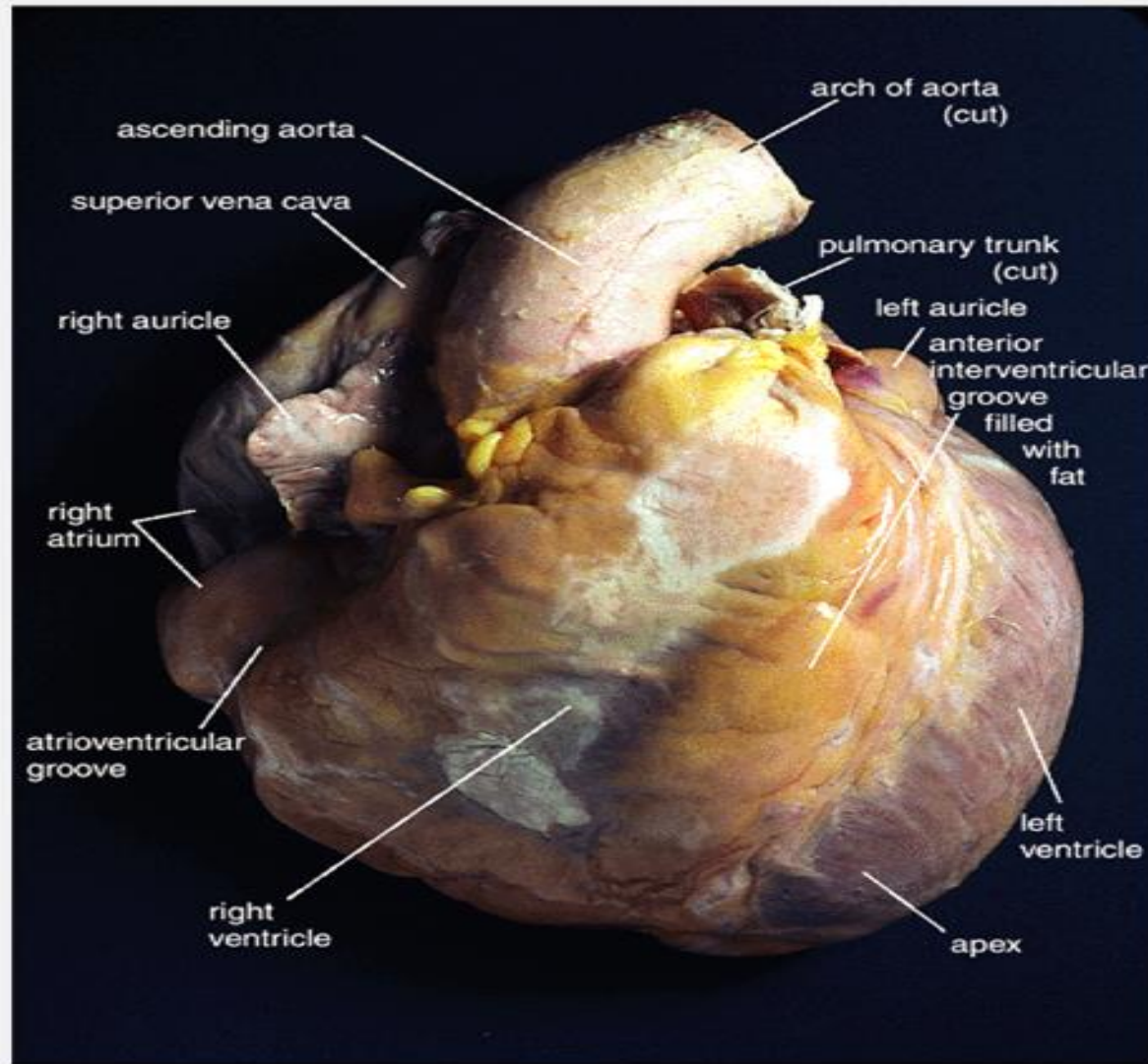


Figure 5-1 General arrangement of abdominal viscera.

3)Cardiac Muscle

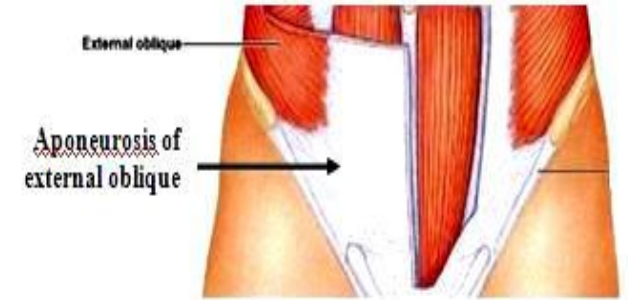
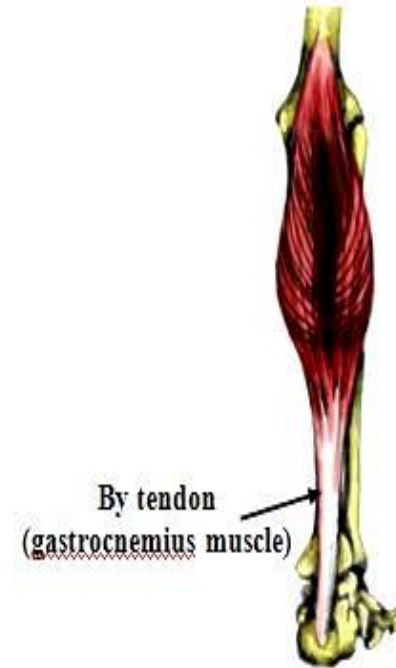


The Skeletal

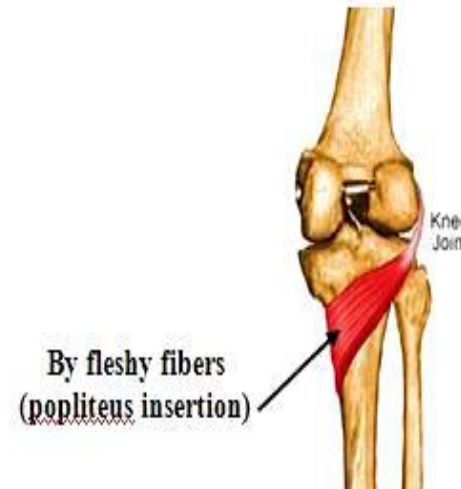
(Striated) Muscles:

Way of attachment of a muscle:

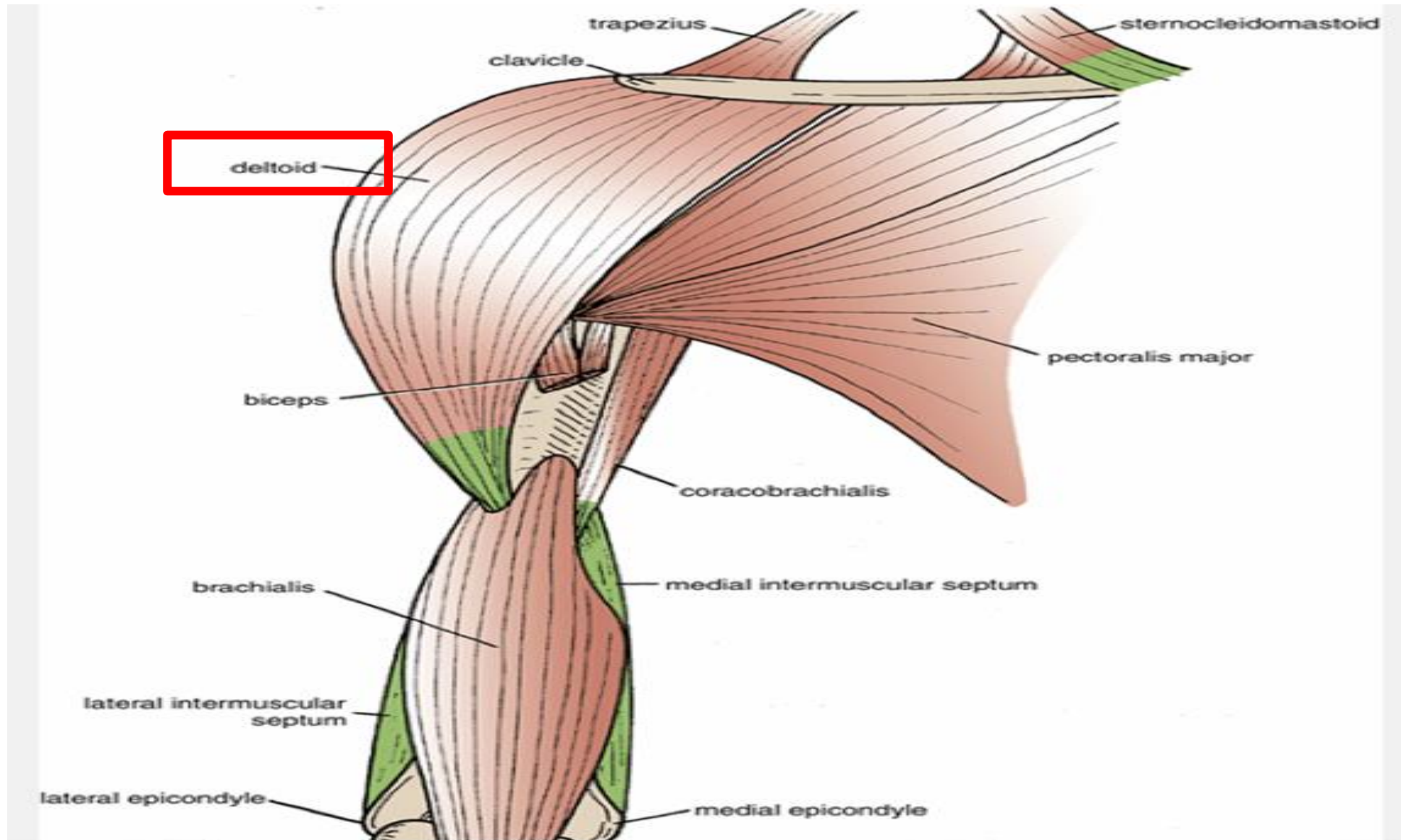
1. ☐ By fleshy fibers
2. ☐ By tendon : the tendon is a long fibrous cord. e.g. Tendocalcaneus.

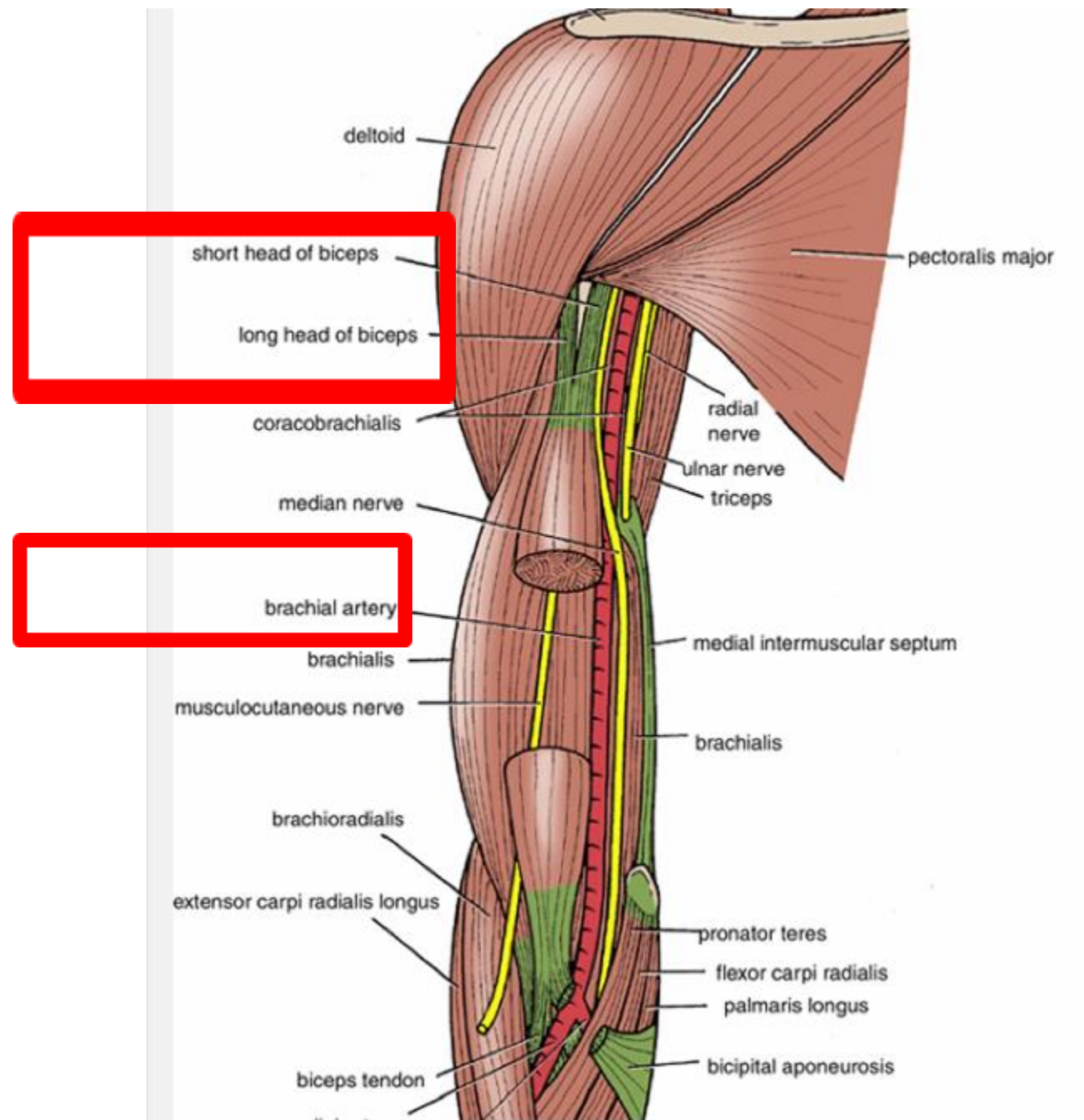


By aponeurosis
(abdominal muscles)

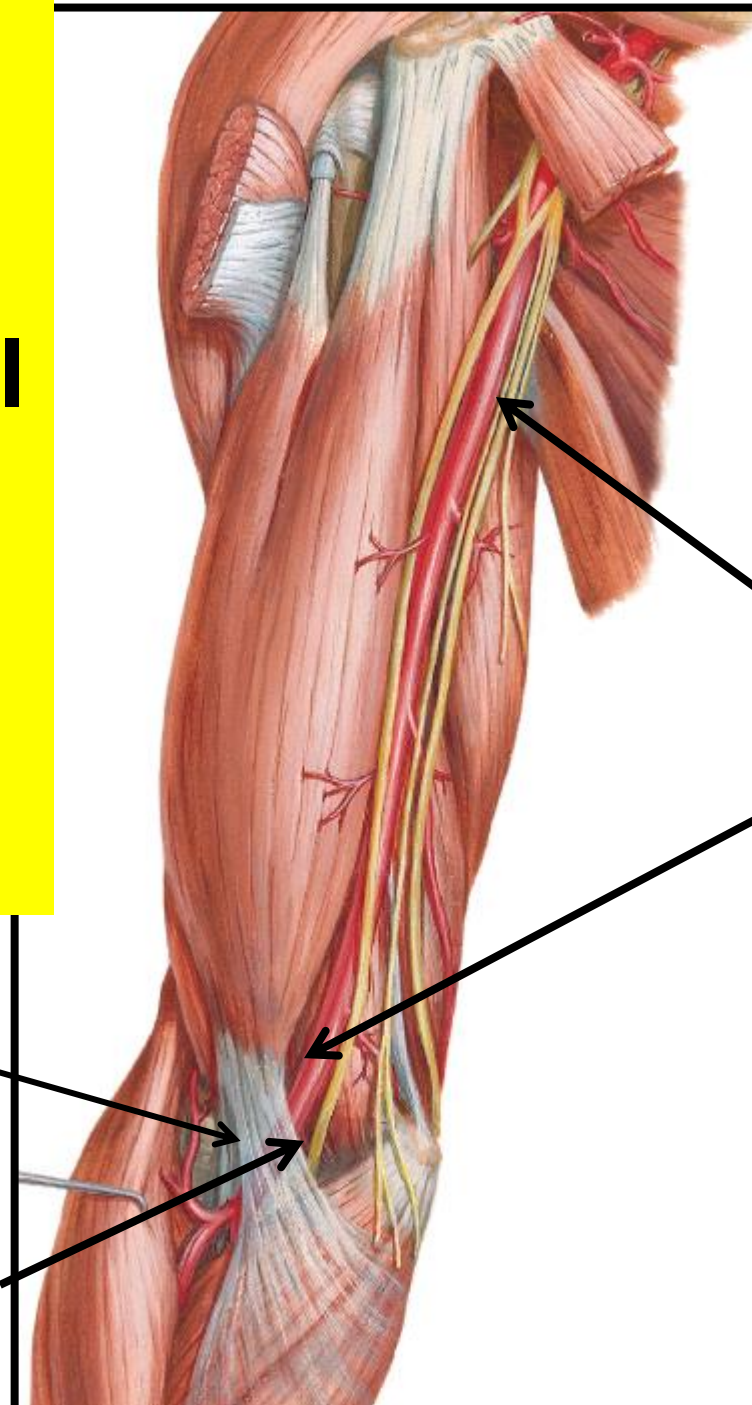


Muscles of the upper limb





**Measure the
blood
pressure by feel
the pulse of
brachial artery
medial to the
biceps tendon**



Brachial artery

Biceps muscle

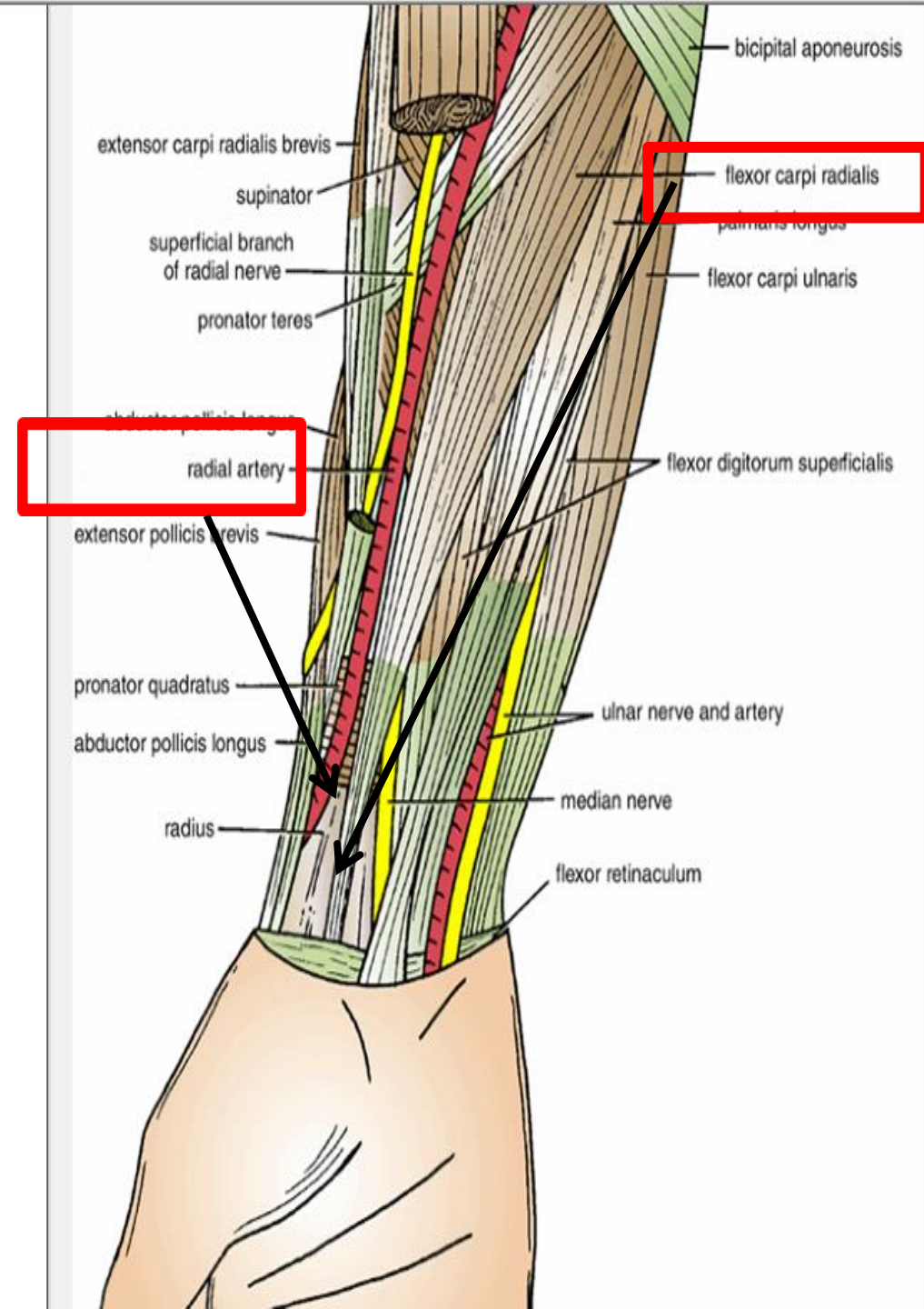
Where to feel the radial pulse ?

- **lateral** to the tendon of flexor carpi radialis (FCR)



Measure the pulse of
radial artery

lateral to the tendon of
flexor carpi radialis
(FCR)

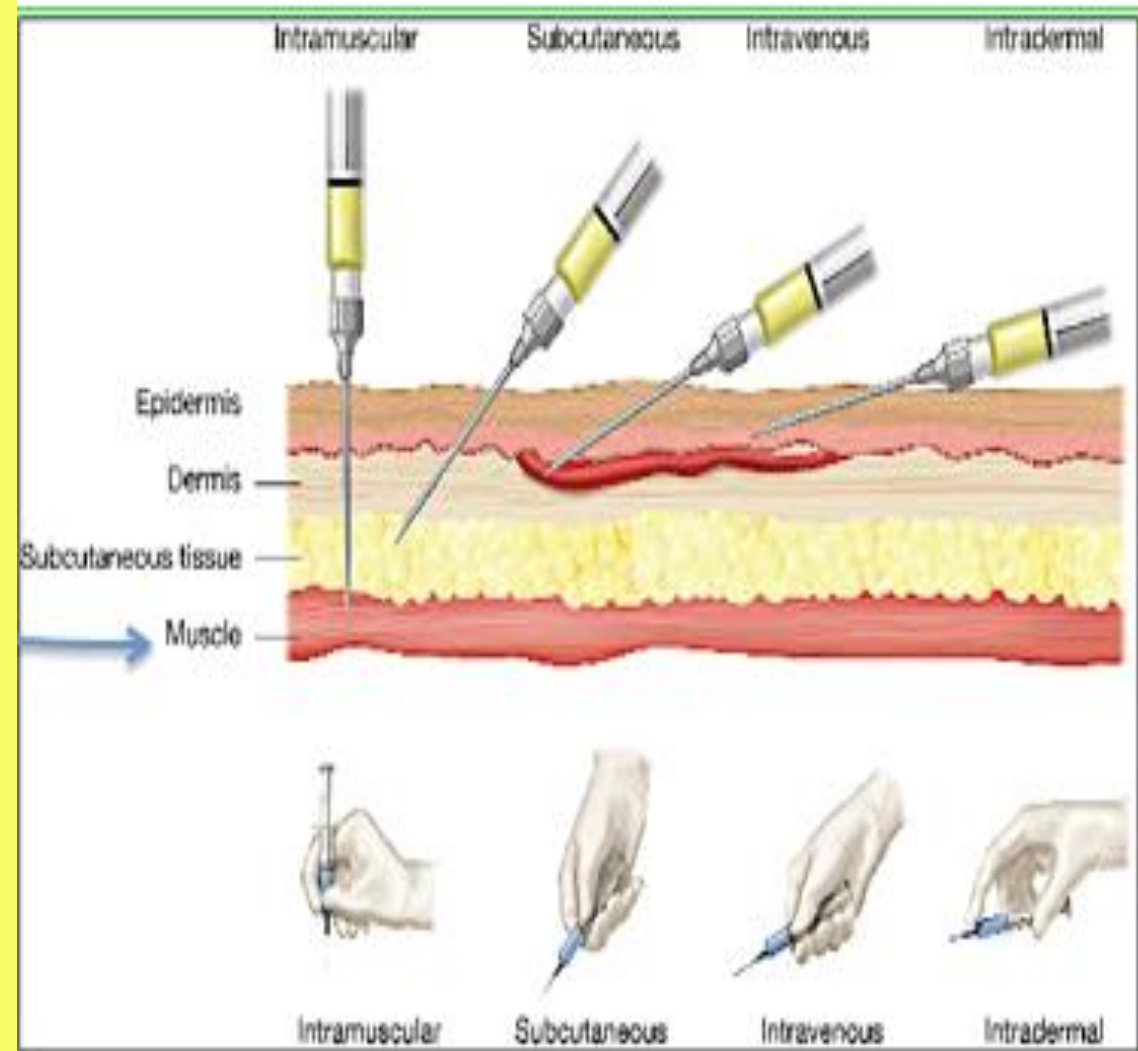


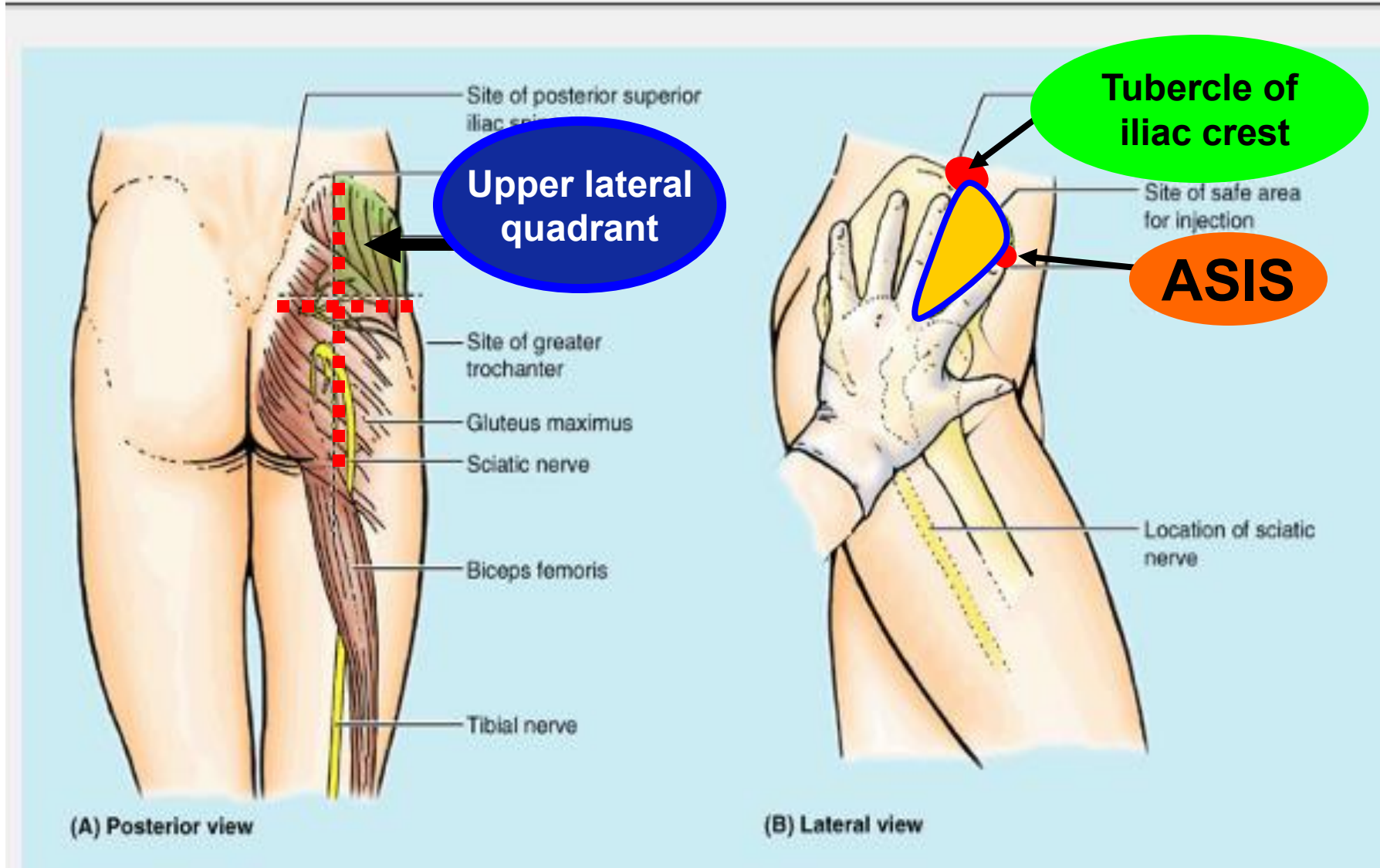
Muscles of lower limbs

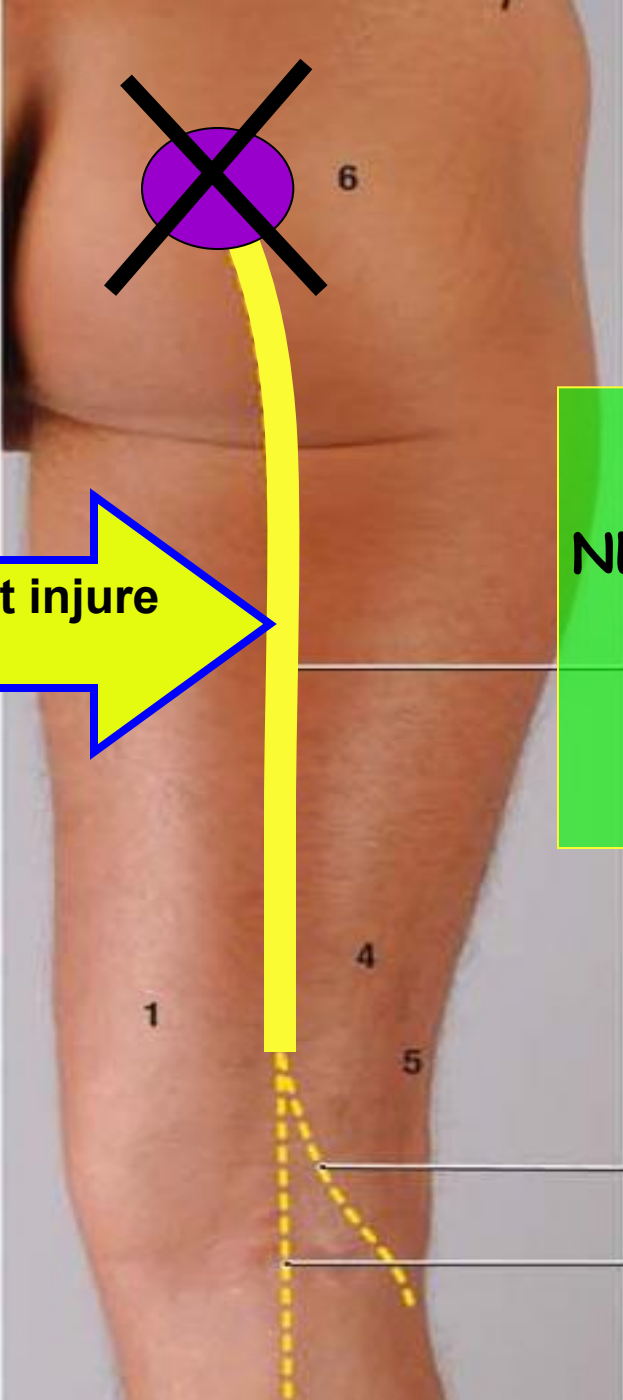
- Intramuscular injection in
- the **gluteus maximus** muscle:
- the upper lateral quadrant to avoid injury to the sciatic nerve

Applied anatomy

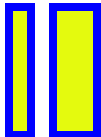
- Because of the rich vascularity of the glutei muscles, it is advisable to do a short aspiration before an intramuscular injection as the tip of the needle may unfortunately lie inside the lumen of a blood vessel.







DANGER!



So you do not injure
sciatic nerve

DANGER!

NEVER GIVE INTRAMUSCULAR
INJECTION IN CENTER
OF BUTTOCK

Sciatic nerve

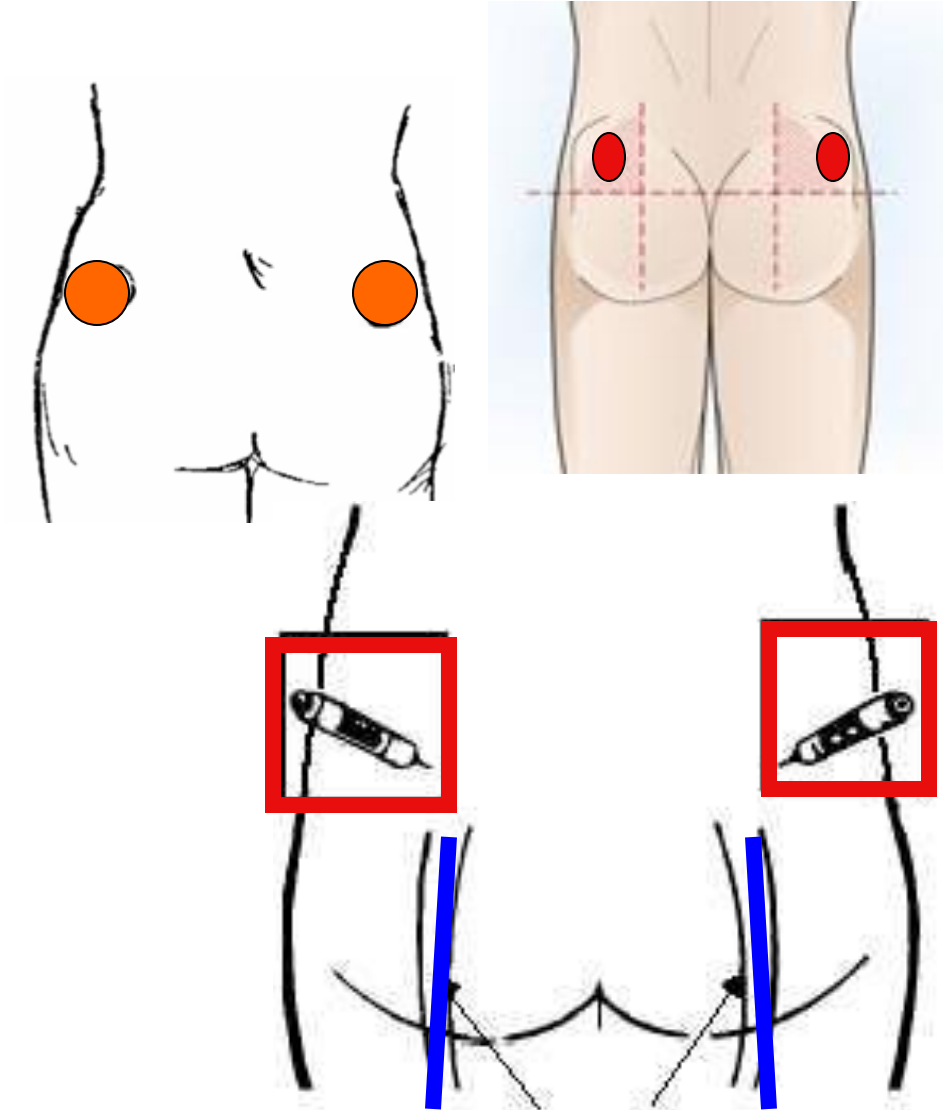
image

Common fibular nerve

Tibial nerve

Safest site for intramuscular injection

- Superolateral part of buttock
- Avoid medial side of buttock so as not to injure the sciatic nerve



Muscles of the lower limb :gluteal muscles in the buttock

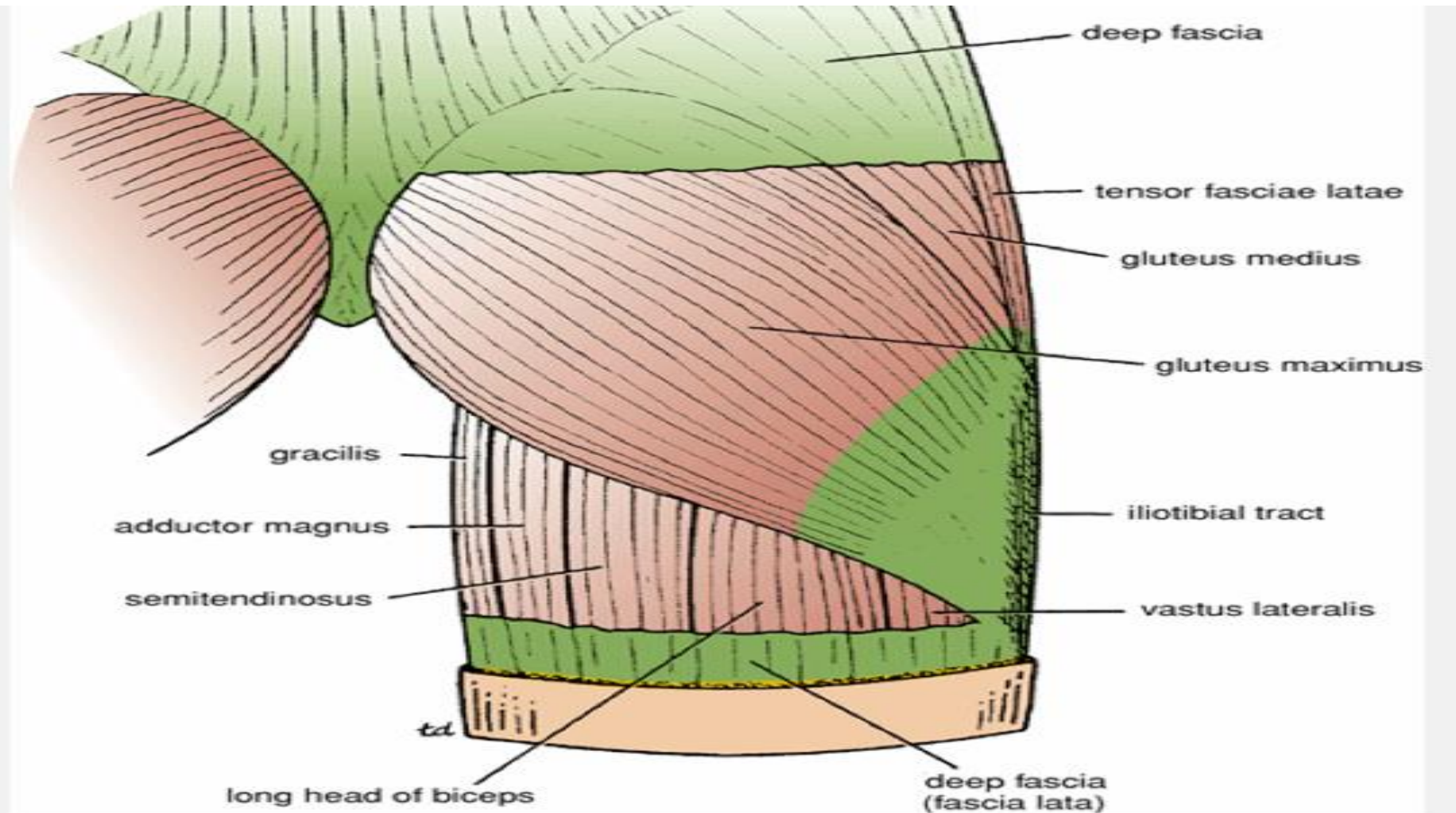
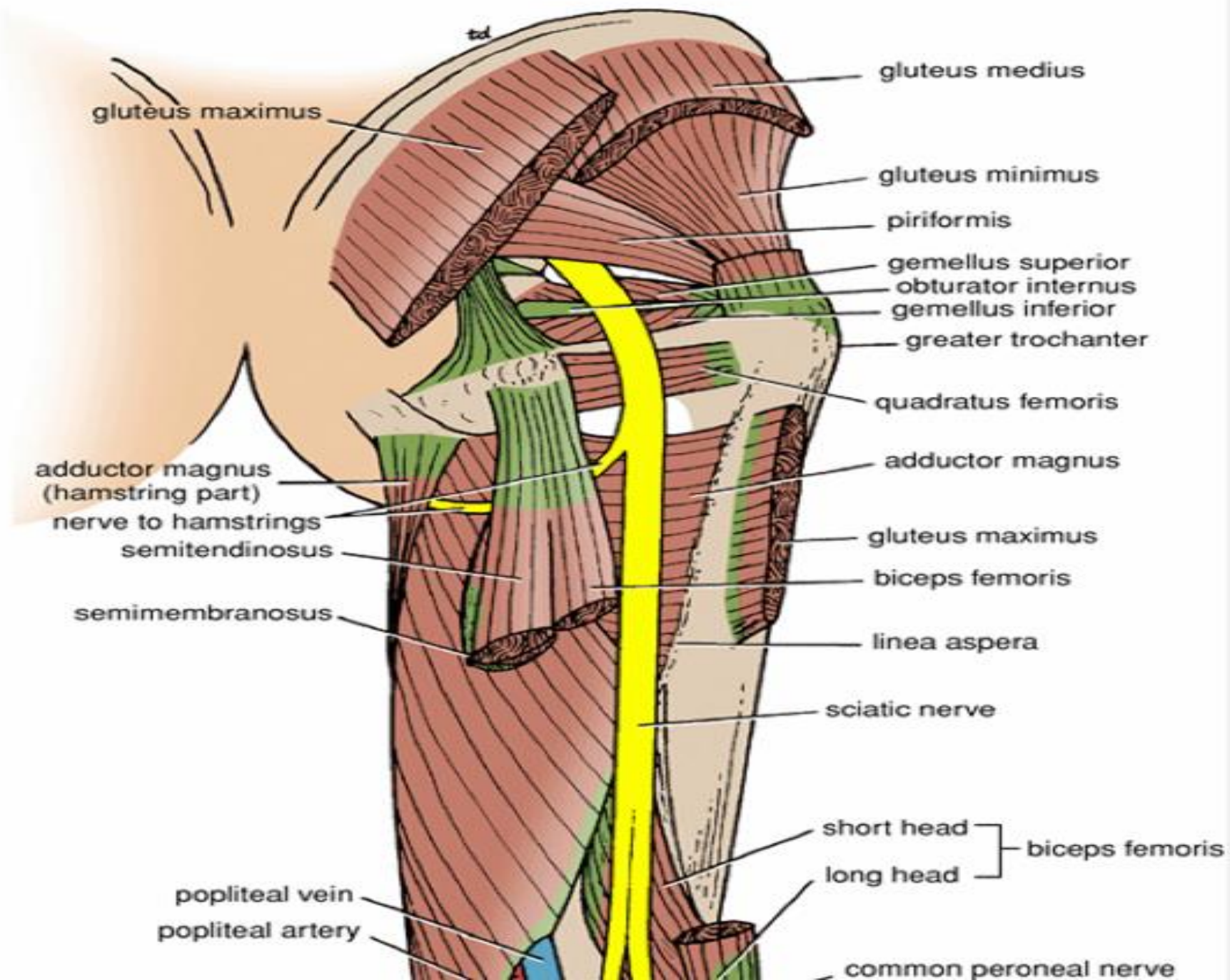
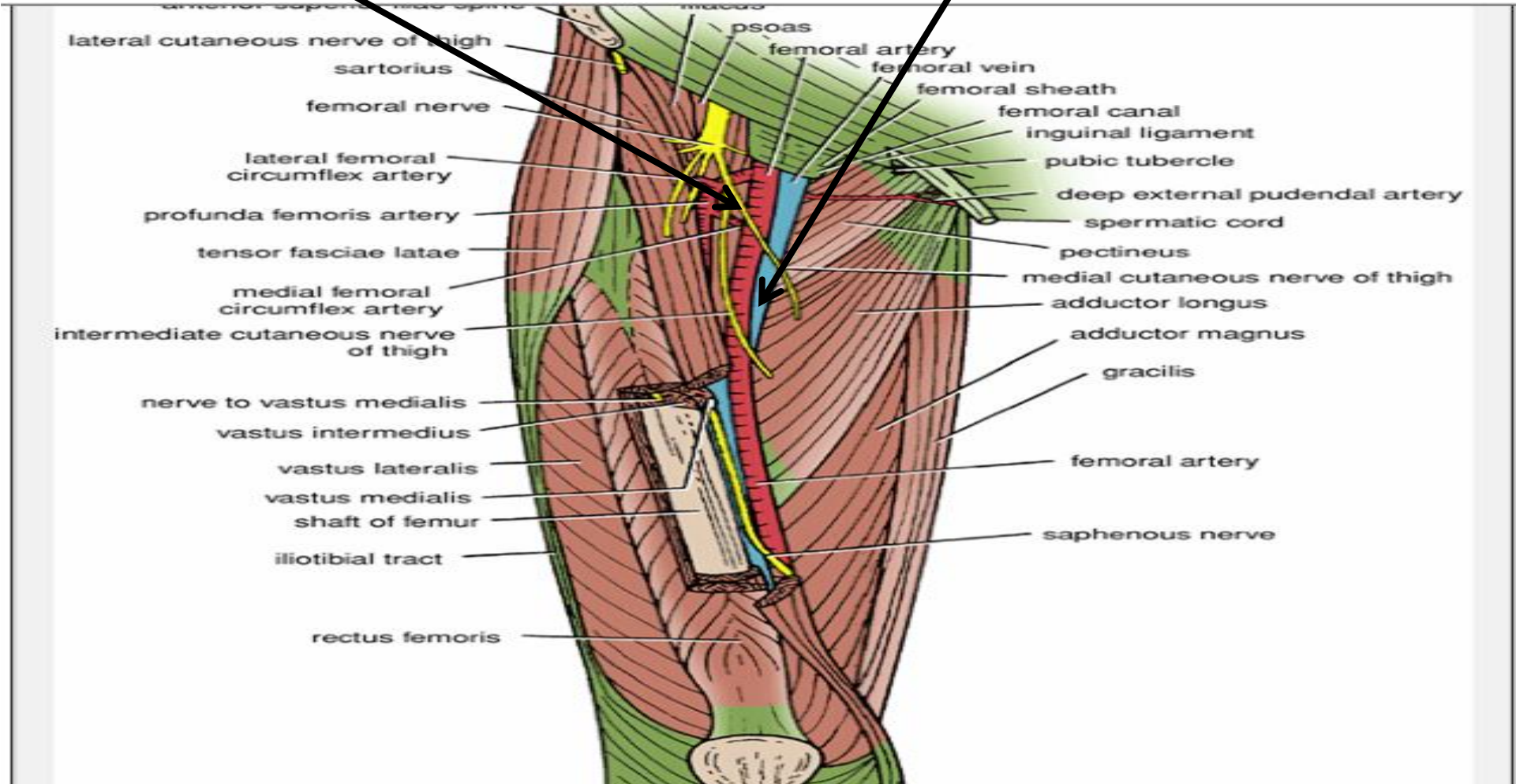


Figure 10-5 Right gluteus maximus muscle.



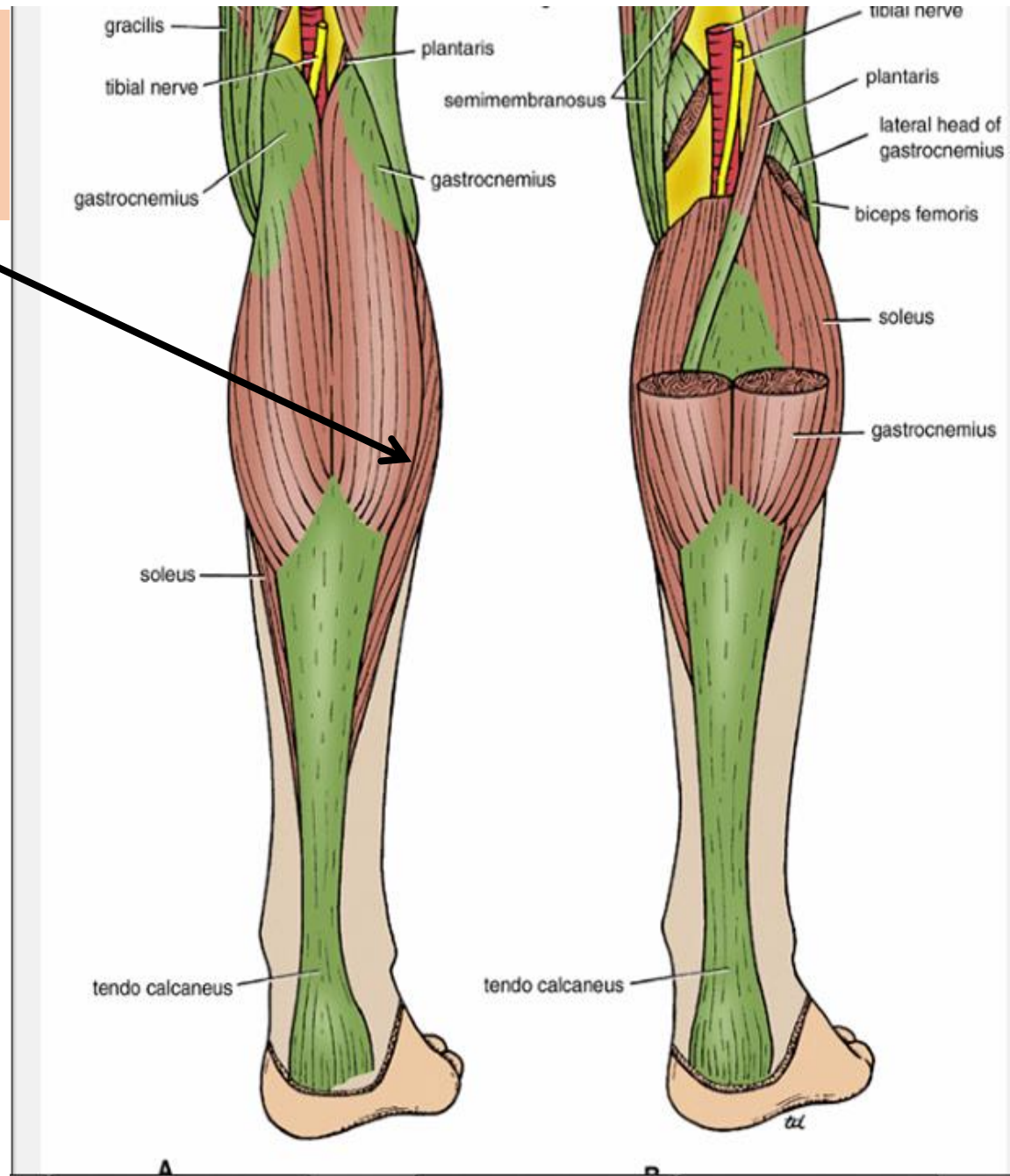
Quadriceps muscle In the front of the thigh & above it femoral artery(pulse) & femoral vein(catheter)



Calf muscle

Gastrocnemius is
the commonest
sit for **DVT**

**Deep Venous
thrombosis**



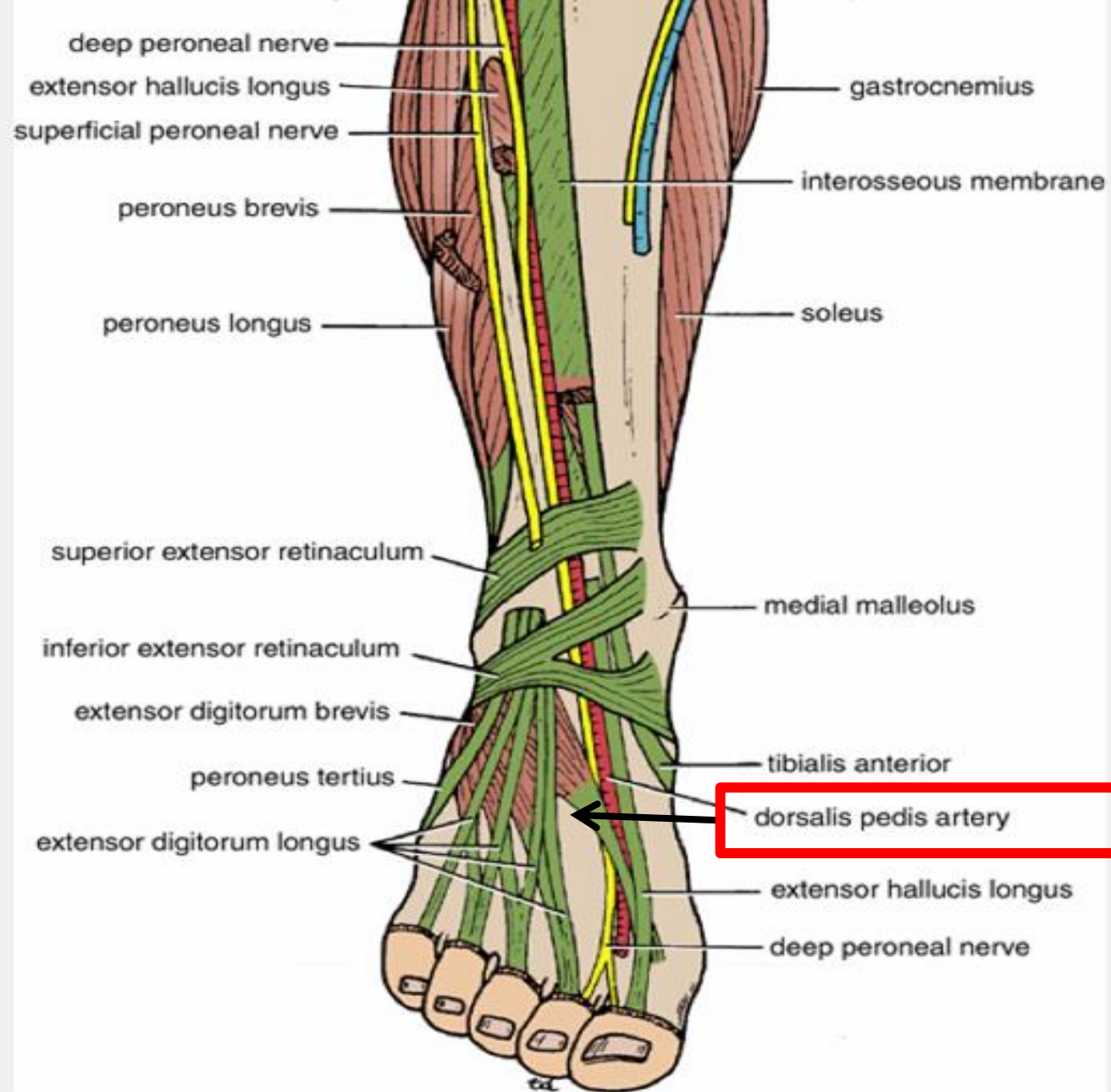
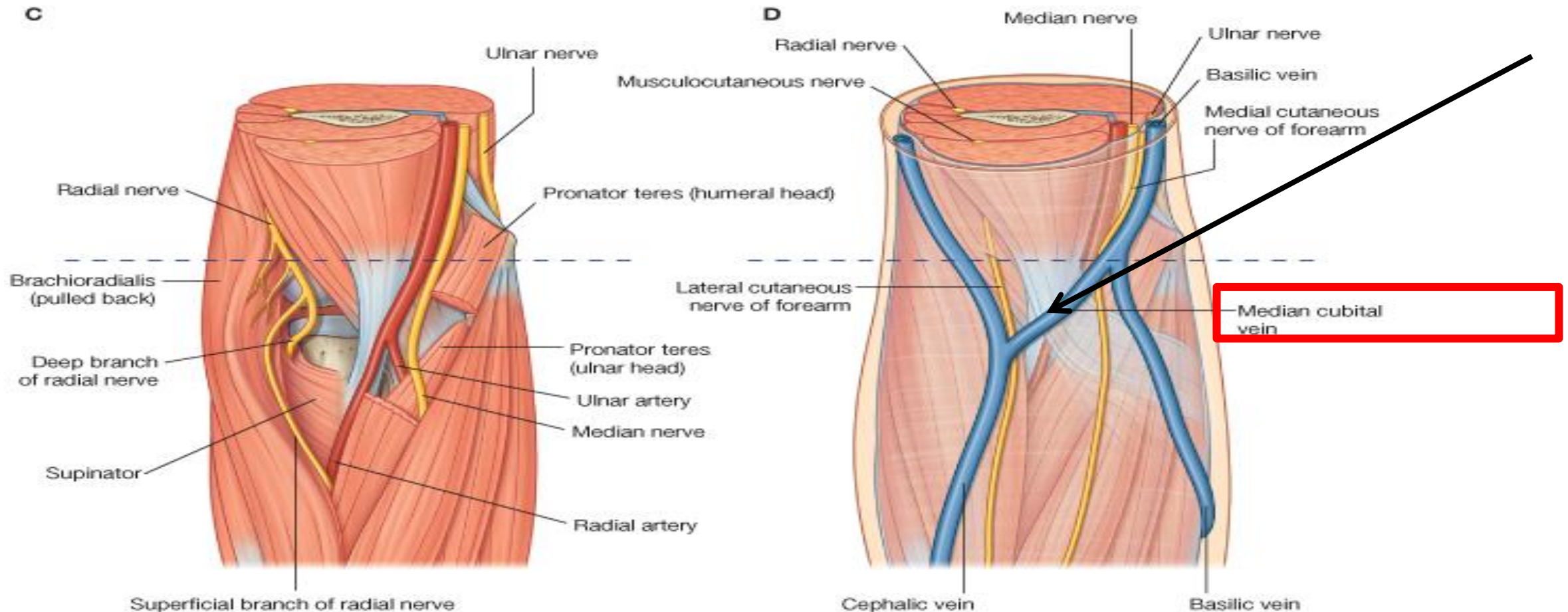
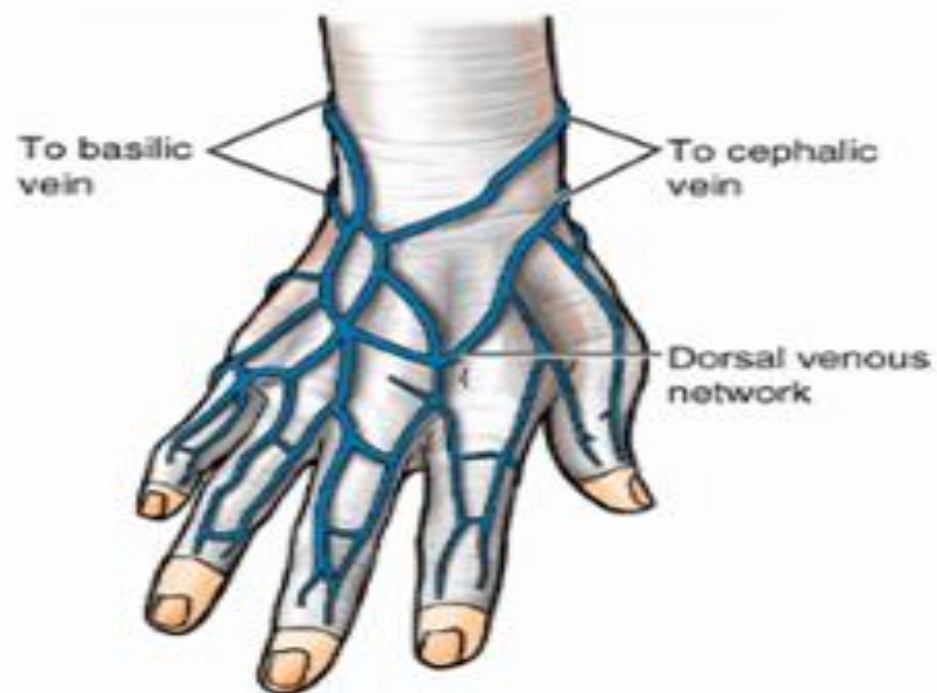


Figure 10-44 Deep structures in the anterior and lateral aspects of the right leg and the dorsum of the foot.

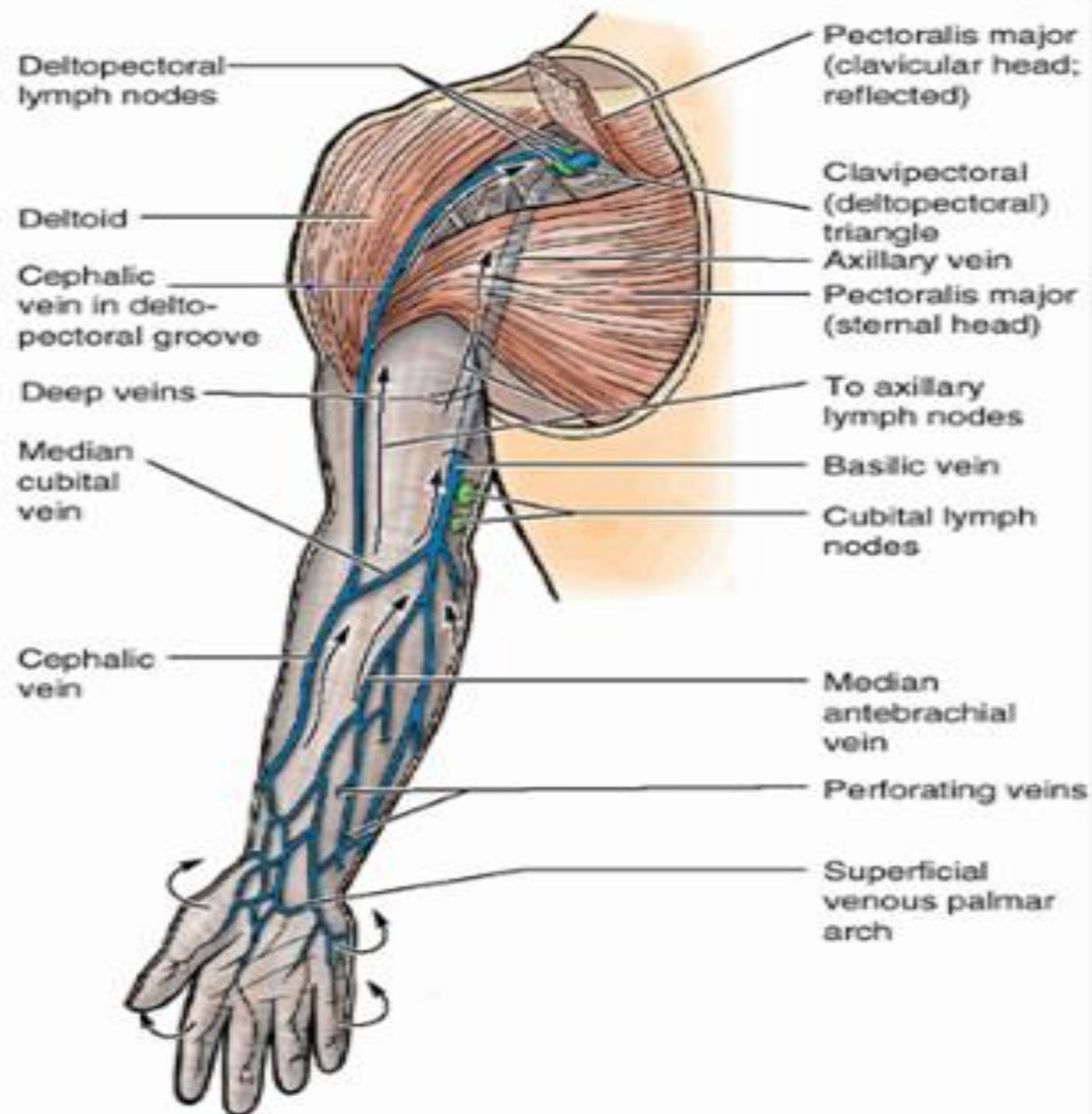
Intra Venous Injection In The Cubital Vein In The Cubital Fossa In Upper Limb

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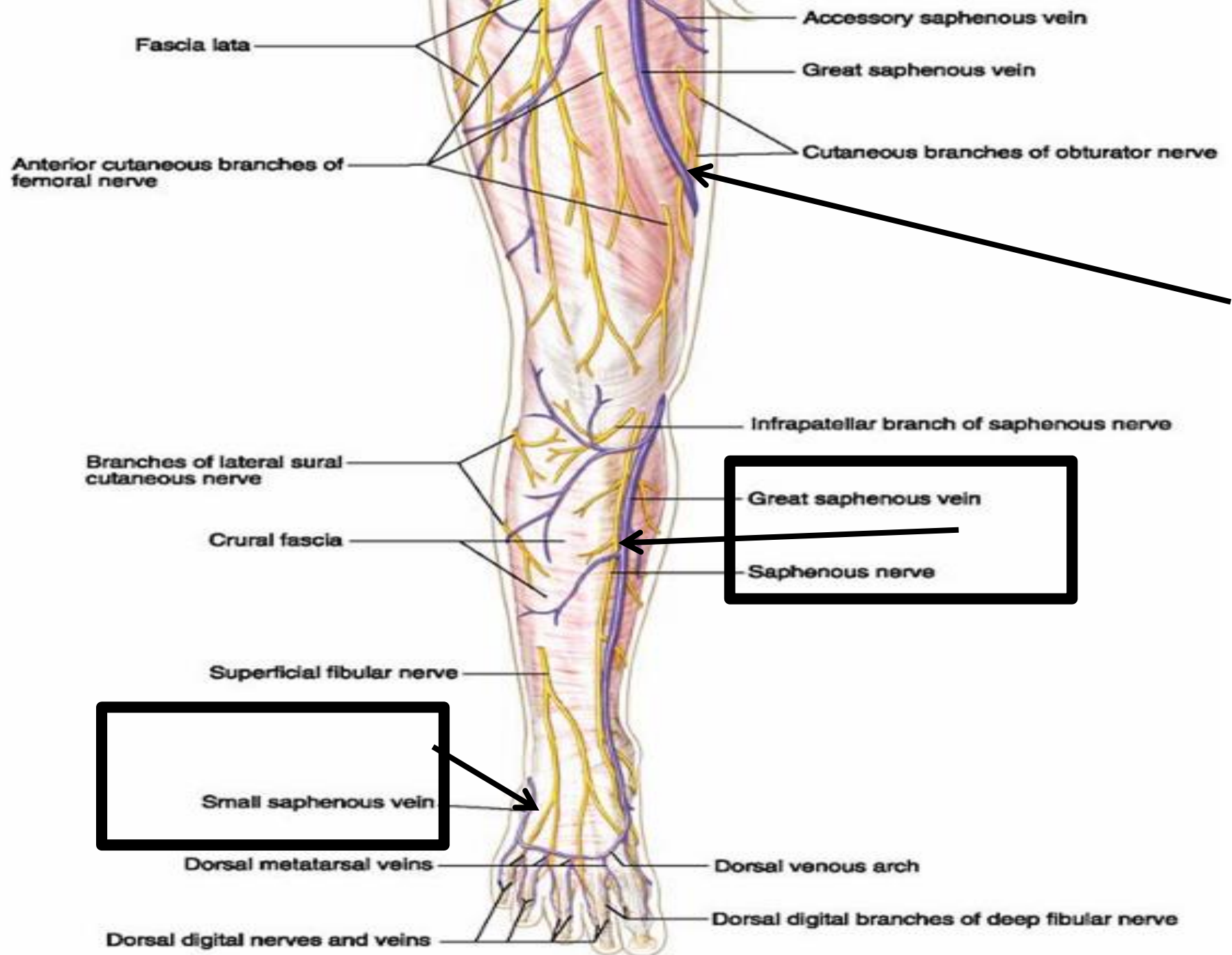




(A) Posterior (dorsal) views



(B) Anterior (palmar) views



Digestive System

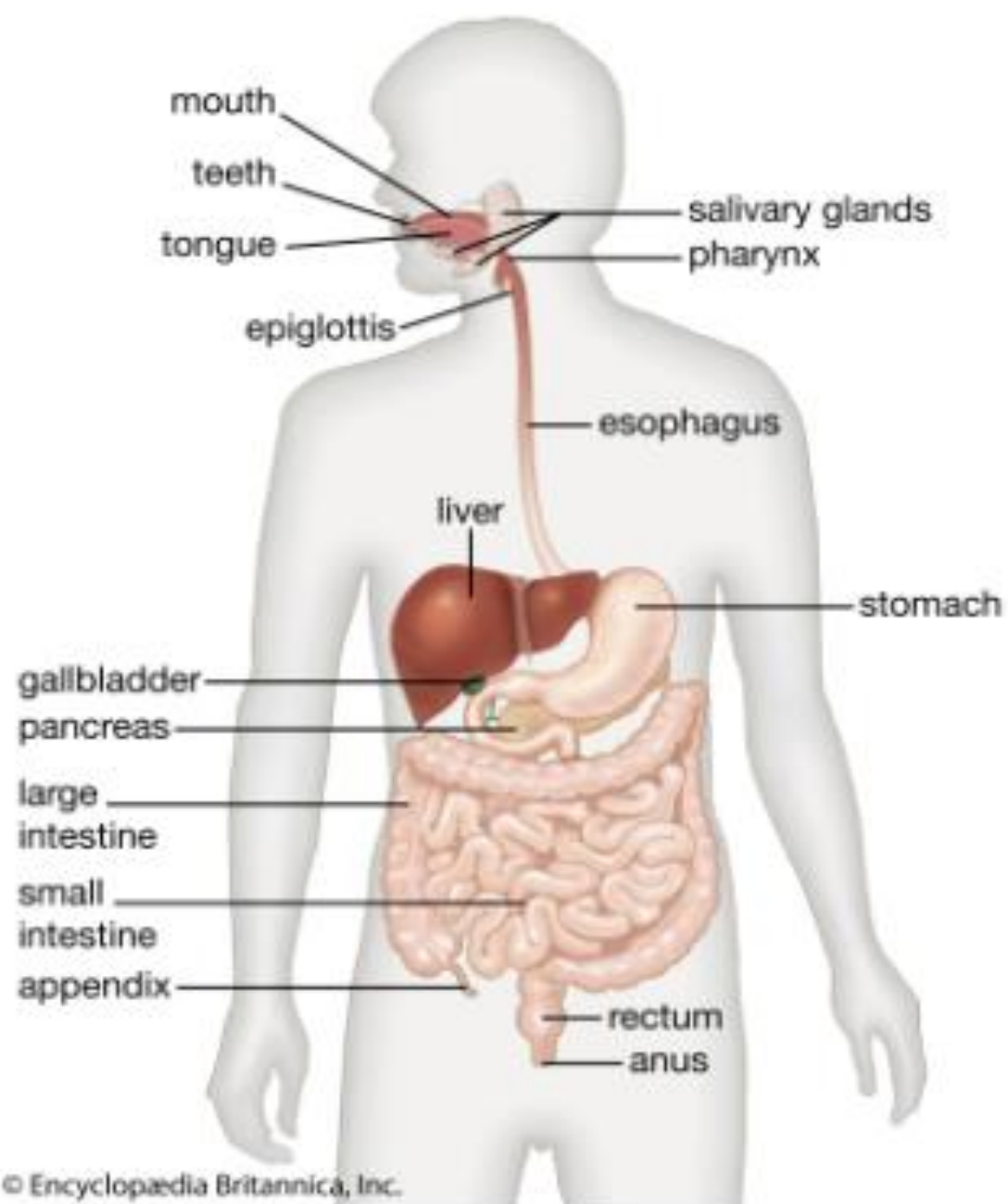
(Gastro Intestinal Tract)

=GIT

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Definition:

- **A long muscular tube with many sections and areas.**
- **Begins with the mouth and ends with the anus.**



Parts of The GIT

- A) Digestive Tract = Parts of the Digestive canal

1. Mouth
2. Pharynx
3. Esophagus
4. Stomach
5. Small Intestine
6. Large Intestine

- B) Accessory Parts: Organs that are not in the digestive tract but helps in the digestion

- Teeth •
- Tongue •
- Salivary glands •
- Liver •
- Gall bladder •
- Pancreas •

Mouth – (Oral Cavity). contain teeth & the tongue

Tongue: is a **skeletal** muscles move to •
assist in **chewing** covered by mucus
membrane

Functions •
chewing and swallowing •
taste. •
Speech •

3.Salivary Glands:

secrete saliva during meal time,.

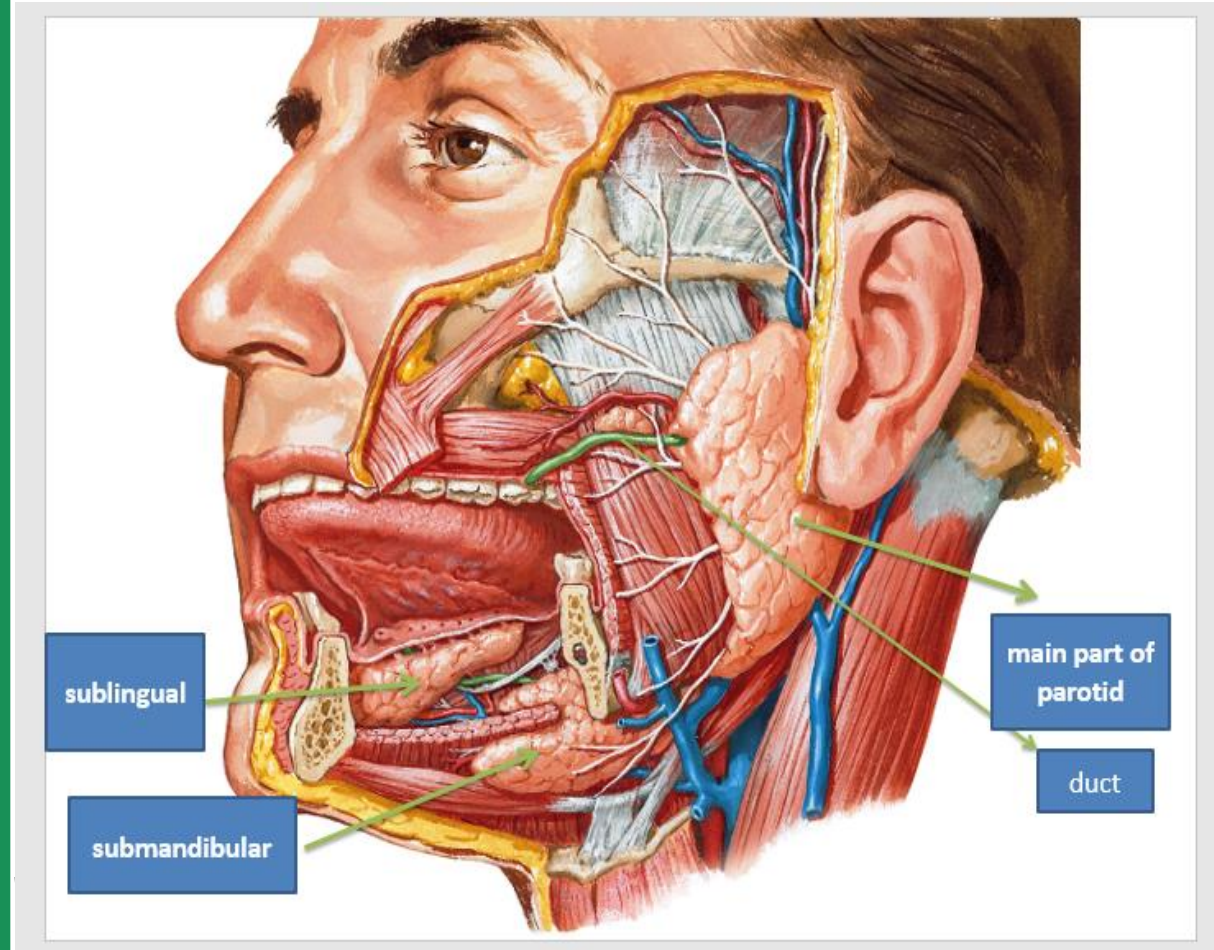
Three pairs of large salivary glands:

1-The Parotid Glands: Largest salivary glands.

below the ear. Its duct opens in the oral cavity. Site of mumps.

2-The Submandibular Glands: deep to the mandible.

3-The Sublingual Glands: deep to the tongue



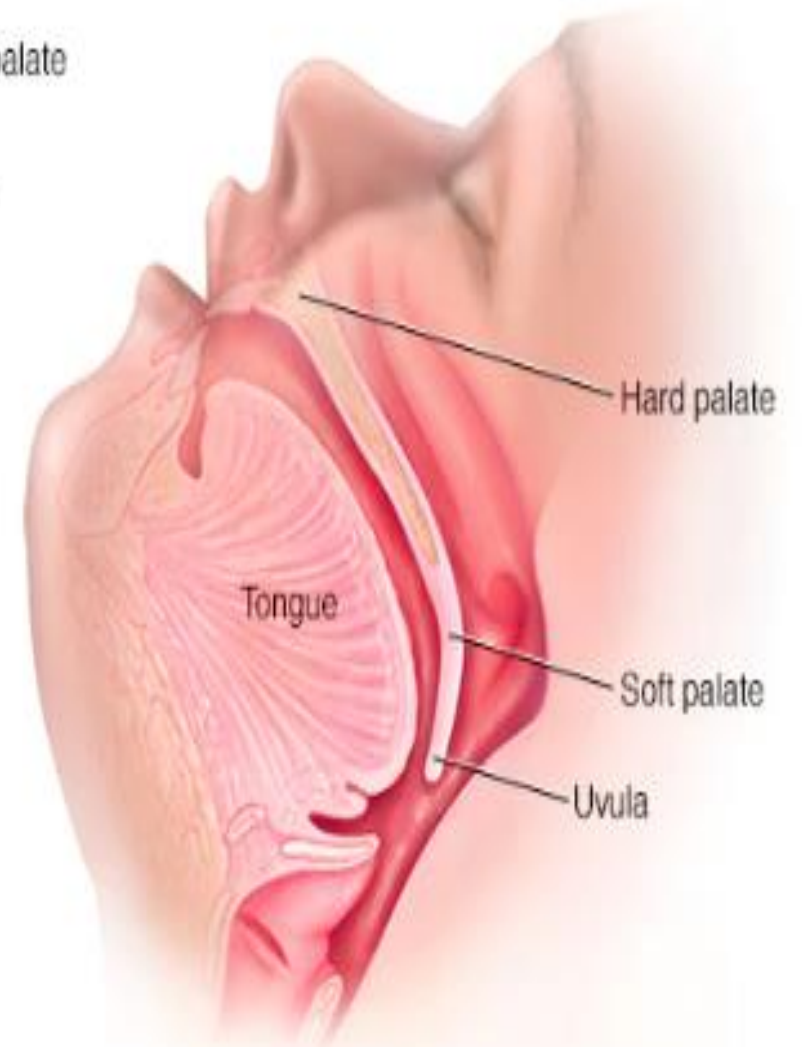
4-The palate:

4-The palate:

It forms the roof of the **mouth** and separates it from the **nose**.

It is formed of:

Hard palate and soft palate



5-Pharynx (Throat)

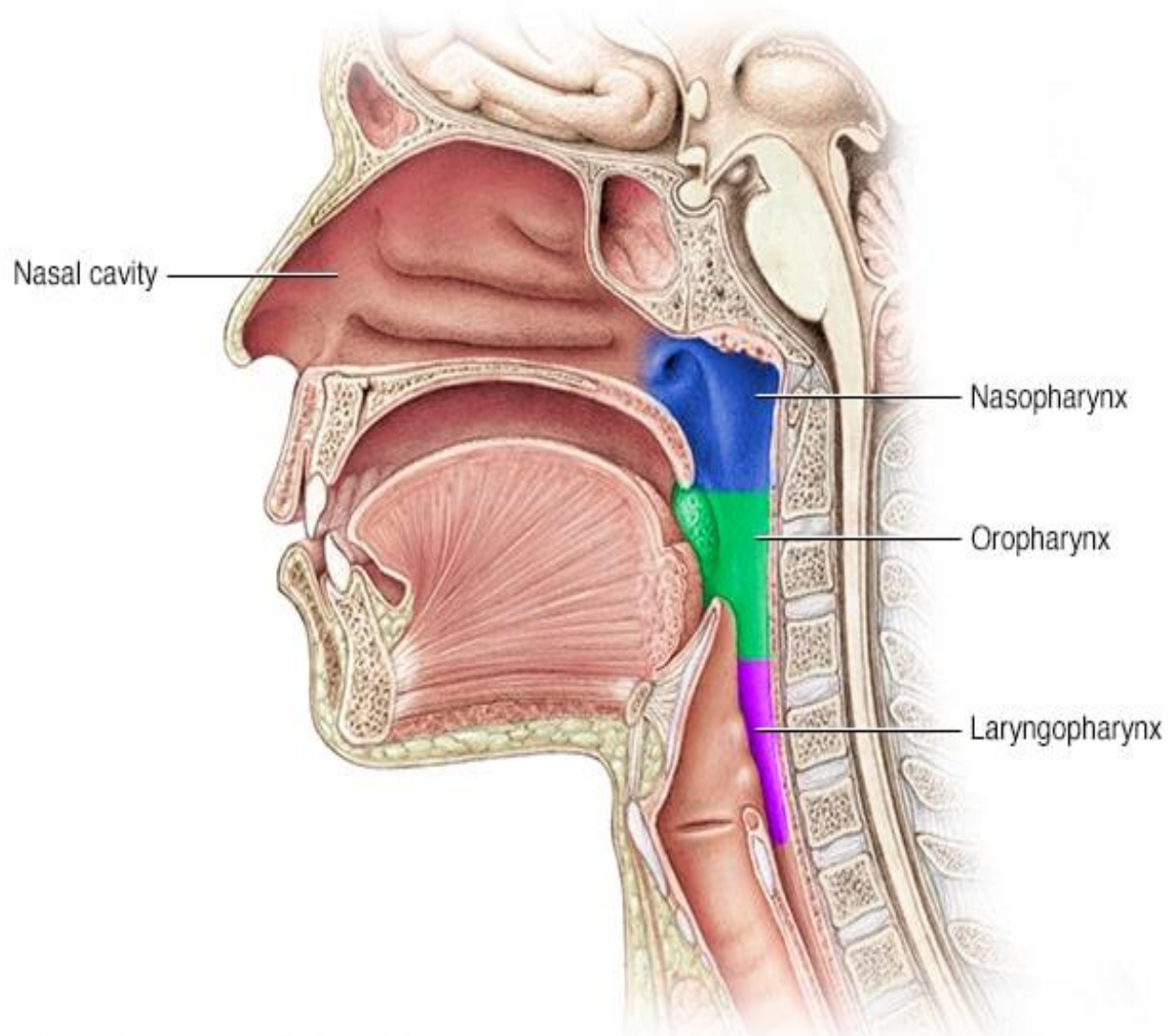
5-Pharynx • (Throat)

3 parts

Nasopharynx .1

oropharynx .2

laryngopharynx
x



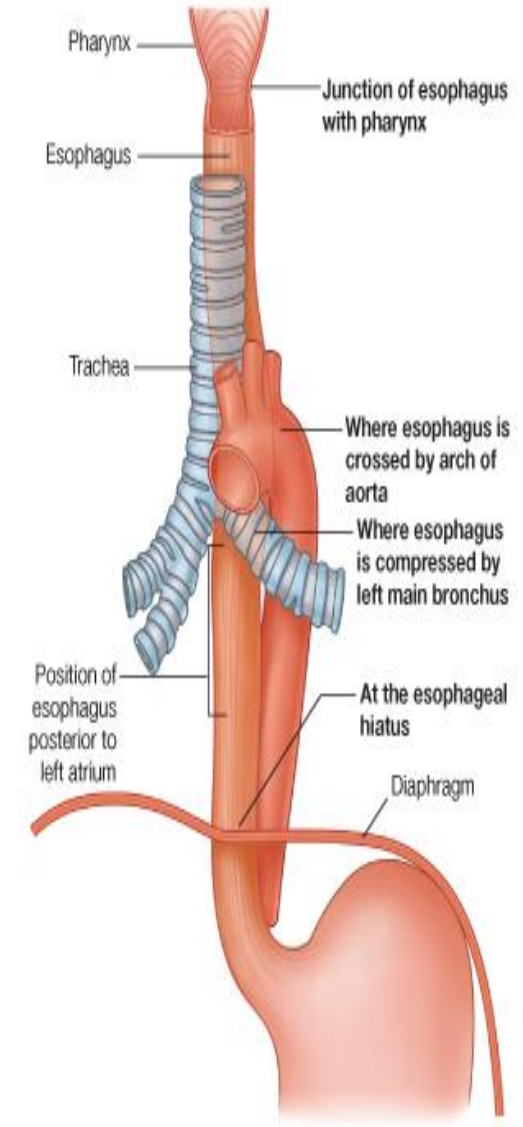
6-Esophagus

**Def : Muscular tube •
=25 cm long.**

**Sit : from pharynx to the •
stomach.**

**function : pushes food •
into the pharynx**

**3 parts (neck , thorax •
&abdomen)**



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Figure 3.90 Sites of normal esophageal constrictions.

7-stomach:

The stomach is described as having:

2 openings, the cardiac and pyloric orifices.

2 borders, the greater and lesser curvatures.

2 surfaces, an anterior and a posterior surface .

The stomach is divided into the following parts:

Fundus: dome-shaped

Body: the largest part .

Pylorus: has
(pyloric sphincter),

2 ends

Cardia: has no sphincter

Pylorus: strong sphincter

4 Parts

Fundus (above cardia)

Body

Pyloric antrum

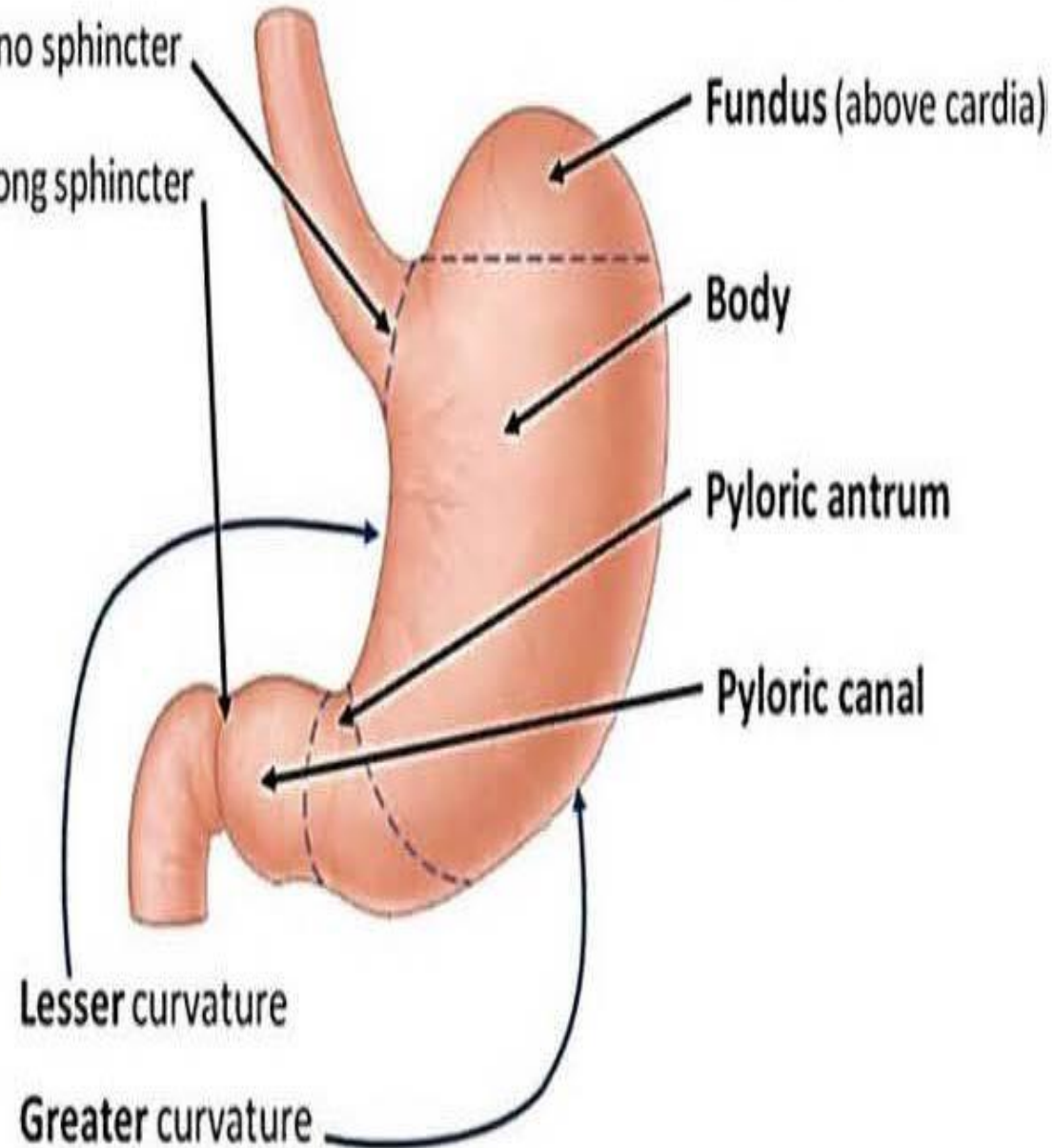
Pyloric canal

2 borders

Lesser curvature

Greater curvature

Fig. A20



Small Intestine

def : Coiled tube

size : 6 meters

Site : **from** stomach **to** the large intestine,

It occupies **a middle portion** of the abdominal cavity.

function : **Finishes the digestion** process and is responsible for **absorbing** most of the nutrients and most of the water.

parts of the Small Intestine

1-The duodenum is C shape.

2-The jejunum

3-The ileum

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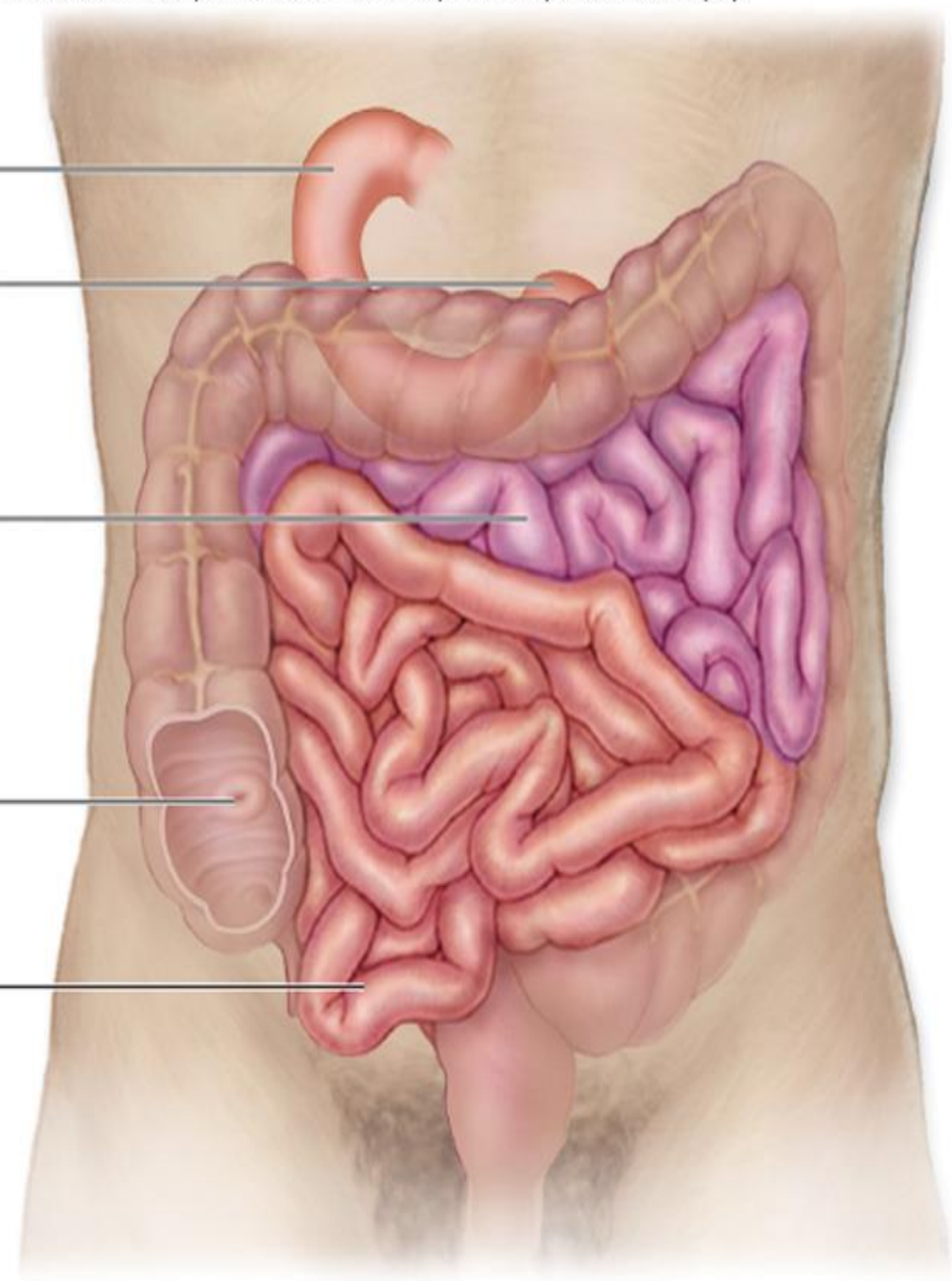
Duodenum

Duodenojejunal
flexure

Jejunum

Ileocecal valve

Ileum



9-Large Intestine

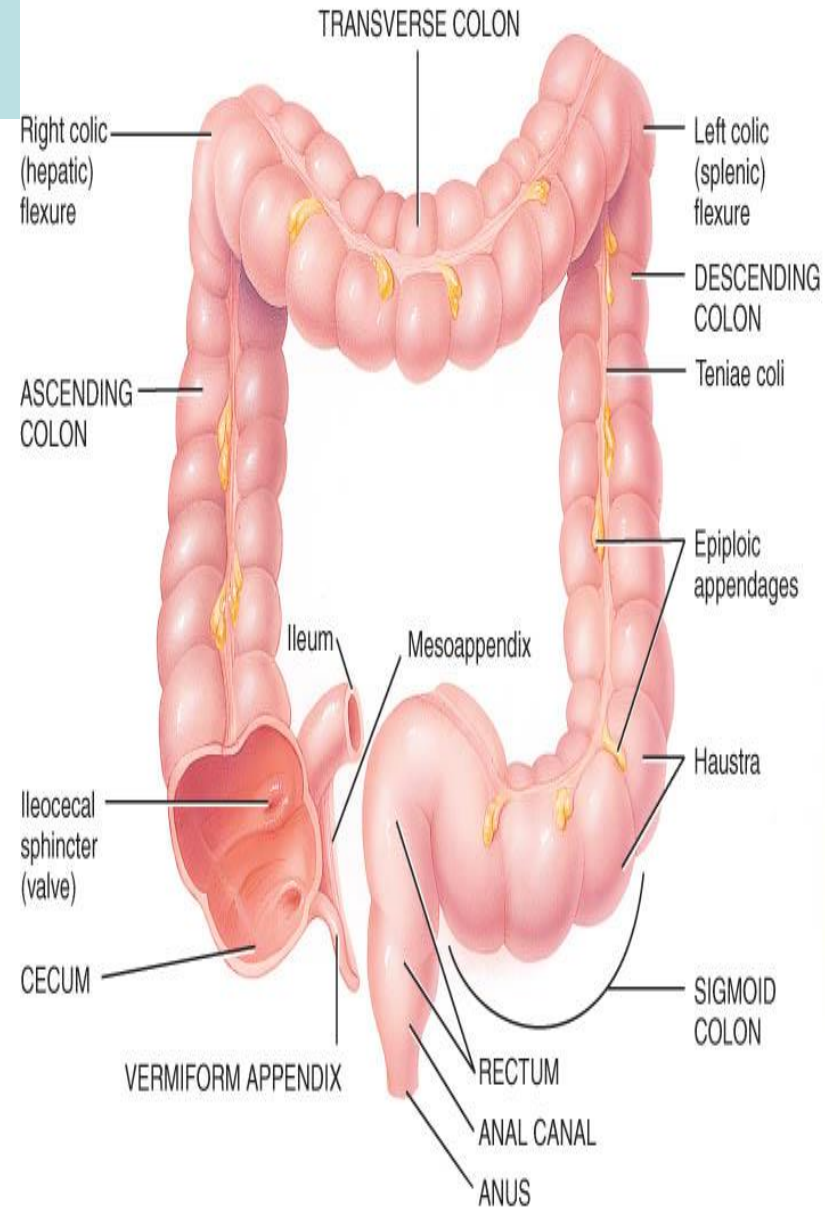
Formed of four parts :
the cecum, colon, rectum,
anal canal

Colon 3 parts
• ascending, transverse
& (ascending colon) + sigmoid
colon

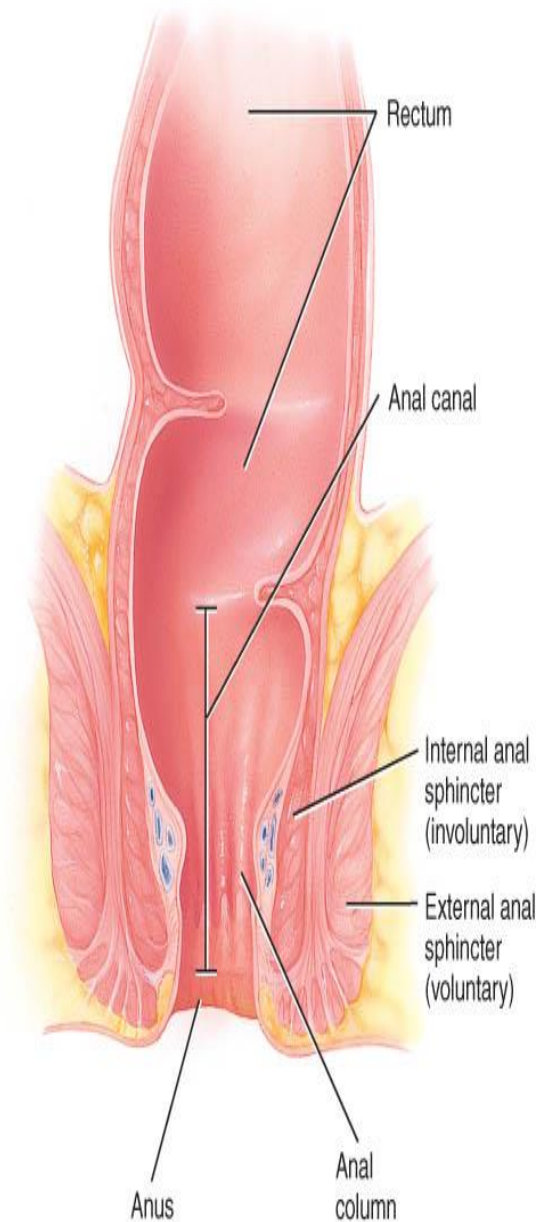
• larger diameter

• **Water** is absorbed from the
undigested food making the
waste

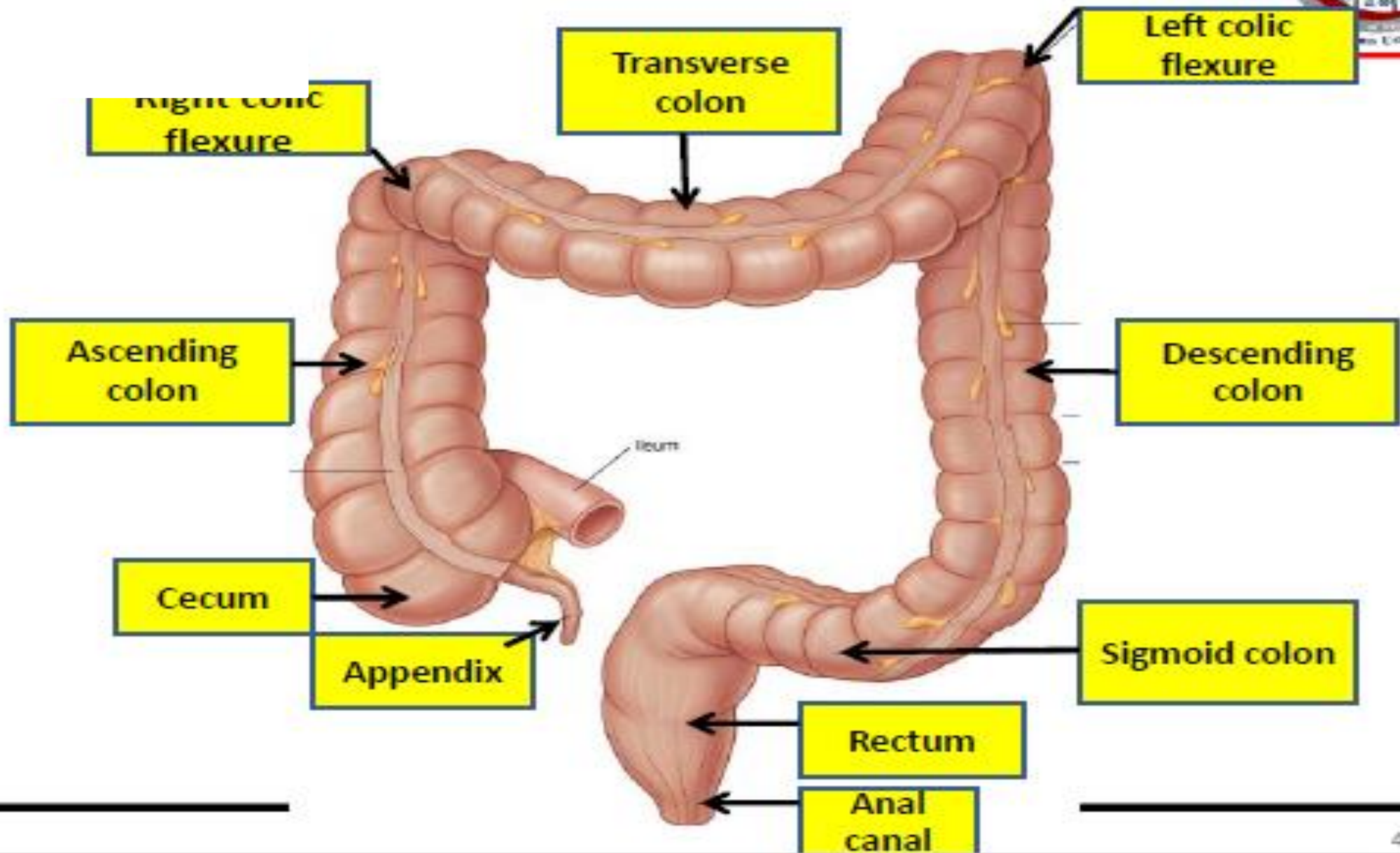
• Solid **waste** stays in the rectum
until it is excreted through the
anus as feces.



(a) Anterior view of large intestine showing major regions



(b) Frontal section of anal canal



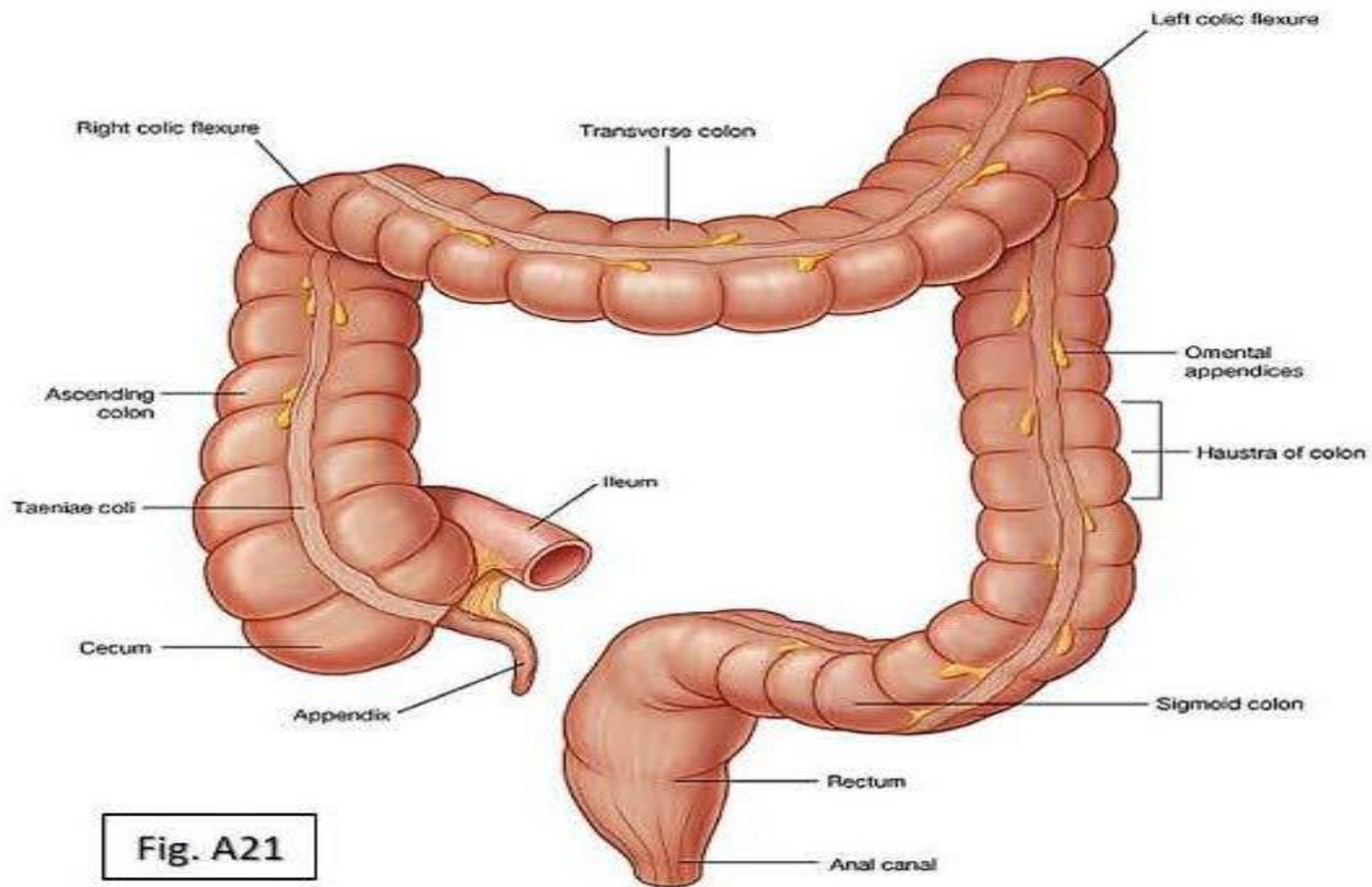
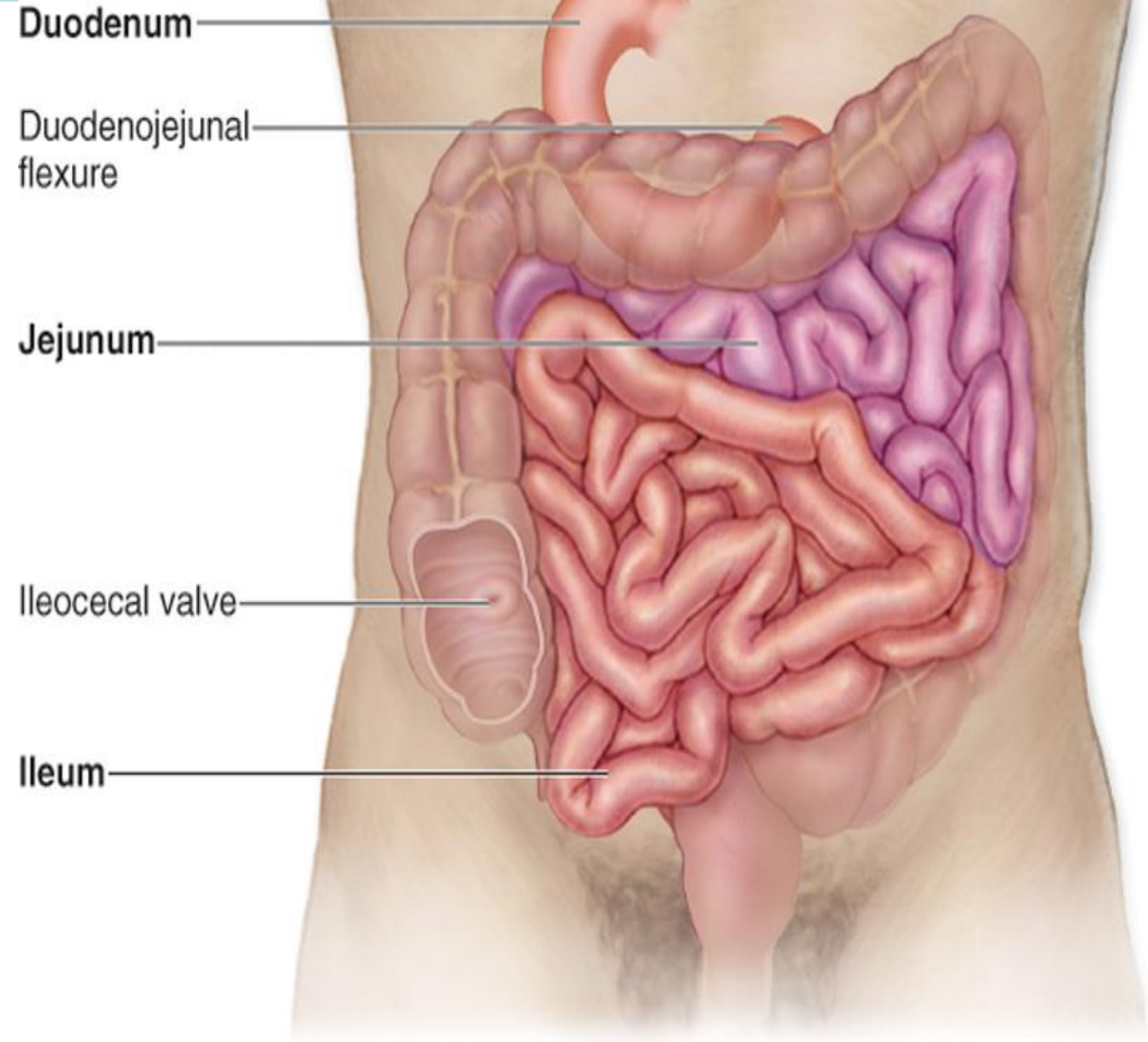


Fig. A21

10-Appendix

- **10-Appendix**
- A finger sized
- at the beginning of the colon.
- blocked by seeds) and then bacteria result in inflammation of the appendix or **appendicitis**.



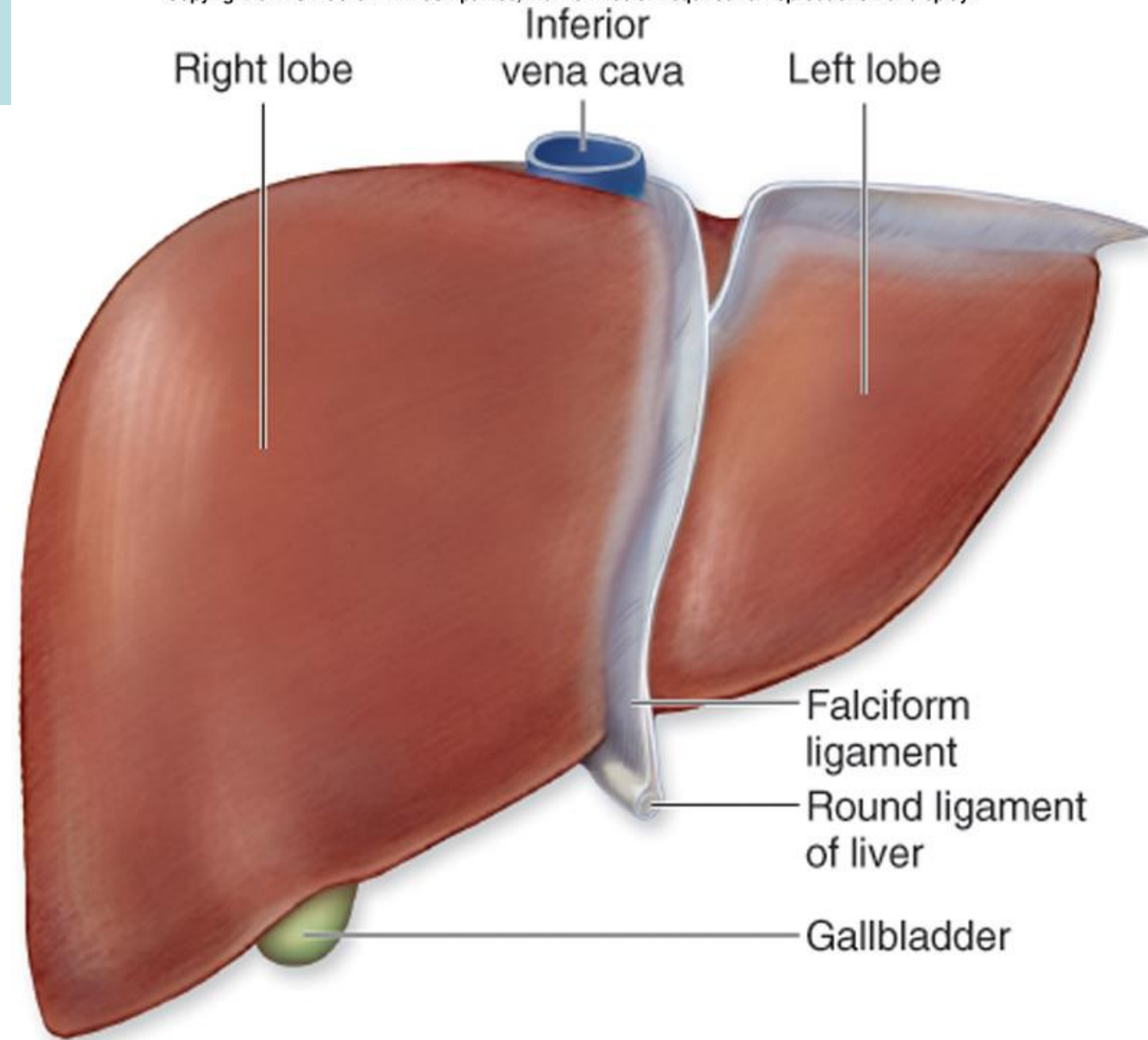
Liver

- Situated on the right side of the abdominal cavity just under the diaphragm.

- It is the largest gland in the body.

- One of its functions is to secrete bile to the gall bladder

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(a) Anterior view

LIVER

Formed of four lobes:

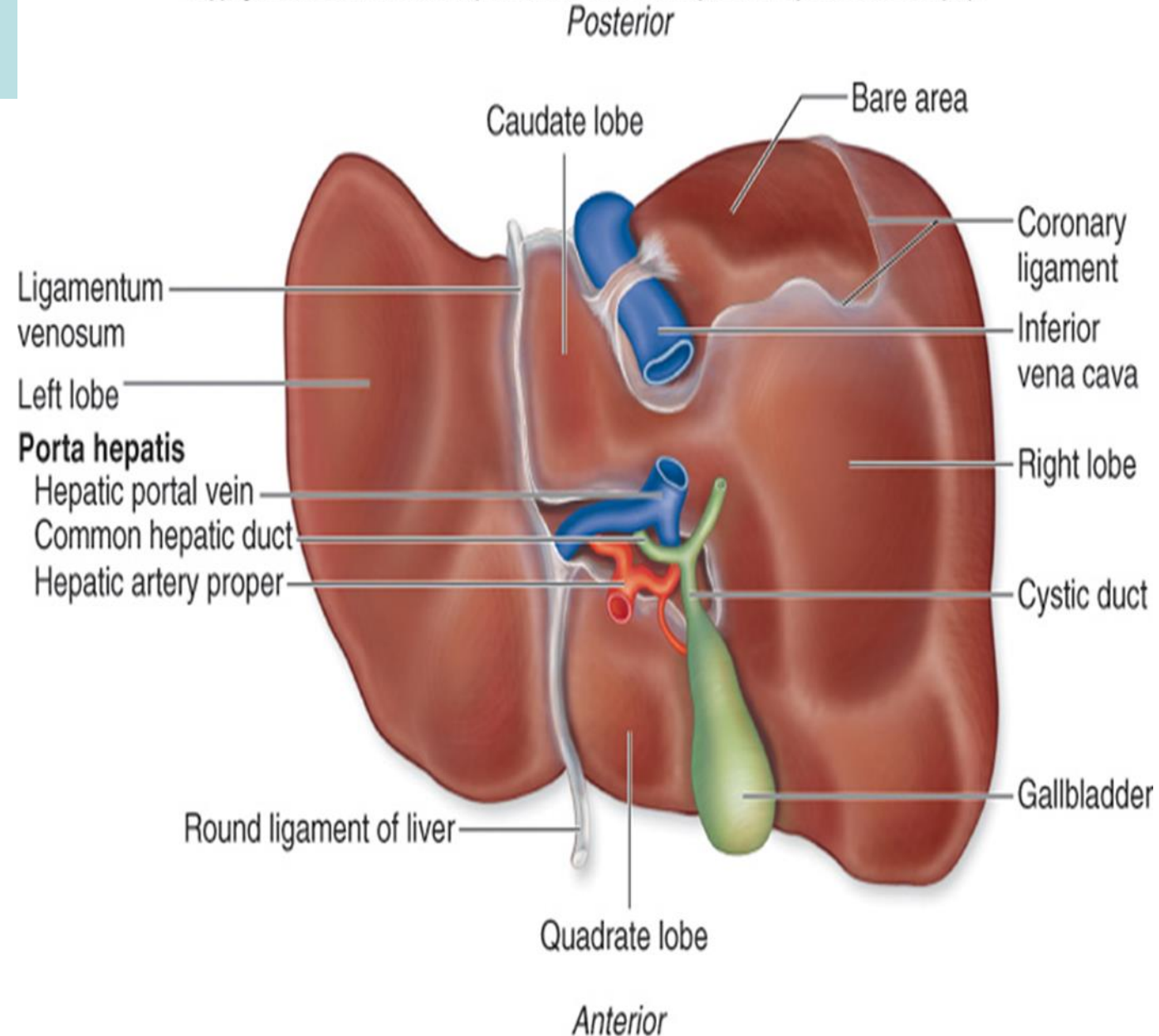
Right lobe .1

Left lobe .2

Caudate lobe .3

Quadrate lobe. .4

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(b) Posteroinferior view

The porta hepatis

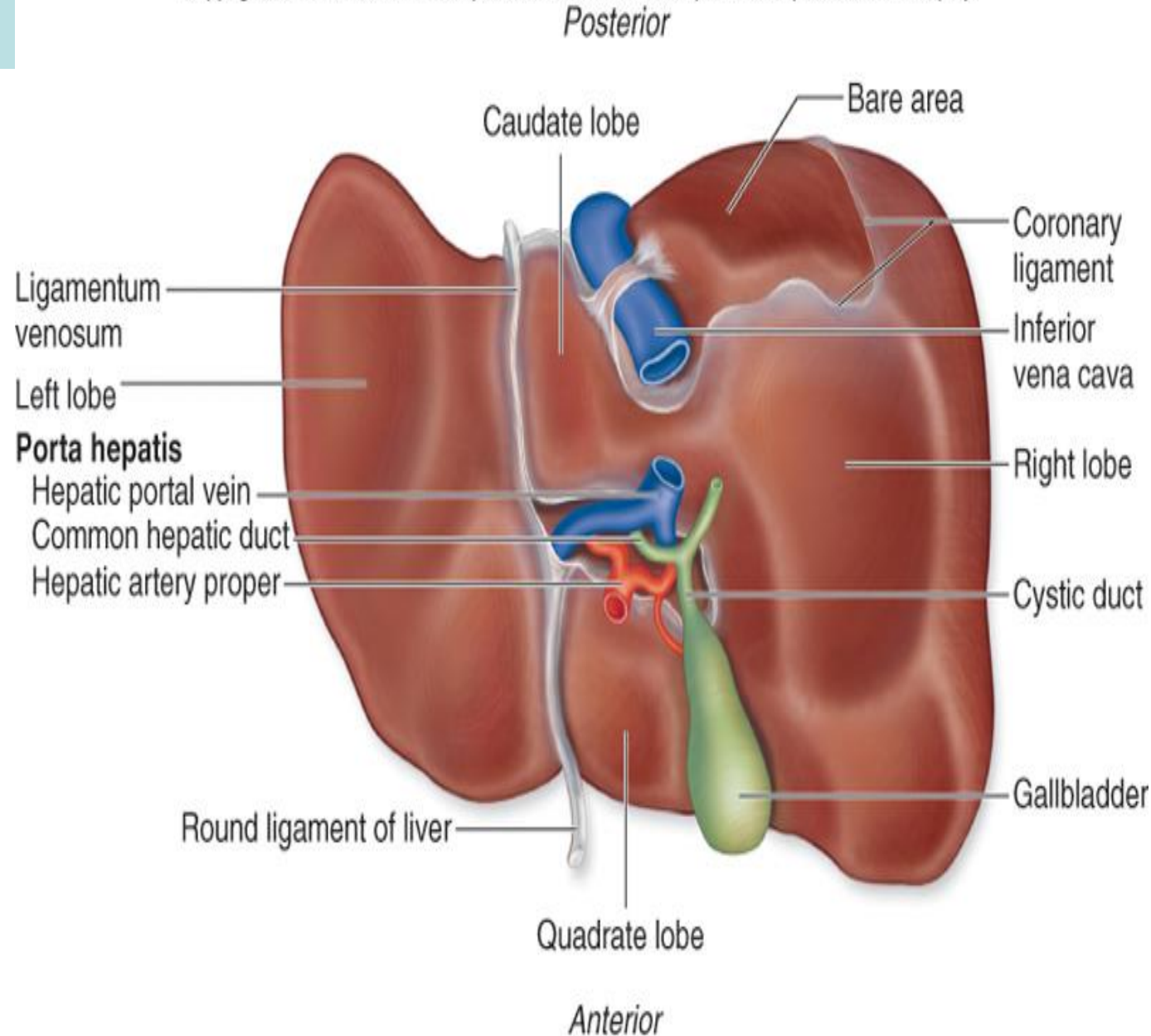
The porta hepatis: •
have 3

the right and left .1
hepatic ducts

the right and left .2
branches of the
hepatic artery

the portal vein .1

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(b) Posteroinferior view

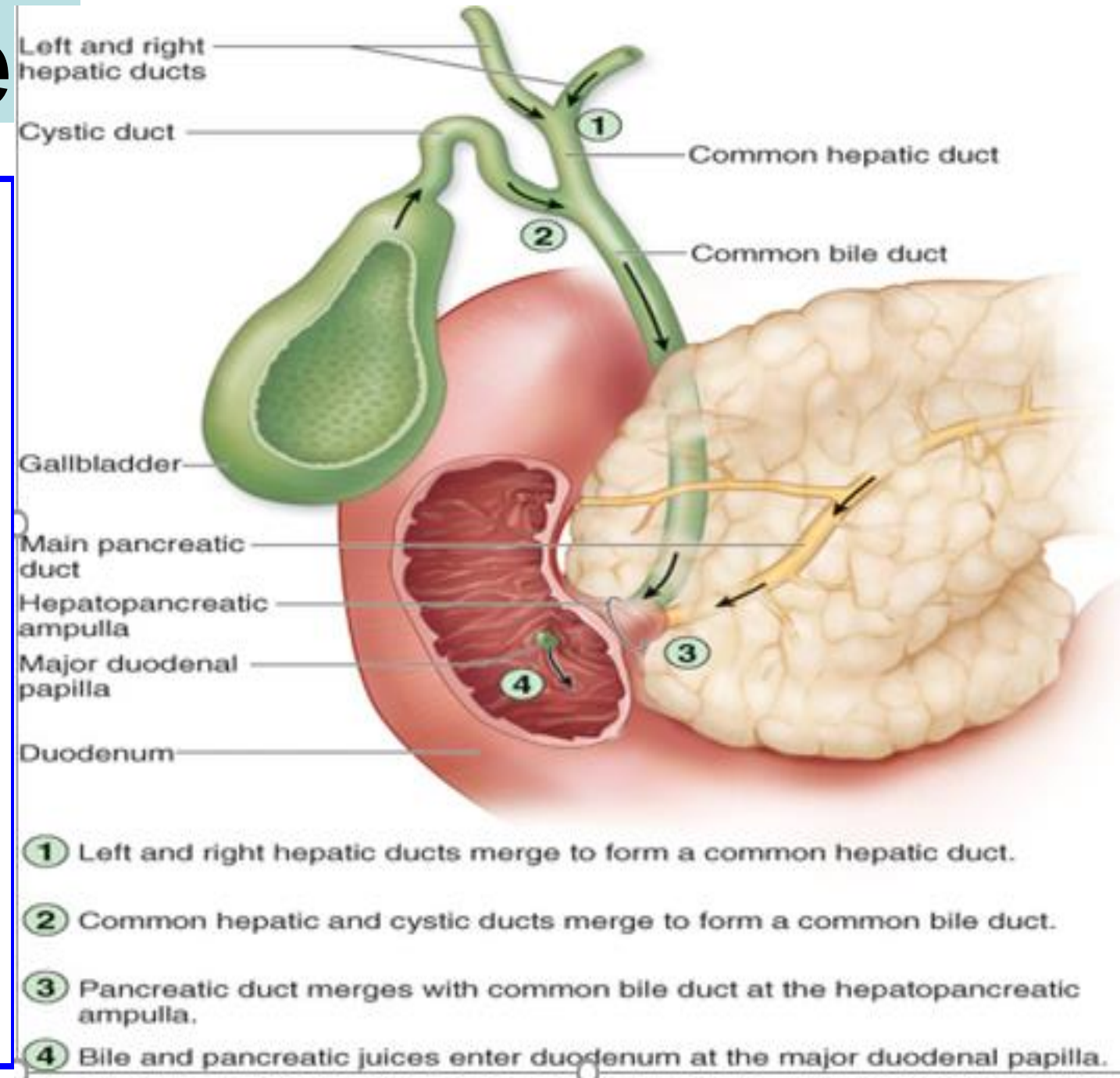
12-Gall Bladder

- A pear shaped sac
- under the liver.

- It is a reservoir for **bile**

- The presence of fat in the (duodenum) stimulates bile.

- Bile is transported from
gallbladder ---
liver ---duodenum



14-Pancreas

- The pancreas is about 15 cm
- It lies **behind the stomach**
- on the **posterior abdominal wall**,
- from the **duodenum to the spleen**
- The pancreas is divided into 4 head, neck, body, and tail
- The pancreas is **both an exocrine and an endocrine gland.**

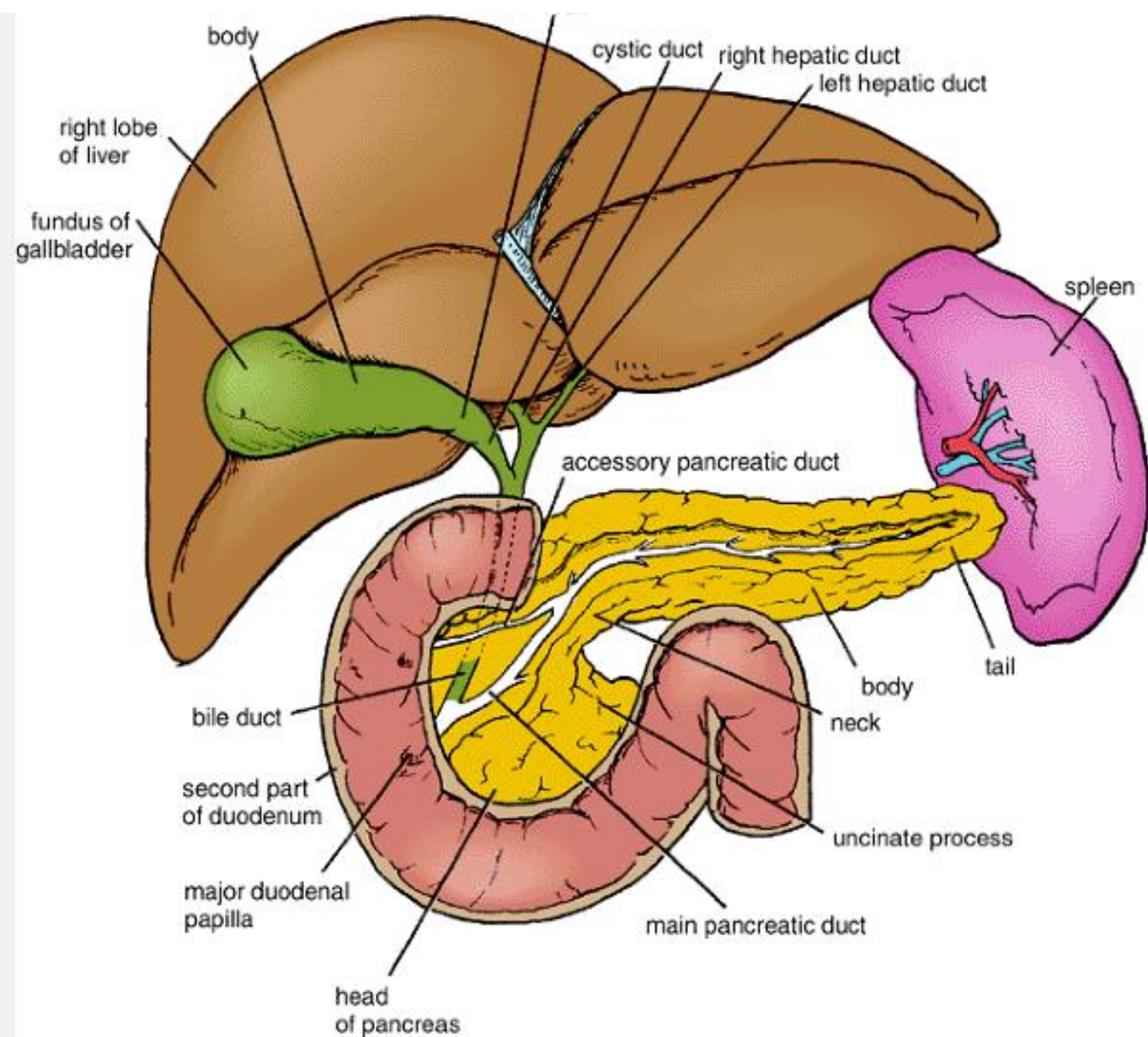
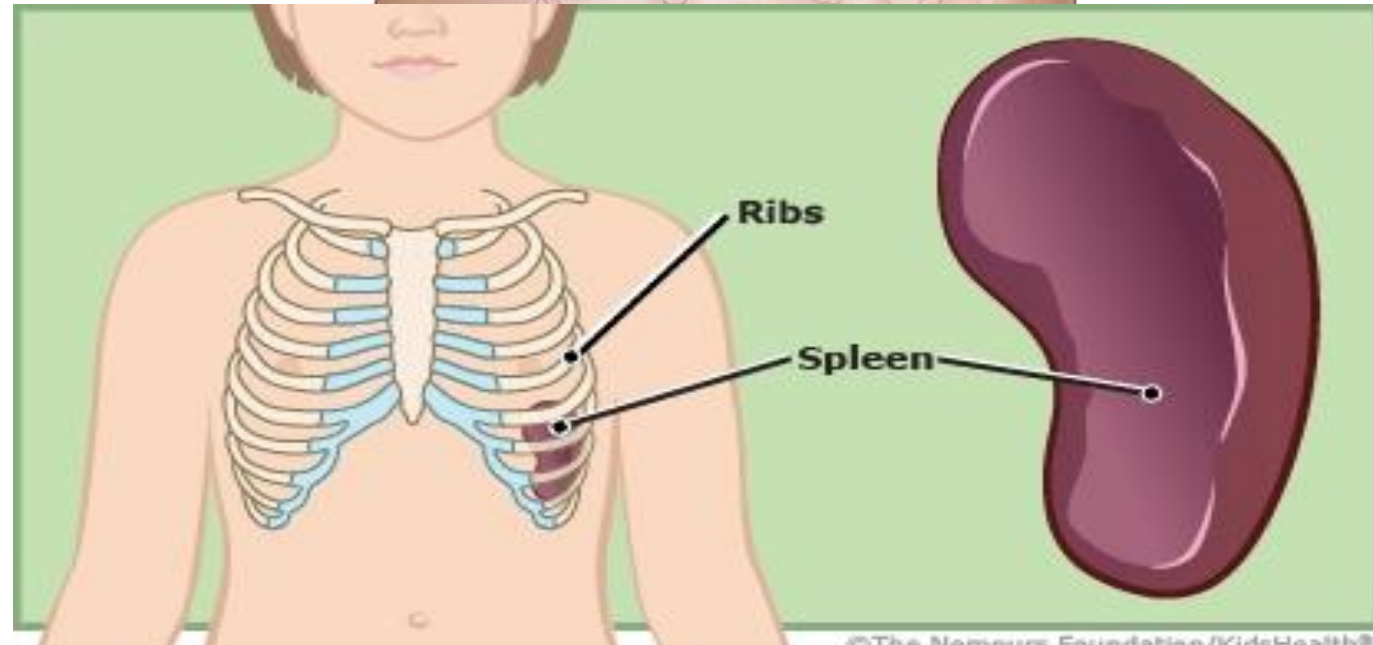
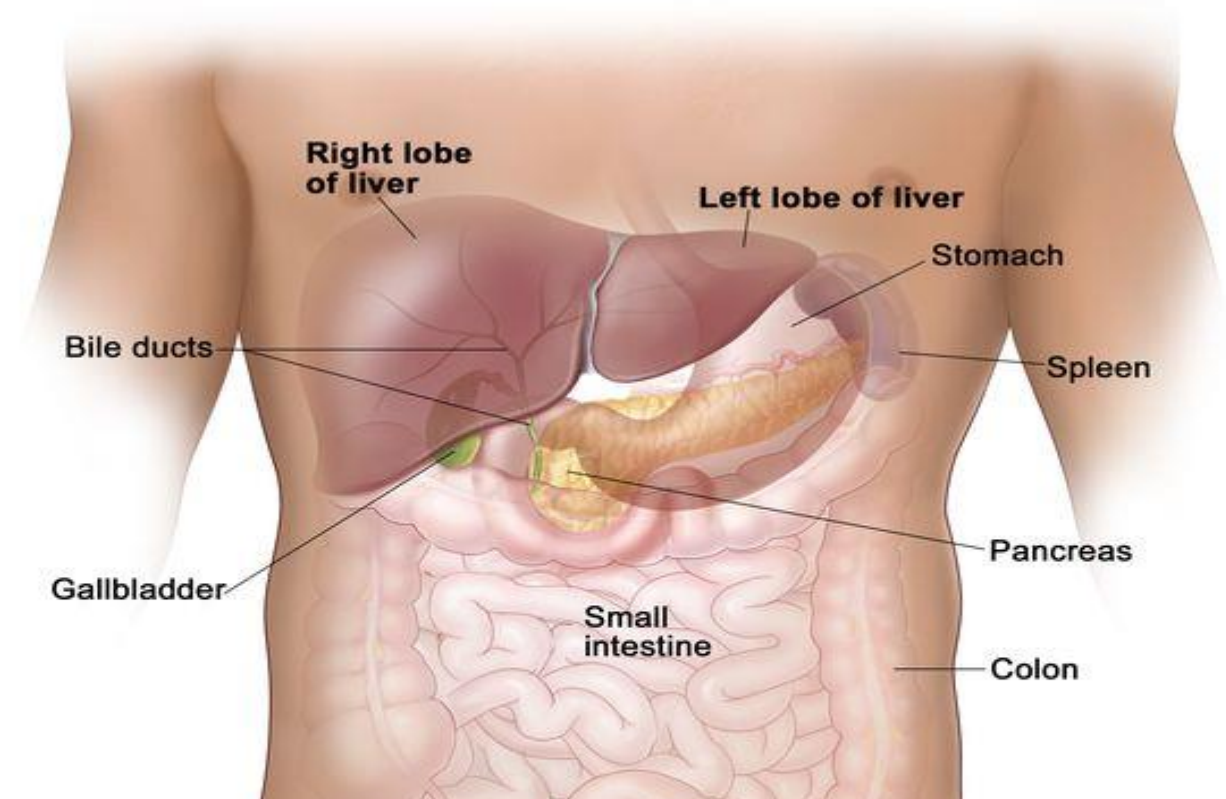


Figure 5-58 Different parts of the pancreas dissected to reveal the duct system.

Spleen

Spleen: in
the left side
of upper
abdomen
Filters and
stores blood.

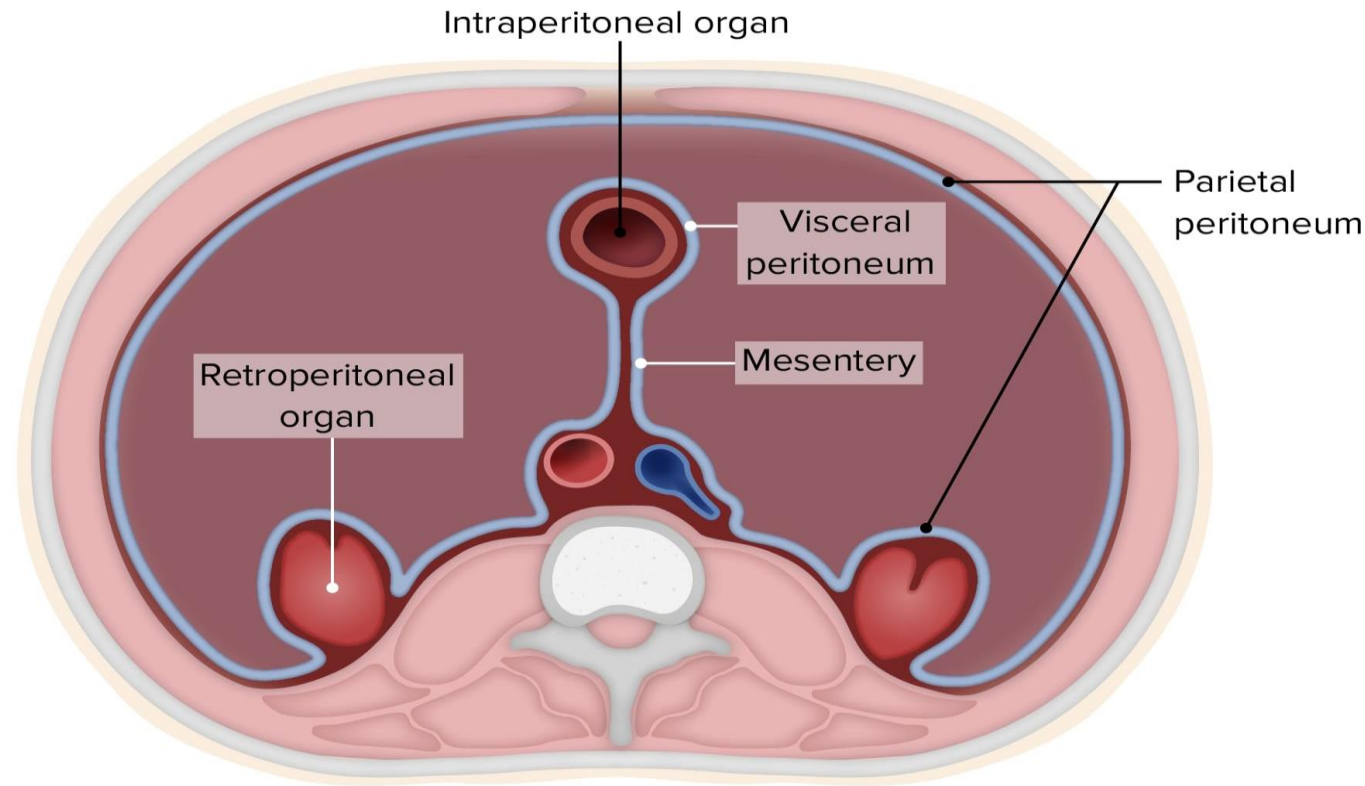


Peritoneum

the organs of GIT •
are **covered with a**
membrane called the
peritoneum.

This membrane .1
secretes **a fluid**
which

lubricates the .2
organs prevents
friction.



Peritoneum

ANATOMY

